



Universe 3 Underwriter Manual

Reinsurance treaty & facultative pricing platform

AUDIENCE	Treaty and facultative underwriters, chief underwriters, and peer reviewers
COVERS	Proportional treaty · Non-proportional treaty · Facultative · Claims · Finance · Administration
SCREENS	68 screens, captured from a running instance with a seeded portfolio
REVISION	Revision 1.0

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1. What this tool is

In one sentence

Universe 3 is the workspace where an underwriter takes a reinsurance submission from the broker's slip all the way to a priced, peer-reviewed offer — and then keeps the record of how that price was arrived at.

It covers three products end to end. **Proportional treaties** (quota share, surplus, and combinations) are priced from development triangles and treaty terms. **Non-proportional treaties** (risk XL, cat XL, aggregate XL, stop loss) are priced layer by layer from burning cost, severity curves and exposure rating. **Facultative risks** are priced per location from a rating engine driven by occupancy, COPE quality and an underwriting-factor score.

Around those three sit the supporting modules: a portfolio dashboard, an approvals queue that enforces authority limits, a claims register, a finance ledger of signed treaties, user and mandate administration, and an actuarial formula workbench holding the benchmark parameters the pricing screens draw on.

The shape of the work

Every treaty and every fac risk is a **wizard**: an ordered set of screens down the left-hand sidebar, grouped into stages. You do not have to walk them in order — the sidebar lets you jump — but the order encodes the dependency chain. Losses are identified before development factors are chosen, because the factor screens offer you a triangle with those losses stripped out. Development factors are chosen before the summaries, because the summaries project on them. Everything lands on a pricing screen, and the pricing screen is the only place an offer can be made.

The platform is deliberately dual-track. A live, bound treaty is a **contract**. A working, pre-bind version is a **quote**. Most screens operate on the same business concept in either mode; the quote workflow is a slimmed-down subset of the treaty workflow.

What it is not

It is not a catastrophe model. It *consumes* vendor model output (AIR, RMS event loss tables) and CRESTA aggregates, and it does deterministic earthquake exposure rating from GEM/HAZUS damage-ratio curves, but it does not run a stochastic cat model itself.

It is not a policy administration system. It prices and records the reinsurance decision; the underlying policy data arrives as risk profiles, claims profiles and loss listings pasted or imported from the cedant's workbooks.

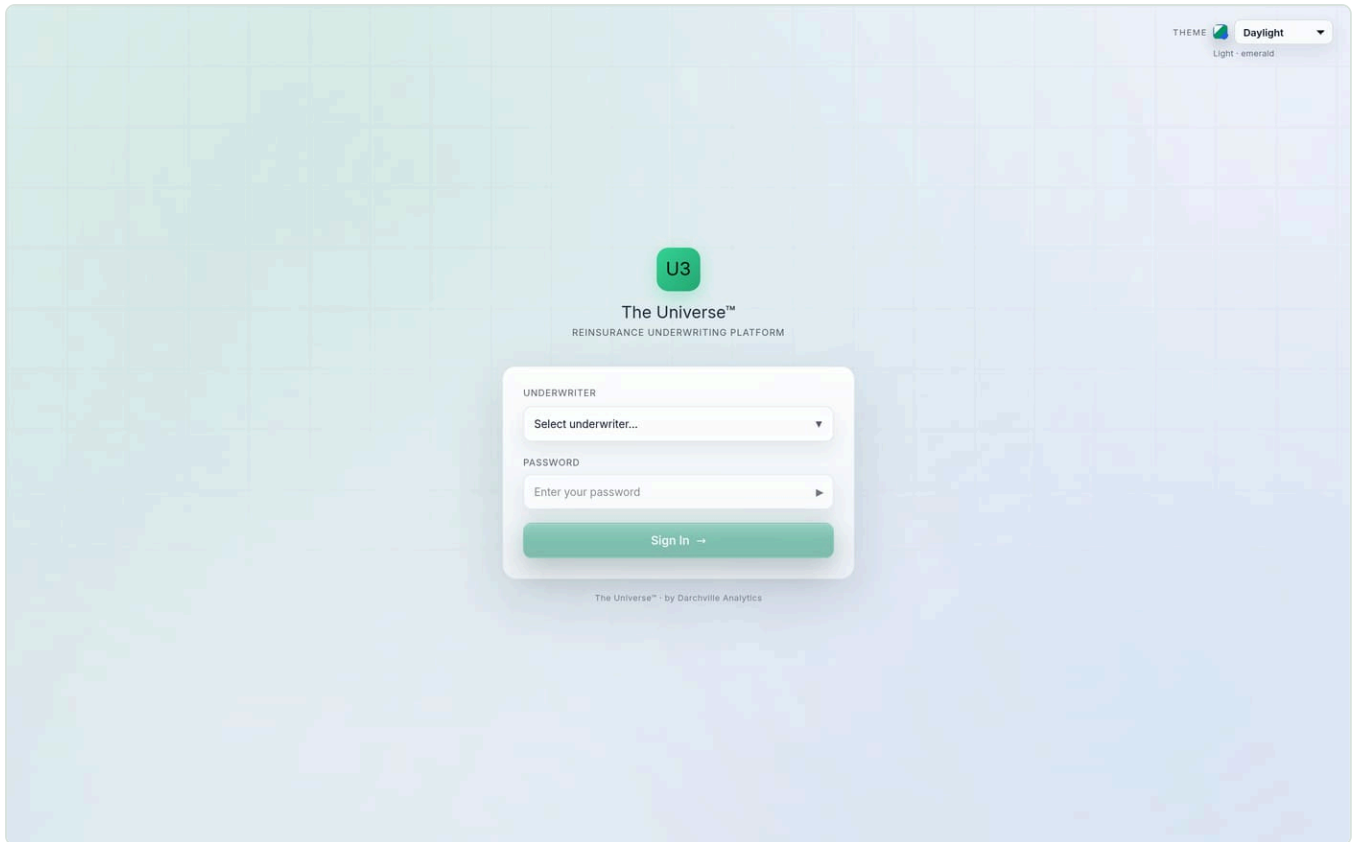
2. Getting in and finding your way

Signing in

The login screen lists people by name only — no roles are shown. Pick your name, enter your password, and sign in. Your role, authority limit and mandate come back from the server with the session and quietly drive what you can see and approve from that point on.

If your organisation has single sign-on enabled, the password box is replaced by an SSO hand-off and you are returned here after your identity provider has authenticated you. A first-time or reset account is forced through a password change before the app will let you past.

The way in. Choose your name from the underwriter list and enter your password.

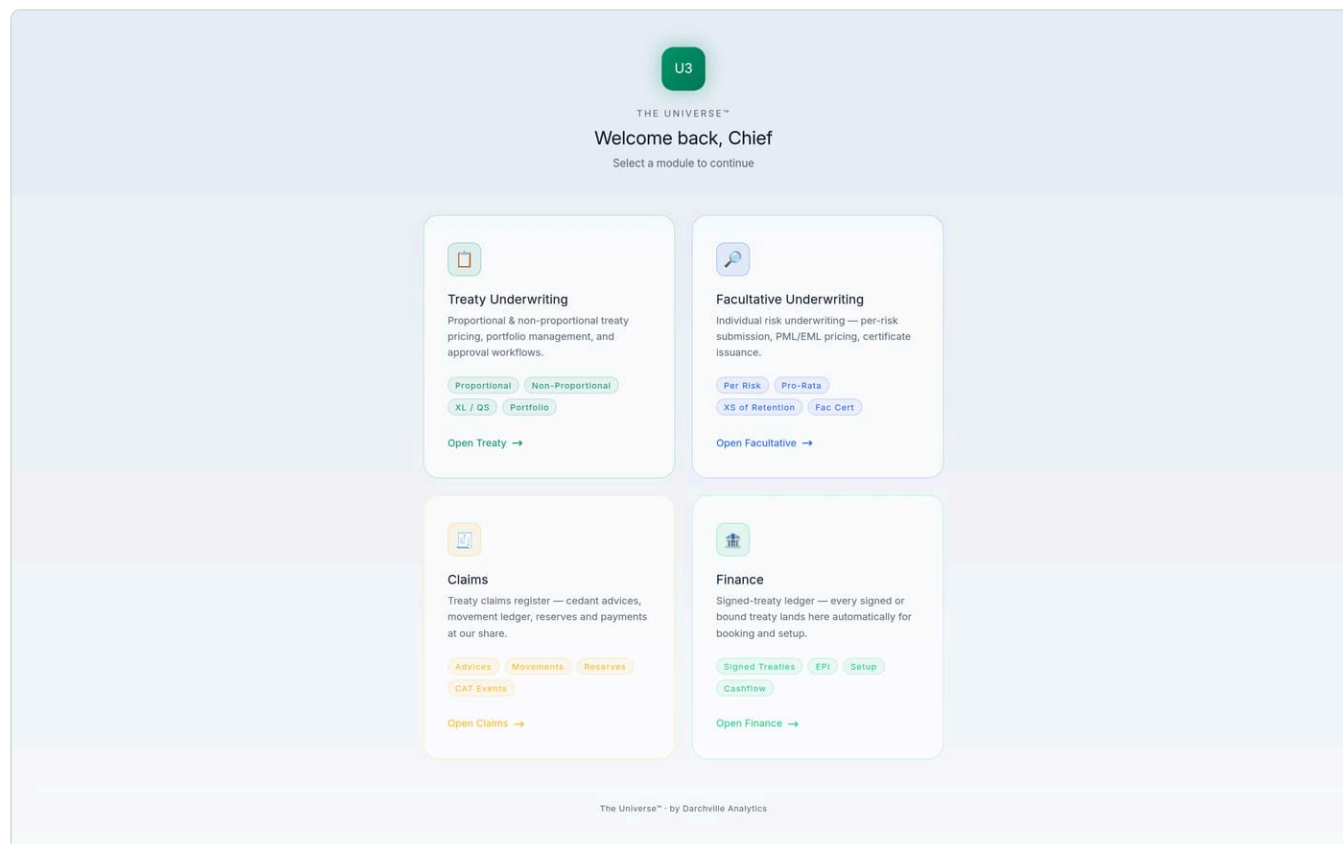


Login — </login>

ON THIS SCREEN

- The theme picker in the corner (Midnight, Ocean, Graphite, Sunset, Daylight) is remembered per browser — the screenshots in this manual use Daylight.
- Five failed attempts in fifteen minutes locks the account for fifteen minutes. The message you get back is deliberately identical to a wrong password, so a lock is never disclosed.
- Sessions last eight hours by default — a working day. You will sign in again the next morning.

The front door once you are authenticated. Four modules, each with its own workspace: Treaty Underwriting, Facultative Underwriting, Claims and Finance.



Module select — /select

ON THIS SCREEN

- **Treaty Underwriting** — proportional and non-proportional pricing, portfolio management and the approval workflow.
- **Facultative Underwriting** — individual risk submissions, PML/EML pricing, certificate issuance.
- **Claims** — the treaty claims register: cedant advices, movement ledger, reserves and payments at our share.
- **Finance** — the signed-treaty ledger. Every treaty that reaches SIGNED lands here automatically for booking.

3. Conventions that apply everywhere

The wizard chrome

Inside a treaty or a fac risk, three things are always on screen. The **topbar** carries Refresh, the autosave indicator, the theme picker, Home and Log out. The **left sidebar** is the wizard: grouped steps, current step highlighted. A floating **Back / Next** dock sits at the bottom of the content lane.

Both the sidebar and the Back/Next dock save the current screen before they navigate. If the save fails you stay put and a red banner tells you why, with a Retry. That is deliberate — the sidebar is nine always-visible buttons and it used to be possible to lose an edit by clicking one.

Saving

Autosave is on by default and the indicator in the topbar shows its state. Screens also save on navigation. Header updates use optimistic locking: if someone else changed the treaty while you had it open, your save is rejected rather than silently overwriting theirs.

Editing is owner-scoped. A treaty or fac risk assigned to someone else opens read-only, and the screen says so — claim it if it is unassigned, or have it allocated to you.

Paste from Excel

Almost every grid in the application accepts a paste straight out of a broker's or cedant's workbook. Click the cell you want to start at and paste; rows are added to fit. The grids tell you what they dropped and why — cells past the grid edge, cells below a triangle's diagonal, non-numeric tokens such as a header row dragged in by accident — rather than silently seeding junk.

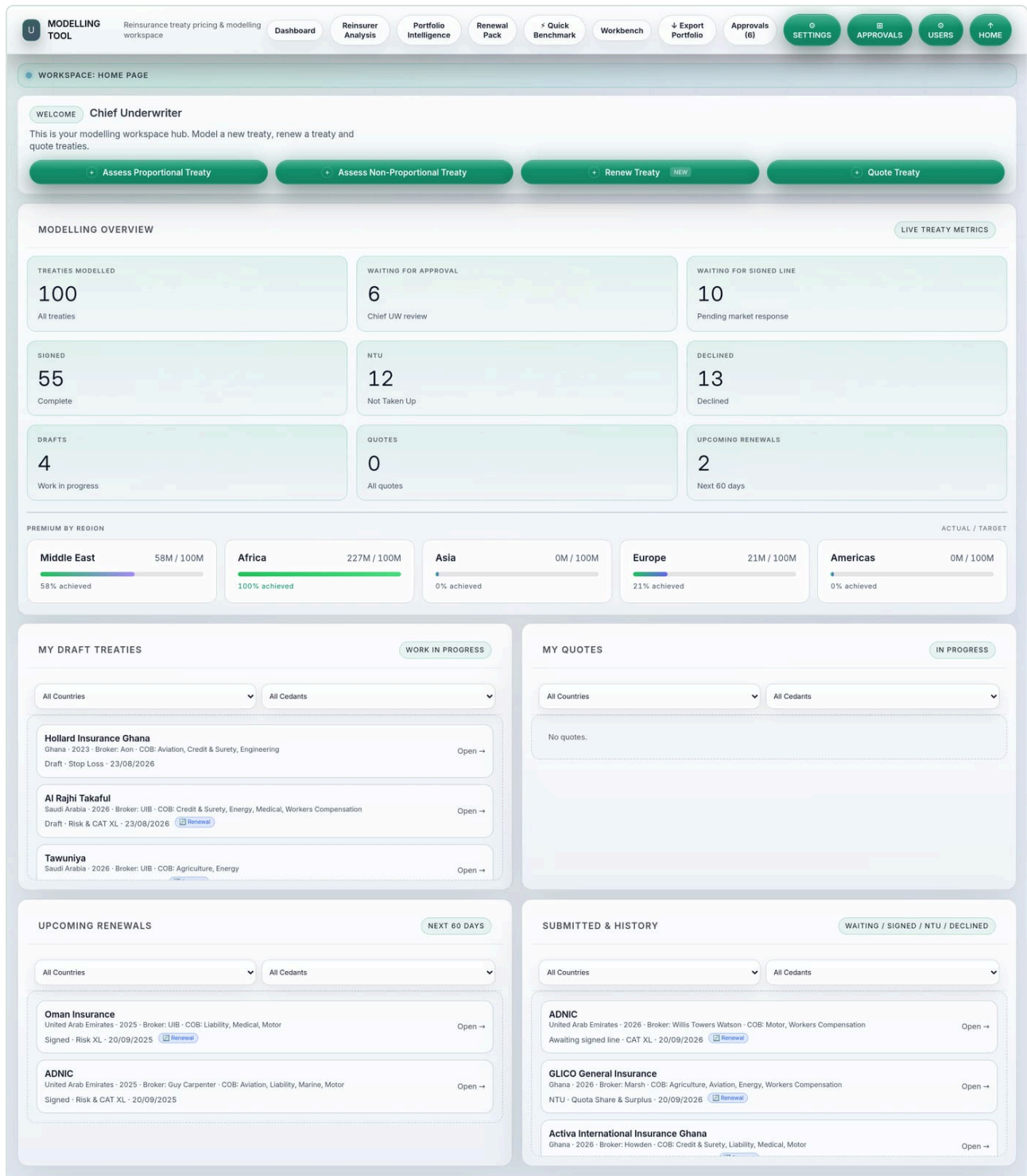
A note on deep links

Several screens depend on state that the Treaty Detail screen publishes when it loads — the triangle year range, the treaty type, whether triangulations are available. Opening one of those screens by pasting its URL into a fresh tab can leave it without that context. Enter a treaty through Treaty Detail (or from Home) and navigate with the sidebar, and everything resolves.

4. The treaty workspace

Home — modelling workspace

Your hub. Four large actions start work: assess a proportional treaty, assess a non-proportional treaty, renew an existing treaty, or quote a treaty. Below them, live counts of the whole modelled book and your own queues.



Home — modelling workspace — /

ON THIS SCREEN

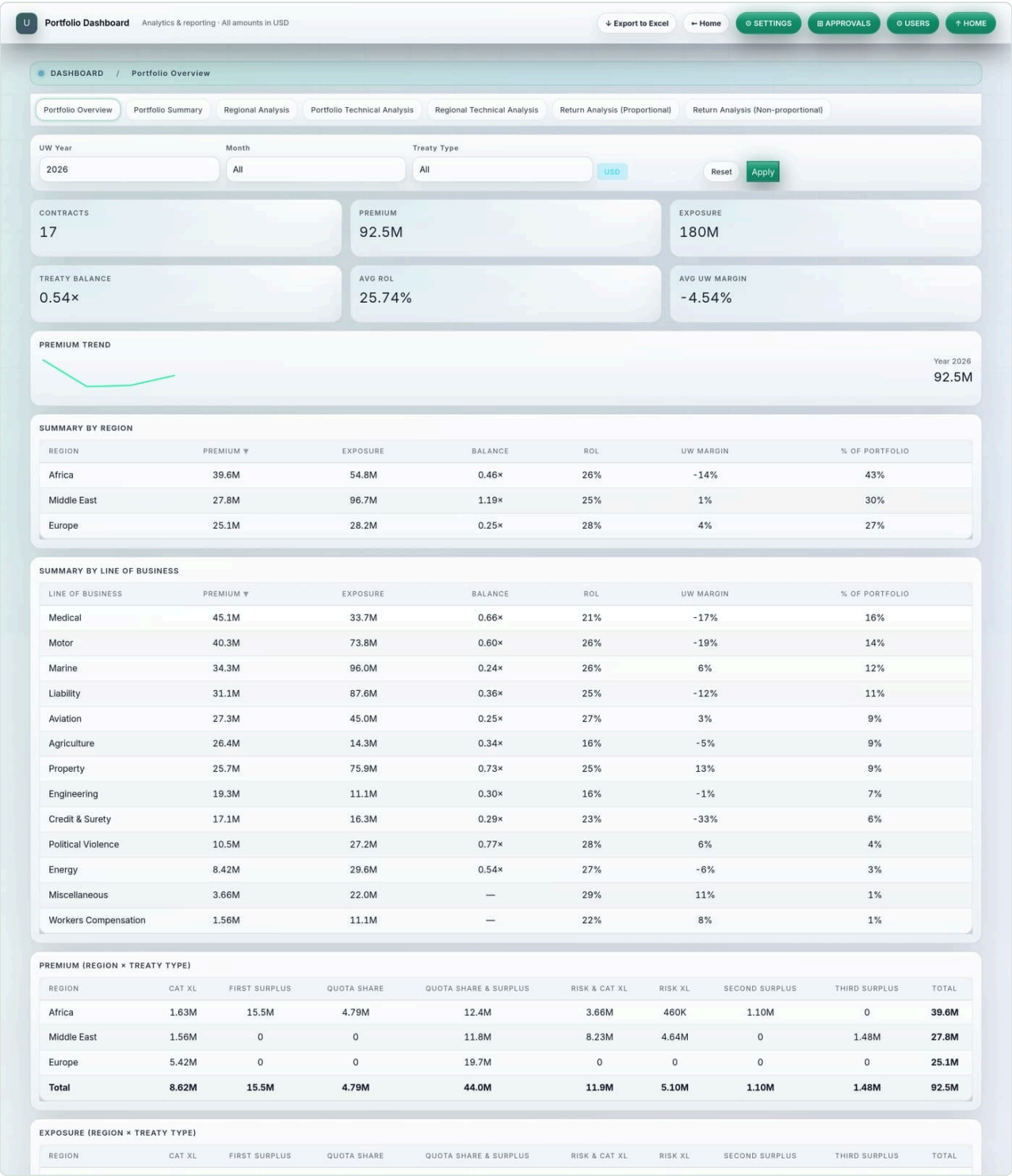
- **Modelling overview** — treaties modelled, waiting for approval, waiting for signed line, signed, NTU, declined, drafts, quotes and upcoming renewals in the next 60 days.
- **Premium by region** — actual against target for Middle East, Africa, Asia, Europe and the Americas.
- **My draft treaties** and **My quotes** — your work in progress, filterable by country and cedant. Each row opens the treaty at its Treaty Detail screen.

- **Upcoming renewals** and **Submitted & history** — what is falling due, and what you have already put into the workflow.
- The topbar also reaches the Dashboard, Reinsurer Analysis, Portfolio Intelligence, Renewal Pack, Quick Benchmark, Workbench, an Excel portfolio export, and the Approvals queue with its pending count.

Portfolio dashboard

/dashboard

Analytics across the whole written book, in USD. Seven tabs — Portfolio Overview, Portfolio Summary, Regional Analysis, Portfolio Technical Analysis, Regional Technical Analysis, and Return Analysis split proportional and non-proportional.



Portfolio dashboard — /dashboard (1 of 4)

REGION	CAT XL	FIRST SURPLUS	QUOTA SHARE	QUOTA SHARE & SURPLUS	RISK & CAT XL	RISK XL	SECOND SURPLUS	THIRD SURPLUS	TOTAL
Middle East	11.1M	0	0	13.1M	44.3M	25.5M	0	2.68M	96.7M
Africa	11.9M	4.43M	1.33M	8.39M	22.0M	5.34M	1.50M	0	54.8M
Europe	23.3M	0	0	4.92M	0	0	0	0	28.2M
Total	46.3M	4.43M	1.33M	26.5M	66.2M	30.9M	1.50M	2.68M	180M

UW MARGIN (REGION × TREATY TYPE)

REGION	CAT XL	FIRST SURPLUS	QUOTA SHARE	QUOTA SHARE & SURPLUS	RISK & CAT XL	RISK XL	SECOND SURPLUS	THIRD SURPLUS	TOTAL
Europe	18.70%	0.00%	0.00%	-0.42%	0.00%	0.00%	0.00%	0.00%	3.70%
Middle East	7.59%	0.00%	0.00%	-13.25%	9.58%	23.73%	0.00%	-13.10%	0.88%
Africa	14.65%	-38.18%	-23.70%	8.40%	10.68%	10.48%	-4.86%	0.00%	-13.58%
Total	15.92%	-38.18%	-23.70%	-1.38%	9.92%	22.53%	-4.86%	-13.10%	-4.54%

PORTFOLIO COMPOSITION (REGION × TREATY TYPE)

REGION	CAT XL	FIRST SURPLUS	QUOTA SHARE	QUOTA SHARE & SURPLUS	RISK & CAT XL	RISK XL	SECOND SURPLUS	THIRD SURPLUS	TOTAL
Africa	1.76%	16.73%	5.18%	13.45%	3.96%	0.50%	1.20%	0.00%	42.78%
Middle East	1.69%	0.00%	0.00%	12.82%	8.90%	5.02%	0.00%	1.60%	30.02%
Europe	5.86%	0.00%	0.00%	21.33%	0.00%	0.00%	0.00%	0.00%	27.20%
Total	9.32%	16.73%	5.18%	47.60%	12.86%	5.52%	1.20%	1.60%	100.00%

PREMIUM (LOB × TREATY TYPE)

LINE OF BUSINESS	CAT XL	FIRST SURPLUS	QUOTA SHARE	QUOTA SHARE & SURPLUS	RISK & CAT XL	RISK XL	SECOND SURPLUS	THIRD SURPLUS	TOTAL
Medical	0	15.5M	3.20M	24.3M	667K	0	0	1.48M	45.1M
Motor	1.56M	15.5M	3.20M	11.8M	8.23M	0	0	0	40.3M
Marine	0	0	0	17.7M	11.9M	4.64M	0	0	34.3M
Liability	1.63M	15.5M	0	0	8.23M	4.64M	1.10M	0	31.1M
Aviation	0	0	0	19.7M	7.56M	0	0	0	27.3M
Agriculture	0	0	4.79M	19.7M	0	460K	0	1.48M	26.4M
Property	7.05M	0	0	12.4M	0	5.10M	1.10M	0	25.7M
Engineering	0	0	0	17.7M	0	460K	1.10M	0	19.3M
Credit & Surety	1.63M	15.5M	0	0	0	0	0	0	17.1M
Political Violence	5.42M	0	1.60M	1.99M	0	0	0	1.48M	10.5M
Energy	0	0	3.20M	0	3.66M	460K	1.10M	0	8.42M
Miscellaneous	0	0	0	0	3.66M	0	0	0	3.66M
Workers Compensation	1.56M	0	0	0	0	0	0	0	1.56M
Total	18.9M	61.9M	16.0M	125M	43.9M	15.8M	4.42M	4.43M	291M

EXPOSURE (LOB × TREATY TYPE)

LINE OF BUSINESS	CAT XL	FIRST SURPLUS	QUOTA SHARE	QUOTA SHARE & SURPLUS	RISK & CAT XL	RISK XL	SECOND SURPLUS	THIRD SURPLUS	TOTAL
Marine	0	0	0	4.22M	66.2M	25.5M	0	0	96.0M
Liability	11.9M	4.43M	0	0	44.3M	25.5M	1.50M	0	87.6M
Property	35.2M	0	0	8.39M	0	30.9M	1.50M	0	75.9M
Motor	11.1M	4.43M	835K	13.1M	44.3M	0	0	0	73.8M
Aviation	0	0	0	4.92M	40.1M	0	0	0	45.0M
Medical	0	4.43M	835K	21.5M	4.18M	0	0	2.68M	33.7M
Energy	0	0	835K	0	22.0M	5.34M	1.50M	0	29.6M
Political Violence	23.3M	0	499K	702K	0	0	0	2.68M	27.2M
Miscellaneous	0	0	0	0	22.0M	0	0	0	22.0M
Credit & Surety	11.9M	4.43M	0	0	0	0	0	0	16.3M

Credit & Surety	11.9M	4.43M	0	0	0	0	0	0	16.3M
Agriculture	0	0	1.33M	4.92M	0	5.34M	0	2.68M	14.3M
Workers Compensation	11.1M	0	0	0	0	0	0	0	11.1M
Engineering	0	0	0	4.22M	0	5.34M	1.50M	0	11.1M
Total	104M	17.7M	4.34M	62.0M	243M	97.9M	6.00M	8.05M	543M

PORTFOLIO MIX (LOB × TREATY TYPE)

LINE OF BUSINESS	CAT XL	FIRST SURPLUS	QUOTA SHARE	QUOTA SHARE & SURPLUS	RISK & CAT XL	RISK XL	SECOND SURPLUS	THIRD SURPLUS	TOTAL
Medical	0.00%	5.32%	1.10%	8.36%	0.23%	0.00%	0.00%	0.51%	15.51%
Motor	0.54%	5.32%	1.10%	4.08%	2.83%	0.00%	0.00%	0.00%	13.86%
Marine	0.00%	0.00%	0.00%	6.10%	4.09%	1.60%	0.00%	0.00%	11.79%
Liability	0.56%	5.32%	0.00%	0.00%	2.83%	1.60%	0.38%	0.00%	10.69%
Aviation	0.00%	0.00%	0.00%	6.78%	2.60%	0.00%	0.00%	0.00%	9.39%
Agriculture	0.00%	0.00%	1.65%	6.78%	0.00%	0.16%	0.00%	0.51%	9.10%
Property	2.43%	0.00%	0.00%	4.28%	0.00%	1.75%	0.38%	0.00%	8.84%
Engineering	0.00%	0.00%	0.00%	6.10%	0.00%	0.16%	0.38%	0.00%	6.64%
Credit & Surety	0.56%	5.32%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	5.88%
Political Violence	1.87%	0.00%	0.55%	0.68%	0.00%	0.00%	0.00%	0.51%	3.61%
Energy	0.00%	0.00%	1.10%	0.00%	1.26%	0.16%	0.38%	0.00%	2.90%
Miscellaneous	0.00%	0.00%	0.00%	0.00%	1.26%	0.00%	0.00%	0.00%	1.26%
Workers Compensation	0.54%	0.00%	0.00%						
Total	6.49%	21.28%	5.50%						

UW MARGIN (LOB × TREATY TYPE)

LINE OF BUSINESS	CAT XL	FIRST SURPLUS	QUOTA SHARE	QUOTA SHARE & SURPLUS	RISK & CAT XL	RISK XL	SECOND SURPLUS	THIRD SURPLUS	TOTAL
Property	17.76%	0.00%	0.00%	8.40%	0.00%	22.53%	-4.86%	0.00%	13.20%
Miscellaneous	0.00%	0.00%	0.00%	0.00%	10.68%	0.00%	0.00%	0.00%	10.68%
Workers Compensation	7.59%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	7.59%
Marine	0.00%	0.00%	0.00%	-0.72%	9.92%	23.73%	0.00%	0.00%	6.28%
Political Violence	18.70%	0.00%	-15.06%	2.23%	0.00%	0.00%	0.00%	-13.10%	5.96%
Aviation	0.00%	0.00%	0.00%	-0.42%	10.82%	0.00%	0.00%	0.00%	2.69%
Engineering	0.00%	0.00%	0.00%	-0.72%	0.00%	10.48%	-4.86%	0.00%	-0.69%
Agriculture	0.00%	0.00%	-23.70%	-0.42%	0.00%	10.48%	0.00%	-13.10%	-5.16%
Energy	0.00%	0.00%	-28.01%	0.00%	10.68%	10.48%	-4.86%	0.00%	-6.05%
Liability	14.65%	-38.18%	0.00%	0.00%	9.58%	23.73%	-4.86%	0.00%	-12.33%
Medical	0.00%	-38.18%	-28.01%	-2.16%	-4.47%	0.00%	0.00%	-13.10%	-16.74%
Motor	7.59%	-38.18%	-28.01%	-13.25%	9.58%	0.00%	0.00%	0.00%	-18.52%
Credit & Surety	14.65%	-38.18%	0.00%	0.00%	0.00%				
Total	15.81%	-38.18%	-25.42%	-1.14%	9.85%				

PREMIUM BY LOB AND REGION

LINE OF BUSINESS	AFRICA	EUROPE	MIDDLE EAST	TOTAL
Medical	31.1M	0	14.0M	45.1M
Motor	18.7M	0	21.6M	40.3M
Marine	3.66M	17.7M	12.9M	34.3M
Liability	18.2M	0	12.9M	31.1M
Aviation	0	19.7M	7.56M	27.3M
Agriculture	5.25M	19.7M	1.48M	26.4M
Property	15.6M	5.42M	4.64M	25.7M

Portfolio dashboard — /dashboard (3 of 4)

Agriculture	0.60M	15.71M	1.40M	20.49M
Property	15.6M	5.42M	4.64M	25.7M
Engineering	1.56M	17.7M	0	19.3M
Credit & Surety	17.1M	0	0	17.1M
Political Violence	1.60M	7.41M	1.48M	
Energy	8.42M	0	0	
Miscellaneous	3.66M	0	0	

UNDERWRITING MARGINS (LOB × REGION)

LINE OF BUSINESS	AFRICA	EUROPE	MIDDLE EAST	TOTAL
Property	8.18%	18.70%	23.73%	13.20%
Miscellaneous	10.68%	0.00%	0.00%	10.68%
Workers Compensation	0.00%	0.00%	7.59%	7.59%
Marine	10.68%	-0.72%	14.68%	6.28%
Political Violence	-15.06%	14.28%	-13.10%	5.96%
Aviation	0.00%	-0.42%	10.82%	2.69%
Engineering	-0.35%	-0.72%	0.00%	-0.69%
Agriculture	-20.70%	-0.42%	-13.10%	-5.16%
Energy	-6.05%	0.00%	0.00%	-6.05%
Liability	-31.43%	0.00%	14.68%	-12.33%
Medical	-18.51%	0.00%	-12.82%	-16.74%
Motor	-36.44%	0.00%	-3.07%	-18.52%
Credit & Surety	-33.14%	0.00%	0.00%	-33.14%
Total	-19.00%	1.88%	3.81%	-6.57%

Portfolio dashboard — /dashboard (4 of 4)

ON THIS SCREEN

- Filters across the top: underwriting year, month, treaty type and reporting currency. Reset and Apply.
- Headline tiles: contracts, premium, exposure, treaty balance, average ROL and average underwriting margin.
- Summary tables by region and by line of business, each carrying premium, exposure, balance, ROL, UW margin and share of portfolio.
- Everything on the dashboard aggregates bound contracts only. Quotes do not contribute to portfolio exposure.
- Export to Excel from the top right.

Approvals queue

/approvals

Where a chief underwriter, treaty director or treaty manager clears offers that exceed the submitting underwriter's authority. Pending items at the top with Approve / Decline / Review; everything already decided below.

Chief Underwriter Approvals

Review offers, approve or decline

Refresh

Log out

PENDING

5

Activa International Insurance Ghana

PROP - Submitted by

2026-09-20 11:00

Awaiting Approval

Approve

Decline

Holland Insurance Ghana

NP - Submitted by

2026-09-20 11:00

Awaiting Approval

Approve

Decline

Activa International Insurance Ghana

PROP - Submitted by

2026-06-20 11:00

Awaiting Approval

Approve

Decline

Kuwait Insurance

PROP - Submitted by

2026-03-20 11:00

Awaiting Approval

Approve

Decline

Holland Insurance Ghana

NP - Submitted by

2025-01-20 11:00

Awaiting Approval

Approve

Decline

ADNIC

PROP - Submitted by

2025-01-20 11:00

Awaiting Approval

Approve

Decline

DECIDED

90

ADNIC

NP - Chief Underwriter

Signed

ADNIC

NP - Chief Underwriter

Awaiting Signed Line

GLICO General Insurance

PROP - Chief Underwriter

NTU

GLICO General Insurance

NP - Chief Underwriter

Awaiting Signed Line

SIC Insurance Company

NP - Chief Underwriter

NTU

Holland Insurance Ghana

PROP - Chief Underwriter

NTU

AXA UK

PROP - Chief Underwriter

Declined

Dubai Insurance

PROP - Chief Underwriter

Signed

Aviva

NP - Chief Underwriter

Signed

Dubai Insurance

NP - Chief Underwriter

Signed

Holland Insurance Ghana

PROP - Chief Underwriter

Signed

Aviva

PROP - Chief Underwriter

Signed

AXA UK

PROP - Chief Underwriter

Signed

Enterprise Insurance

PROP - Chief Underwriter

Signed

Activa International Insurance Ghana

PROP - Chief Underwriter

Signed

Oman Insurance

NP - Chief Underwriter

Signed

Star Assurance

PROP - Chief Underwriter

Awaiting Signed Line

ADNIC

NP - Chief Underwriter

Signed

Activa International Insurance Ghana

PROP - Chief Underwriter

Signed

GLICO General Insurance

PROP - Chief Underwriter

Signed

Approvals queue — /approvals

ON THIS SCREEN

- Each pending row shows the cedant, whether it is proportional or non-proportional, who submitted it and when.

- **Review** → opens the treaty at its pricing screen so the decision is made against the numbers, not the summary line.
- Visible only to roles that carry approval authority (CE, CU, TD, TM). Everyone else is routed away.
- The decided list is the audit surface: every offer, its decision and the decider.

5. Pricing a proportional treaty

The route through

Twenty-three screens in six groups: **Setup** (treaty detail, documents), **Triangles** (premium, paid, outstanding, incurred), **Large losses** and **Cat losses** (list, selection, Pareto for each), **Dev factors** (one per triangle), **Summaries** (projected, quick), **Profiles** (risk, claims), **Exposure** (CRESTA, event loss tables) and finally **Pricing**.

If the cedant cannot supply triangles, answer *No* to "Triangulations available" on Treaty Detail. The triangle, dev-factor and projected-summary steps disappear from the sidebar and are replaced by a single Straight Stats screen that projects off benchmark development curves instead.

Treaty Detail

/prop/treaty-detail

The header of the treaty and the root of everything downstream. Parties, period, currency, classes, and the structural limits that the pricing engine and the exposure screens read.

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PROPORTIONAL TREATY: TREATY DETAIL

Cedant: Tawuniya

Country: Saudi Arabia

Broker: UIB

Currency: USD

Treaty Type: Quota Share

COB: Energy, Agriculture

Renewal — linked to prior year contract

Triangulations available: YES NO

Contract: 2026 Tawuniya Quota Share (Energy, Agriculture) SA

CONTRACT DETAILS

INPUT

Upload Slip

Country: Saudi Arabia

Cedant Name: Tawuniya

Treaty Type: Quota Share

Line of Business: Energy, Agriculture

Broker: UIB

Contract ID: 88d20b77-edf4-4e47-8236-9c1baa262132

Alt. Contract ID: SA/PR/2026/015

Currency: USD

Treaty Inception Date: 01/01/2026

Treaty Renewal Date: 01/01/2027

UW Year: 2026

Experience Start Year: 2020

Contract Description: 2026 Tawuniya Quota Share (Energy, Agriculture) SA

LIMIT DETAILS

CAPACITY

QS 100% Limit: 19,500,000

Retention %: 35.26%

Retention Amount: 6,875,700

Cession %: 64.74%

Cession Amount: 12,624,300

Surplus Max Retention: 4,400,000

Number of Lines: 12

Total Treaty Capacity: 19,500,000

Event Limit: 70,009,895

AAL: 2,555,982

COMMISSIONS

FIXED COMMISSION

SLIDING SCALE

Choose between a single fixed commission or a sliding scale commission structure.

FIXED COMMISSION

QS Commission %: 22.36%

Surplus Commission %: 19.61%

SLIDING SCALE

Min Loss Ratio %: 49.07%

Max Loss Ratio %: 74.96%

Min Commission %: 16.08%

Max Commission %: 28.18%

Enter slide manually: 8 row(s)

ADDITIONAL

Management Expenses %: 2.19%

Profit Commission %: 15.6%

Loss Carry Forward (LCF): 4 years

LOSS PARTICIPATION

YES

NO

Capture loss participation corridors where the reinsurer share changes above a given loss ratio.

Min Loss Ratio %: 72.07%

Max Loss Ratio %: 100%

Reinsurer Share %: 64.53%

Enter Slides Manually: 2 corridor(s)

EPI

Quota Share EPI: 2,915,000

Surplus EPI: 2,585,000

EPI Split: 2 class(es)

BROKERAGE & TAXES

Brokerage %: 6.63%

Taxes %: 2.06%

Loss Cap %: 231.09%

Treaty Detail — /prop/treaty-detail

ON THIS SCREEN

- **Contract details** — country, cedant, treaty type, lines of business (multi-select), broker, currency, inception and renewal dates, underwriting year, experience start year.
- **Limit details** — QS 100% limit, retention % and amount, cession % and amount, surplus maximum retention, number of lines, total treaty capacity, event limit and AAL. Greyed fields are derived, not typed.

- **Triangulations available** — the Yes/No that reshapes the whole wizard.
- **Upload Slip** hands a broker slip to the ingestion agent, which reads it and pre-fills what it can.
- A renewal is flagged in the header and linked to the prior-year contract, which is what makes year-on-year comparison possible further down.

Everything attached to the treaty: slips, wordings, accounts, bordereaux, loss listings, presentations and cat modelling output.

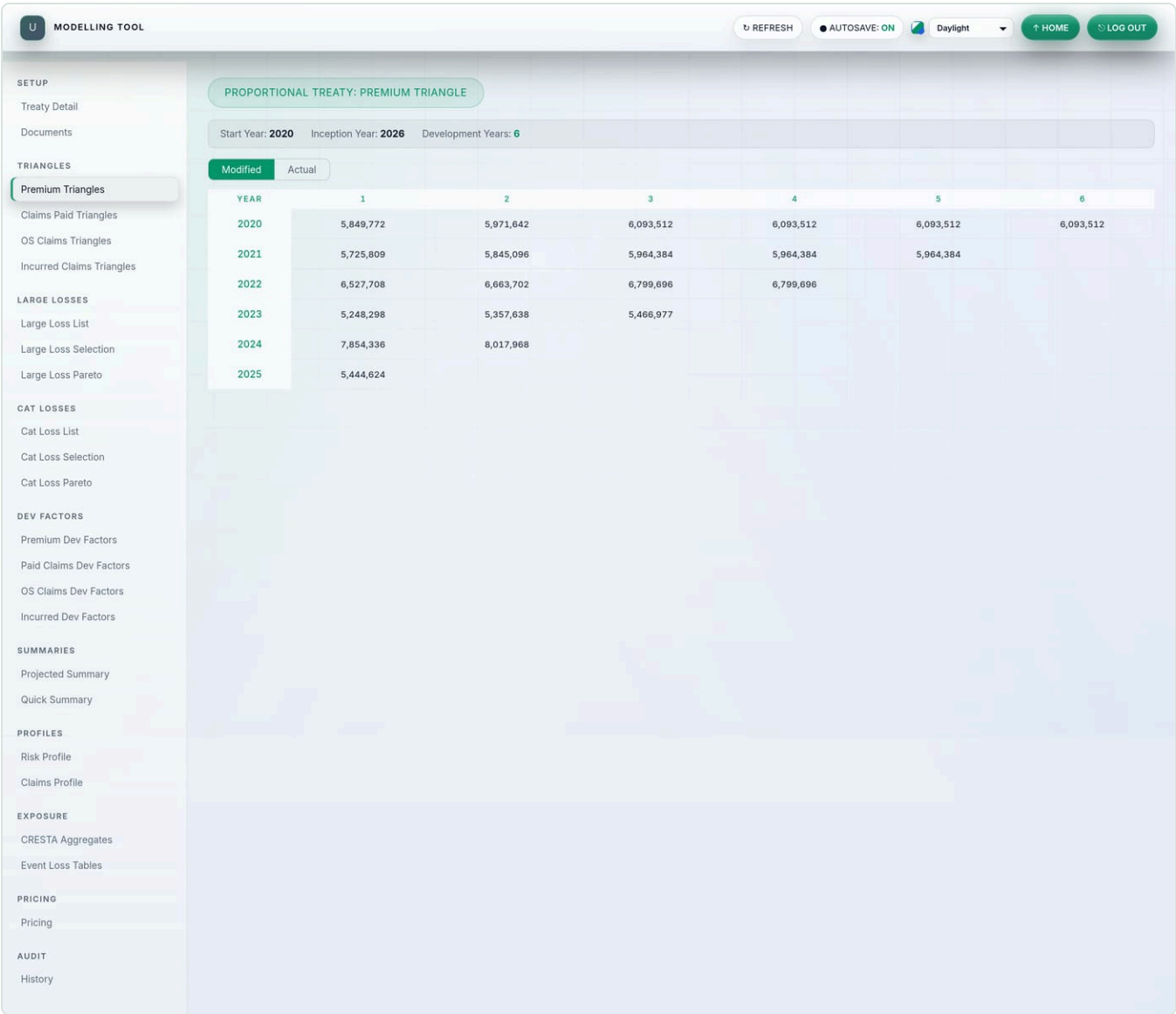
The screenshot shows the 'MODELLING TOOL' interface with a sidebar on the left containing various menu items like 'SETUP', 'TRIANGLES', 'LARGE LOSSES', 'CAT LOSSES', 'DEV FACTORS', 'SUMMARIES', 'PROFILES', 'EXPOSURE', 'PRICING', and 'AUDIT'. The main area is titled 'PROPORTIONAL TREATY: DOCUMENTS' and includes a 'FILES FOR TREATY' section with a contract ID. Below this is a file upload area with instructions to 'Drag & drop files here or browse' and a '+ Select Files' button. A form below the upload area has fields for 'DOCUMENT TYPE' (with a dropdown set to 'Final Slip'), 'DOCUMENT TITLE' (with 'Final Signed Slip' entered), and 'DESCRIPTION (OPTIONAL)' (with 'Notes...' entered). An 'Upload' button is to the right of the description field. Below the form is an 'Uploaded Documents' section with a 'Reload' button and the text 'No documents uploaded yet.'

Documents — /prop/documents

ON THIS SCREEN

- Drag and drop or browse. PDF, XLSX, DOCX and CSV up to 50 MB each.
- Each upload is tagged with a document type — Final Slip, Draft Slip, Expiring Slip, Renewal Pack, Large Loss List, Risk Profiles, Claims Profile, Presentation, CAT Modelling, Bordereaux, Accounts, Other — plus a title and optional description.
- Files are stored server-side against this treaty and are viewable inline or downloadable from the list.

The written/earned premium development triangle. Origin years down, development months across, cumulative values in the upper triangle.



Premium triangles — /prop/premium-triangles

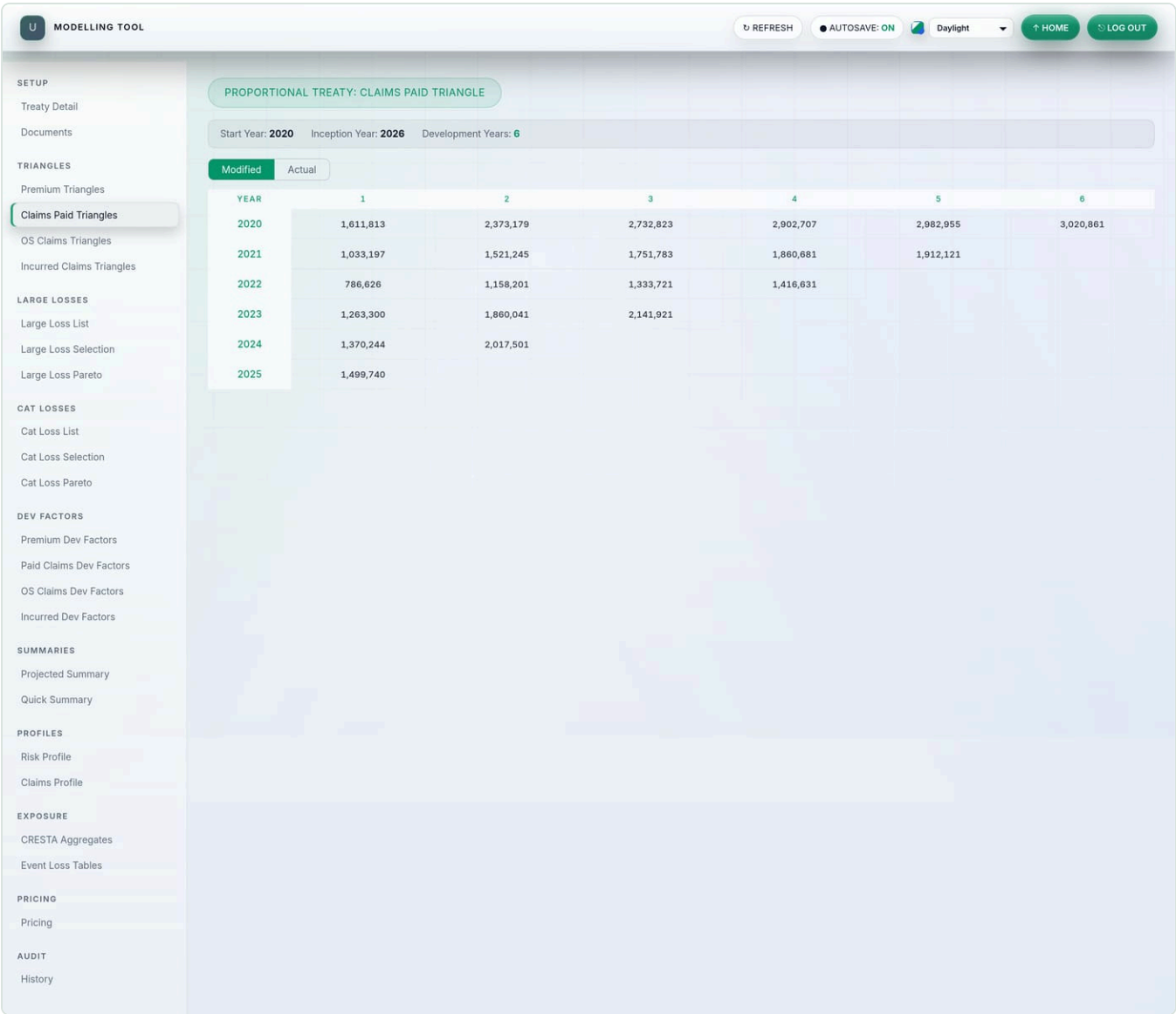
ON THIS SCREEN

- The year range and number of development years come from the treaty header — start year and inception year are shown in the info bar so you can see what the grid is built on.
- Paste a block straight from Excel. Cells outside the grid or below the diagonal are skipped and reported rather than written.
- Each triangle carries variants (as-reported and modified) on tabs, so a cleaned view can sit alongside the raw one.

Claims paid triangles

/prop/claims-paid-triangles

Cumulative paid claims by origin year and development month. Same grid mechanics as the premium triangle.

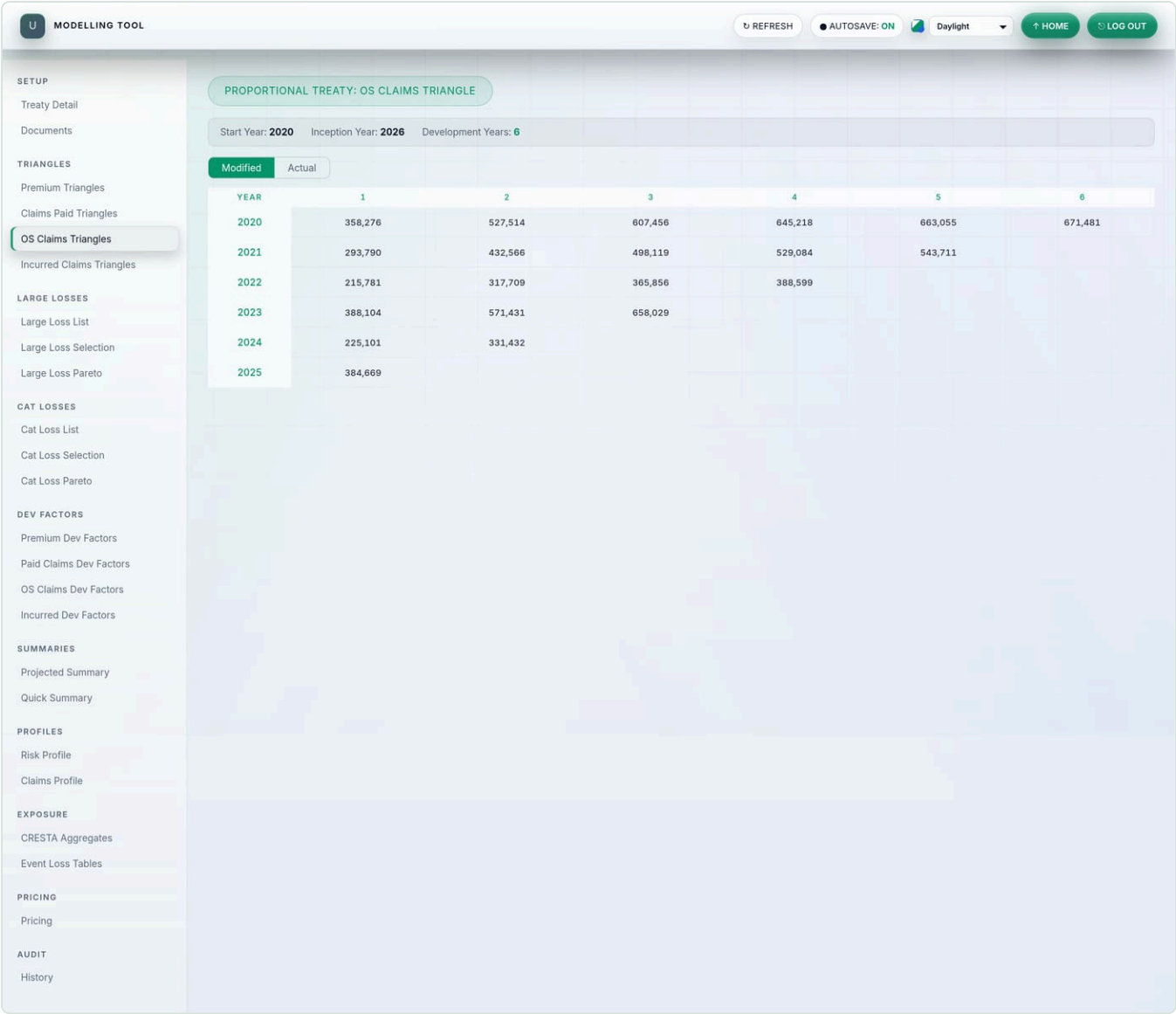


Claims paid triangles — /prop/claims-paid-triangles

ON THIS SCREEN

- Paid and outstanding are captured separately so that incurred can be derived rather than typed — which is what stops the three from disagreeing.

Cumulative outstanding (case reserve) claims by origin year and development month.

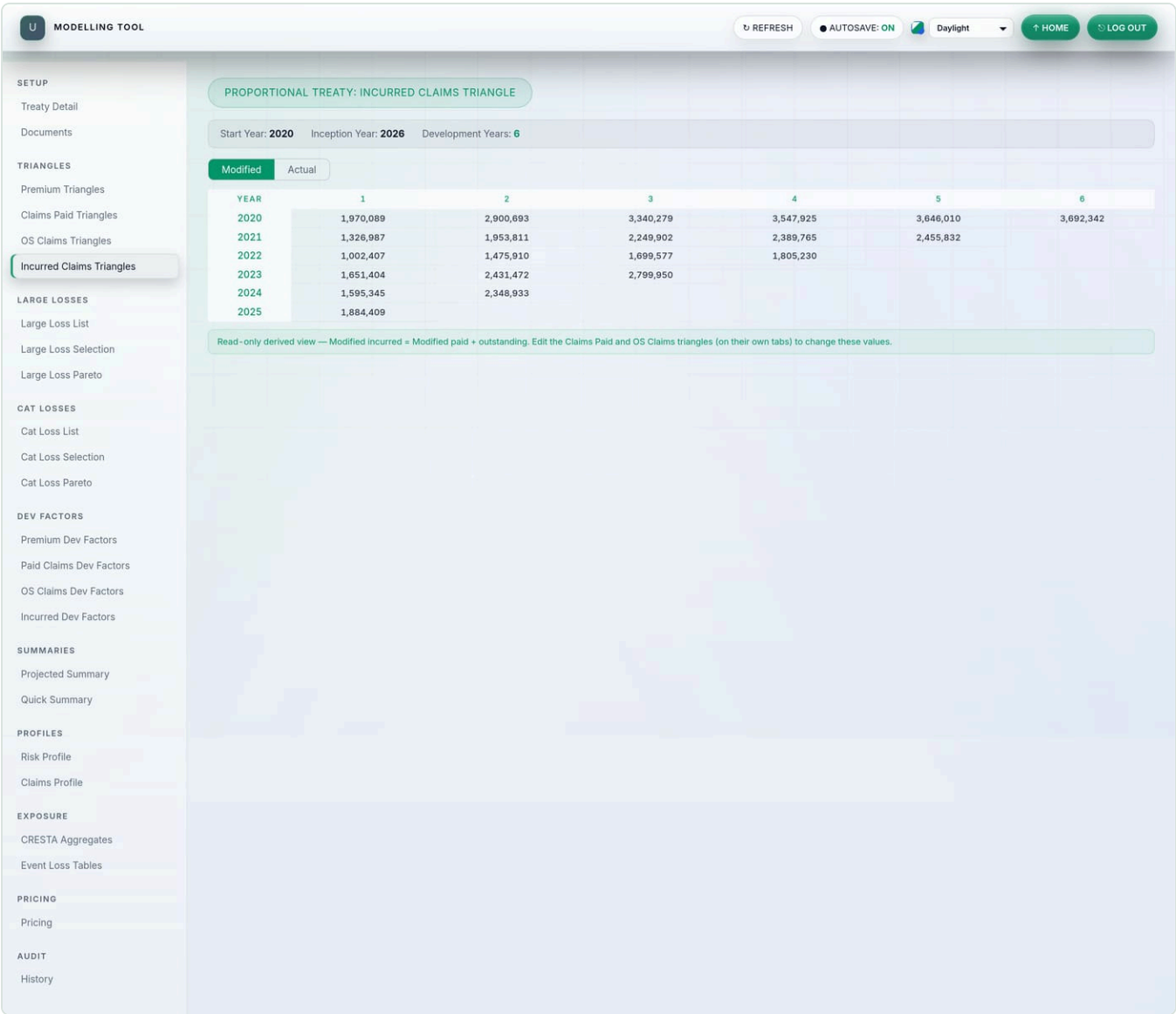


OS claims triangles — /prop/os-claims-triangles

ON THIS SCREEN

- Feeds the incurred triangle and, with paid, the Munich chain-ladder option on the dev-factor screens.

Paid plus outstanding. This triangle is derived: the grid shows the sum in the modified slot rather than letting you type a figure that contradicts its two inputs.



Incurred claims triangles — /prop/incurred-claims-triangles

ON THIS SCREEN

- This is the triangle most of the projection work runs on.

Large loss list

/prop/large-loss-list

The cedant's large-loss listing, one row per loss: underwriting year, insured, event name, date of loss, class, paid, outstanding and incurred.

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PROPORTIONAL TREATY: LARGE LOSSES

REPORT DATE: mm/dd/yyyy 10 records

Loss Analysis Load Clear Save

#	UW YEAR	INSURED NAME	LOSS / EVENT	DATE OF LOSS	CLASS	PAID	O/S	INCURRED	SAVED
1	2020	Kumasi Brewery	Agriculture loss 2020-8	02/02/2020	Agriculture	1,139,753	715,332	1,855,085	2020-03-02
2	2020	Nestlé Ghana Tema Plant	Energy loss 2020-5	08/22/2020	Energy	878,884	245,986	1,124,870	2020-09-22
3	2020	Takoradi Port Terminal	Agriculture loss 2020-7	08/27/2020	Agriculture	923,612	611,880	1,535,492	2020-09-27
4	2020	Ghana Grid Company Subst.	Energy loss 2020-2	10/28/2020	Energy	459,817	47,425	507,242	2020-11-28
5	2020	Volta Aluminium	Energy loss 2020-6	11/24/2020	Energy	502,541	303,931	806,472	2020-12-24
6	2022	Unilever Ghana	Agriculture loss 2022-1	06/06/2022	Agriculture	542,094	254,952	797,046	2022-07-06
7	2024	Tema Oil Refinery	Agriculture loss 2024-9	02/12/2024	Agriculture	573,053	318,635	891,688	2024-03-12
8	2025	Ghana Grid Company Subst.	Energy loss 2025-3	02/12/2025	Energy	226,504	77,752	304,256	2025-03-12
9	2025	Golden Star Resources	Agriculture loss 2025-10	06/25/2025	Agriculture	389,617	235,935	625,552	2025-07-25
10	2025	Ghana Grid Company Subst.	Agriculture loss 2025-4	10/03/2025	Agriculture	350,376	156,061	506,437	2025-11-03
TOTAL							5,986,251	2,967,889	8,954,140

Large loss list — /prop/large-loss-list

ON THIS SCREEN

- Paste the whole loss run from the broker's workbook; rows are added to fit.
- Totals for paid, outstanding and incurred sit at the foot of the table.
- **Loss Analysis** opens the analytical view; Load / Clear / Save manage the listing itself.
- Large and cat losses are captured *before* the dev-factor screens so those screens can offer a triangle with these losses stripped out — the attritional basis.

Choose which large losses belong in the pricing basis, and trend them to today's money. This is where the loss list becomes a loss load.

PROPORTIONAL TREATY: LARGE LOSS SELECTION

Losses have been added or edited since the last selection was saved — review and re-save your selection.

TOTAL: 10, SELECTED: 10, INCURRED: 8,954,140, INFLATION-ADJ: 18,005,875, AVG FACTOR: 1.8343, INCEPTION YR: 2026, CPI SOURCE: Saudi Arabia

THRESHOLD: e.g. 500,000. Apply. Inflation (Saudi Arabia). Factor +10%. Loss Loading. Yearly Aggregates. All. None. Save.

✓	YEAR	INSURED / EVENT	LOSS NAME	DATE	PAID	OS	INCURRED	FACTOR	INFLATED
☒	2020	Kumasi Brewery	Agriculture loss 2020-8	2020-02-02	1,139,753	715,332	1,855,085	2.0058	3,720,929
☒	2020	Nestle Ghana Tema Plant	Energy loss 2020-5	2020-08-22	878,884	245,986	1,124,870	2.3378	2,629,721
☒	2020	Takoradi Port Terminal	Agriculture loss 2020-7	2020-08-27	923,612	611,880	1,535,492	2.5197	3,868,979
☒	2020	Ghana Grid Company Substation	Energy loss 2020-2	2020-10-28	459,817	47,425	507,242	1.5199	770,957
☒	2020	Volta Aluminium	Energy loss 2020-6	2020-11-24	502,541	303,931	806,472	3.0942	2,495,386
☒	2022	Unilever Ghana	Agriculture loss 2022-1	2022-06-06	542,094	254,952	797,046	2.1561	1,718,511
☒	2024	Tema Oil Refinery	Agriculture loss 2024-9	2024-02-12	573,053	318,635	891,688	1.3014	1,160,443
☒	2025	Ghana Grid Company Substation	Energy loss 2025-3	2025-02-12	226,504	77,752	304,256	1.0932	332,613
☒	2025	Golden Star Resources	Agriculture loss 2025-10	2025-06-25	389,617	235,935	625,552	1.1395	712,817
☒	2025	Ghana Grid Company Substation	Agriculture loss 2025-4	2025-10-03	350,376	156,061	506,437	1.1759	595,519

ADDITIONAL LOADINGS

No additional loadings. Click + Add to include IBNR, trend, or other adjustments.

Large loss selection — /prop/large-loss-selection

ON THIS SCREEN

- A threshold filter, then tick or untick individual losses. Counters show total, selected, incurred, inflation-adjusted and the average trend factor applied.
- Inflation is drawn from the country CPI series for the cedant's country, up to the inception year. A manual +10% factor override is available.
- Each row shows its own factor and inflated amount, so you can see exactly what the trending did to a 2020 loss.
- **Loss loading** and **Yearly aggregates** open the derived views; additional loadings (IBNR, trend, other adjustments) can be added on top.

Fit a severity distribution to the selected large losses and read the tail. Pareto, lognormal, exponential and Weibull are fitted and ranked by Kolmogorov–Smirnov goodness of fit.

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PROPORTIONAL TREATY: LARGE LOSS PARETO

Severity Distribution Fit

Fit parametric distributions to large losses. Adjust threshold and limit. Compare Pareto, Lognormal, Exponential, Weibull via KS goodness-of-fit.

✓ Saved

Save Curve

View Sev-Freq Curve

PARAMETERS

Inputs auto-recompute stats + pricing

PARETO MIN (THRESHOLD)

332,613

Only losses $\geq$ threshold drive the tail.

TREATY LIMIT (MAX)

19,500,000

Auto-set from total Risk/Cat layer limits in Structure. Editable override.

ALPHA (A)

0.713

Higher $\alpha \rightarrow$ lighter tail. Editable.

OBSERVATION YEARS

10

Frequency = tail count + years.

GOODNESS OF FIT

Select a distribution to use for return periods and pricing. KS statistic: lower = better fit.

USE	DISTRIBUTION	PARAMETERS	N	KS STAT	P-VALUE	VERDICT
☒	Pareto	$\alpha=0.713$	10	0.2398	0.5643	Fair
☐	Lognormal	$\mu=14.12 \sigma=0.80$	10	0.1781	0.9950	Good
☐	Exponential BEST	$\lambda=0.000001$	10	0.1708	1.0000	Good
☐	Weibull	$k=1.311 \lambda=1769929$	9	0.2518	0.4956	Fair

LARGE LOSS STATISTICS

	COUNT	10
TOTAL INCURRED	18,005,875	
AVERAGE	1,800,587	
MAX / MIN	3,868,979 / 332,613	
STD DEVIATION	1,241,097	
TOP 5 CONCENTRATION	80.2%	
AVG YEARLY LOSS	1,800,587	
FREQUENCY ($\geq$ XM)	1.00 / yr	
RISK PURE PREMIUM	0	

$\Delta A \leq 1$: INFINITE MEAN

RETURN PERIODS (PARETO)

Implied severity from fitted tail

RETURN PERIOD	ESTIMATED LOSS
1 IN 500 YR	2,033,316,254 Δ
1 IN 250 YR	768,958,043 Δ
1 IN 100 YR	212,642,806 Δ
1 IN 50 YR	80,417,204 Δ
1 IN 25 YR	30,412,159 Δ
1 IN 20 YR	22,238,038 Δ
1 IN 10 YR	8,409,976
1 IN 5 YR	3,180,483
1 IN 2 YR	879,509

PARETO RANKING

#	UW YR	INSURED / EVENT	LOSS	PAID	OS	INCURRED	INFL.	INFLATED	CUM%	RETURN PERIOD
1	2020	Takoradi Port Terminal	Agriculture loss 2020-7	923,612	611,880	1,535,492	2.52	3,868,979	21.5%	5.7y
2	2020	Kumasi Brewery	Agriculture loss 2020-8	1,139,753	715,332	1,855,085	2.01	3,720,929	42.2%	5.6y
3	2020	Nestlé Ghana Tema Plant	Energy loss 2020-5	878,884	245,986	1,124,870	2.34	2,629,721	56.8%	4.4y
4	2020	Volta Aluminium	Energy loss 2020-6	502,541	303,931	806,472	3.09	2,495,386	70.6%	4.2y
5	2022	Unilever Ghana	Agriculture loss 2022-1	542,094	254,952	797,046	2.16	1,718,511	80.2%	3.2y
6	2024	Tema Oil Refinery	Agriculture loss 2024-9	573,053	318,635	891,688	1.30	1,160,443	86.6%	2.4y
7	2020	Ghana Grid Company Substation	Energy loss 2020-2	459,817	47,425	507,242	1.52	770,957	90.9%	1.8y
8	2025	Golden Star Resources	Agriculture loss 2025-10	389,617	235,935	625,552	1.14	712,817	94.8%	1.7y
9	2025	Ghana Grid Company Substation	Agriculture loss 2025-4	350,376	156,061	506,437	1.18	595,519	98.2%	1.5y
10	2025	Ghana Grid Company Substation	Energy loss 2025-3	226,504	77,752	304,256	1.09	332,613	100.0%	1.0y

Large loss Pareto — /prop/large-loss-pareto

ON THIS SCREEN

- Parameters** — Pareto minimum (only losses at or above the threshold drive the tail), treaty limit (auto-set from the structure, editable), alpha, and the number of observation years that turns the tail count into a

frequency.

- **Goodness of fit** — each candidate distribution with its parameters, N, KS statistic, p-value and a plain verdict (Good / Fair / Poor). The best fit is badged. You choose which one drives return periods and pricing.
- **Large loss statistics** — count, total incurred, average, max and min, standard deviation, top-5 concentration, average yearly loss, frequency above the threshold and the resulting pure premium.
- Save Curve pins the chosen fit so the pricing screens use it.

Cat loss list

/prop/cat-loss-list

The catastrophe loss listing, kept separate from large losses because it is trended, selected and fitted separately.

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PROPORTIONAL TREATY: CAT LOSSES

REPORT DATE: mm/dd/yyyy 4 records

Loss Analysis Load Clear Save

#	UW YEAR	INSURED NAME	LOSS / EVENT	DATE OF LOSS	CLASS	PAID	O/S	INCURRED	SAVED	
1	2022	Kumasi Brewery	Earthquake event 2022-	04/28/2022	Energy	2,852,162	261,021	3,113,183	2022-05-28	
2	2023	Ghana Grid Company Subs	Riot & Strike event 2023	07/24/2023	Energy	1,223,744	191,398	1,415,142	2023-08-24	
3	2025	Sunon Asogli Power	Flood event 2025-3	03/03/2025	Energy	974,126	477,307	1,451,433	2025-04-03	
4	2025	Tema Oil Refinery	Fire following event 202	09/17/2025	Energy	1,857,638	271,306	2,128,944	2025-10-17	
5				mm/dd/yyyy	Select...	0	0	-		
TOTAL								6,907,670	1,201,032	8,108,702

Cat loss list — /prop/cat-loss-list

ON THIS SCREEN

- Same grid and paste behaviour as the large-loss list; rows carry the event name (earthquake, flood, riot and strike, fire following, and so on).

Select and trend the cat losses that belong in the pricing basis.

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PROPORTIONAL TREATY: CAT LOSS SELECTION

Losses have been added or edited since the last selection was saved — review and re-save your selection.

TOTAL4

SELECTED4

INCURRED8,108,702

INFLATION-ADJ10,818,675

AVG FACTOR1.2816

INCEPTION YR2026

CPI SOURCESaudi Arabia

THRESHOLD

e.g. 500,000

Apply

Inflation (Saudi Arabia)

Factor +10%

Growth

Loss Loading

Yearly Aggregates

All

None

Save

✓	YEAR	INSURED / EVENT	LOSS NAME	DATE	PAID	OS	INCURRED	FACTOR	INFLATED
☒	2022	Kumasi Brewery	Earthquake event 2022-1	2022-04-28	2,852,162	261,021	3,113,183	1.5901	4,950,272
☒	2023	Ghana Grid Company Substation	Riot & Strike event 2023-4	2023-07-24	1,223,744	191,398	1,415,142	1.2100	1,712,322
☒	2025	Sunon Asogli Power	Flood event 2025-3	2025-03-03	974,126	477,307	1,451,433	1.1756	1,706,305
☒	2025	Tema Oil Refinery	Fire following event 2025-2	2025-09-17	1,857,638	271,306	2,128,944	1.1507	2,449,776

ADDITIONAL LOADINGS

No additional loadings. Click + Add to include IBNR, trend, or other adjustments.

+ Add

Cat loss selection — /prop/cat-loss-selection

ON THIS SCREEN

- Same mechanics as large-loss selection — threshold, tick list, CPI trending, factor override, additional loadings.
- A **Growth** adjustment is available in addition to inflation, for exposure growth between the loss year and now.

Severity fit for the cat losses. Identical machinery to the large-loss Pareto screen, run on the cat basis.

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PROPORTIONAL TREATY: CAT LOSS PARETO

Severity Distribution Fit

Fit parametric distributions to cat losses. Adjust threshold and limit. Compare Pareto, Lognormal, Exponential, Weibull via KS goodness-of-fit.

✓ Saved

Save Curve

View Sev-Freq Curve

PARAMETERS

Inputs auto-recompute stats + pricing

PARETO MIN (THRESHOLD)

1,706,305

Only losses $\geq$ threshold drive the tail.

TREATY LIMIT (MAX)

70,009,895

Auto-set from total Risk/Cat layer limits in Structure. Editable override.

ALPHA (A)

2.797

Higher $\alpha \rightarrow$ lighter tail. Editable.

OBSERVATION YEARS

10

Frequency = tail count + years.

GOODNESS OF FIT

Select a distribution to use for return periods and pricing. KS statistic: lower = better fit.

USE	DISTRIBUTION	PARAMETERS	N	KS STAT	P-VALUE	VERDICT
☒	Pareto	$\alpha=2.797$	4	0.4902	0.2059	Poor
☐	Lognormal BEST	$\mu=14.71 \sigma=0.43$	4	0.2926	0.8896	Fair
☐	Exponential	$\lambda=0.000001$	4	0.4940	0.1988	Poor
☐	Weibull	$k=0.510 \lambda=846853$	3	0.4230	0.3681	Poor

CAT LOSS STATISTICS

	COUNT	4
TOTAL INCURRED	10,818,675	
AVERAGE	2,704,669	
MAX / MIN	4,950,272 / 1,706,305	
STD DEVIATION	1,331,277	
TOP 5 CONCENTRATION	100.0%	
AVG YEARLY LOSS	1,081,867	
FREQUENCY ($\geq X$ M)	0.40 / yr	
RISK PURE PREMIUM	379,433	

RETURN PERIODS (PARETO)

Implied severity from fitted tail

COMPARE WITH TP $\rightarrow$

RETURN PERIOD	ESTIMATED LOSS
1 IN 500 YR	15,744,916
1 IN 250 YR	12,288,520
1 IN 100 YR	8,855,361
1 IN 50 YR	6,911,392
1 IN 25 YR	5,394,171
1 IN 20 YR	4,980,491
1 IN 10 YR	3,887,151
1 IN 5 YR	3,033,826
1 IN 2 YR	2,186,237

PARETO RANKING

#	UW YR	INSURED / EVENT	LOSS	PAID	OS	INCURRED	INFL.	INFLATED	CUM%	RETURN PERIOD
1	2022	Kumasi Brewery	Earthquake event 2022-1	2,852,162	261,021	3,113,183	1.59	4,950,272	45.8%	49y
2	2025	Tema Oil Refinery	Fire following event 2025-2	1,857,638	271,306	2,128,944	1.15	2,449,776	68.4%	6.9y
3	2023	Ghana Grid Company Substation	Riot & Strike event 2023-4	1,223,744	191,398	1,415,142	1.21	1,712,322	84.2%	2.5y
4	2025	Sunon Asogli Power	Flood event 2025-3	974,126	477,307	1,451,433	1.18	1,706,305	100.0%	2.5y

CAT MODEL OEP BURNING COST ▼

Enter OEP curve from cat model (RMS / AIR / Verisk). Layer burning cost = $\int [ID, D+L] P(\text{occ loss} > x) dx$.

Cat loss Pareto — /prop/cat-loss-pareto

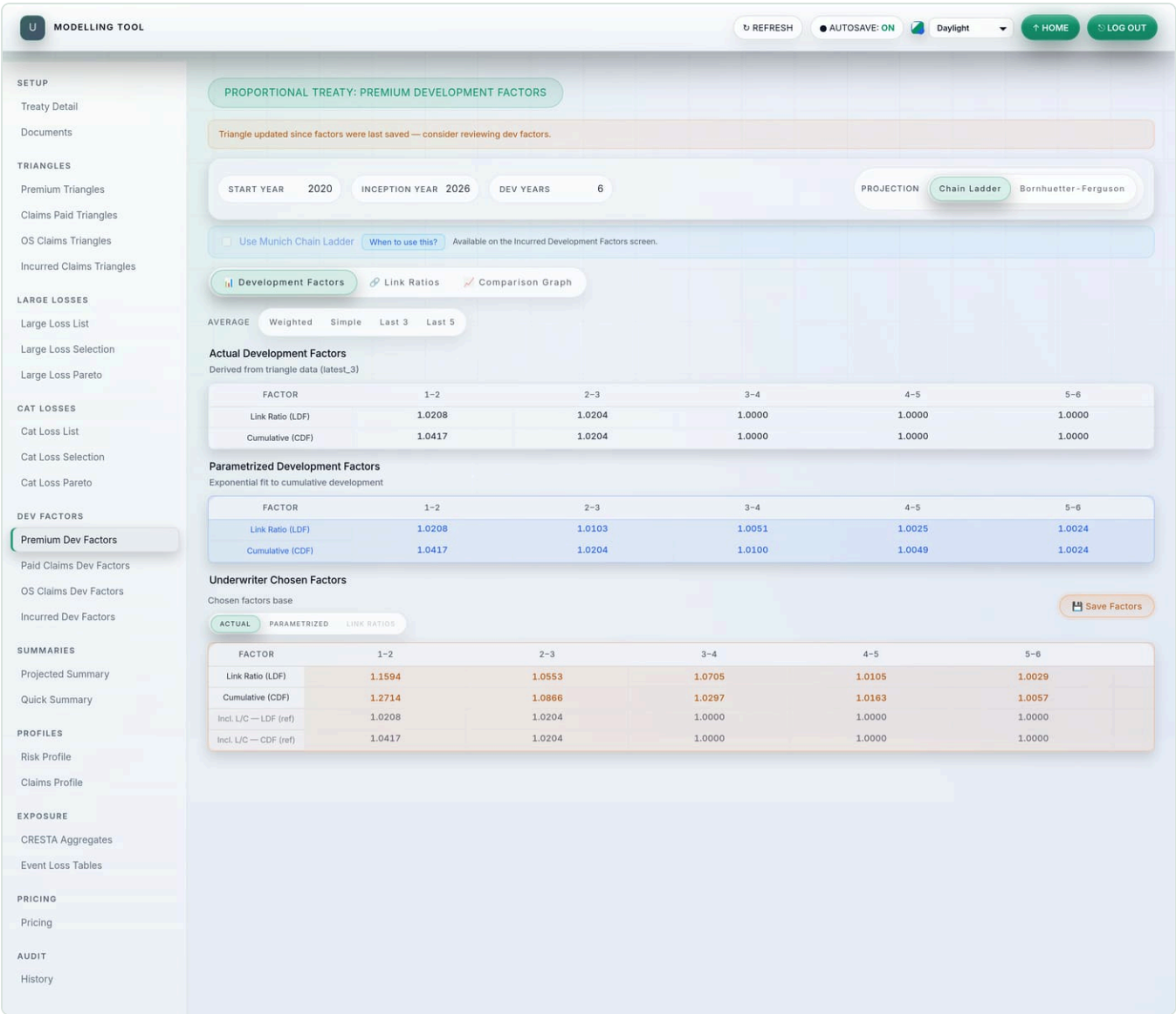
ON THIS SCREEN

- Cat samples are usually small, and the screen is honest about it — the KS verdicts and p-values will tell you when a fit is not carrying much information.

Universe 3 — Underwriter Manual

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Choose the development pattern for premium. Three views: the factor table, the link-ratio triangle, and a comparison graph against benchmarks.



Premium dev factors — /prop/premium-dev-factors

ON THIS SCREEN

- **Factor table** — volume-weighted, simple and other averages of the observed link ratios, the parametrized (exponential) fit, and the row of factors you actually select. LDFs and the cumulative CDFs are shown together.
- **Link ratios** — the ratio triangle with per-cell outlier exclusion; excluding a cell recomputes the averages immediately.
- **Comparison graph** — your selected curve against country, regional and all-market benchmark curves from the formula workbench.
- **Bornhuetter-Ferguson** — for premium this is the "% achieved" bar; the projection table sits underneath.

- A triangle-basis toggle switches between the full triangle and the attritional (large/cat stripped) triangle. When you are on the stripped basis the table also greys in the full-basis factors for reference and amber-flags any selected LDF that exceeds its full-basis counterpart, which is unusual and worth a second look.

The same factor-selection surface run on the paid-claims triangle.

U

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PROPORTIONAL TREATY: PAID CLAIMS DEV FACTORS

Triangle updated since factors were last saved — consider reviewing dev factors.

Triangle basis

Full

Stripped of Large/Cat Losses

Stripped is the attritional basis — recommended. Saved per treaty: choosing Full uses the original triangle and folds large/cat into attritional (shown as nil) on the summaries.

10 large losses and 4 cat losses have been excluded from this triangle. Selected factors reflect the underlying attritional experience.

2 losses placed by loss date (actuarial reported date not set) — their development period is an estimate. Set a reported date on the loss screens to place them precisely.

START YEAR2020INCEPTION YEAR2026DEV YEARS6

PROJECTIONChain LadderBornhuetter-Ferguson

Use Munich Chain LadderWhen to use this?Available on the Incurred Development Factors screen.

Development FactorsLink RatiosComparison Graph

AVERAGEWeightedSimpleLast 3Last 5

Actual Development Factors

Derived from triangle data (latest_3)

FACTOR	1-2	2-3	3-4	4-5	5-6
Link Ratio (LDF)	6.4568	1.2973	1.0622	1.0276	1.0000
Cumulative (CDF)	9.1428	1.4160	1.0915	1.0276	1.0000

Parametrized Development Factors

Exponential fit to cumulative development

FACTOR	1-2	2-3	3-4	4-5	5-6
Link Ratio (LDF)	3.3513	1.5801	1.0996	1.0158	1.0029
Cumulative (CDF)	5.9319	1.7700	1.1202	1.0188	1.0029

Underwriter Chosen Factors

Chosen factors base

ACTUALPARAMETRIZEDLINK RATIOS

Save Factors

FACTOR	1-2	2-3	3-4	4-5	5-6
Link Ratio (LDF)	1.1576	1.0392	1.0197	1.0521	1.0081
Cumulative (CDF)	1.2719	1.0987	1.0573	1.0329	1.0151
Incl. L/C — LDF (ref)	1.4724	1.1515	1.0622	1.0276	1.0127
Incl. L/C — CDF (ref)	1.8742	1.2729	1.1054	1.0407	1.0127

Paid claims dev factors — /prop/paid-claims-dev-factors

ON THIS SCREEN

- BF here takes an initial expected loss ratio (IELR); the per-class defaults come from the formula workbench.
- **Munich chain ladder** can be switched on where paid and incurred are both available — it corrects the classic bias of projecting the two independently.

Development pattern for outstanding claims.

U

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PROPORTIONAL TREATY: OS CLAIMS DEV FACTORS

Triangle updated since factors were last saved — consider reviewing dev factors.

Triangle basis

Full

Stripped of Large/Cat Losses

Stripped is the attritional basis — recommended. Saved per treaty: choosing Full uses the original triangle and folds large/cat into attritional (shown as nil) on the summaries.

10 large losses and 4 cat losses have been excluded from this triangle. Selected factors reflect the underlying attritional experience.

2 losses placed by loss date (actuarial reported date not set) — their development period is an estimate. Set a reported date on the loss screens to place them precisely.

START YEAR2020INCEPTION YEAR2026DEV YEARS6

PROJECTIONChain LadderBornhuetter-Ferguson

Use Munich Chain LadderWhen to use this?Available on the Incurred Development Factors screen.

Development FactorsLink RatiosComparison Graph

AVERAGEWeightedSimpleLast 3Last 5

Actual Development Factors

Derived from triangle data (latest_3)

FACTOR	1-2	2-3	3-4	4-5	5-6
Link Ratio (LDF)	1.7022	1.1897	1.0622	1.0276	1.0000
Cumulative (CDF)	2.2104	1.2986	1.0915	1.0276	1.0000

Parametrized Development Factors

Exponential fit to cumulative development

FACTOR	1-2	2-3	3-4	4-5	5-6
Link Ratio (LDF)	1.6129	1.2125	1.0647	1.0188	1.0076
Cumulative (CDF)	2.1374	1.3252	1.0930	1.0266	1.0076

Underwriter Chosen Factors

Chosen factors base

ACTUALPARAMETRIZEDLINK RATIOS

Save Factors

FACTOR	1-2	2-3	3-4	4-5	5-6
Link Ratio (LDF)	1.3402	1.0233	1.0408	1.0124	1.0123
Cumulative (CDF)	1.5628	1.1892	1.0947	1.0518	1.0298
Incl. L/C — LDF (ref)	1.4724	1.1515	1.0622	1.0276	1.0127
Incl. L/C — CDF (ref)	1.8742	1.2729	1.1054	1.0407	1.0127

OS claims dev factors — /prop/os-claims-dev-factors

ON THIS SCREEN

- Same three views, same averaging choices, same benchmark comparison.

Development pattern for incurred claims — in practice the most consequential of the four, because the ultimate loss projection runs on it.

U

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PROPORTIONAL TREATY: INCURRED DEV FACTORS

Triangle updated since factors were last saved — consider reviewing dev factors.

Triangle basis

Full

Stripped of Large/Cat Losses

Stripped is the attritional basis — recommended. Saved per treaty: choosing Full uses the original triangle and folds large/cat into attritional (shown as nil) on the summaries.

Stripped Incurred Triangle

10 large losses and 4 cat losses have been excluded from this triangle. Selected factors reflect the underlying attritional experience.

2 losses placed by loss date (actuarial reported date not set) — their development period is an estimate. Set a reported date on the loss screens to place them precisely.

START YEAR

2020

INCEPTION YEAR

2026

DEV YEARS

6

PROJECTION

Chain Ladder

Bornhuetter-Ferguson

Use Munich Chain Ladder

When to use this?

Development Factors

Link Ratios

Comparison Graph

AVERAGE

Weighted

Simple

Last 3

Last 5

Actual Development Factors

Derived from triangle data (manual)

FACTOR	1-2	2-3	3-4	4-5	5-6
Link Ratio (LDF)	2.6150	1.2571	1.0622	1.0276	1.0000
Cumulative (CDF)	3.5881	1.3721	1.0915	1.0276	1.0000

Parametrized Development Factors

Exponential fit to cumulative development

FACTOR	1-2	2-3	3-4	4-5	5-6
Link Ratio (LDF)	2.1174	1.3313	1.0796	1.0181	1.0052
Cumulative (CDF)	3.1144	1.4708	1.1048	1.0233	1.0052

Underwriter Chosen Factors

Chosen factors base

ACTUAL

PARAMETRIZED

LINK RATIOS

Save Factors

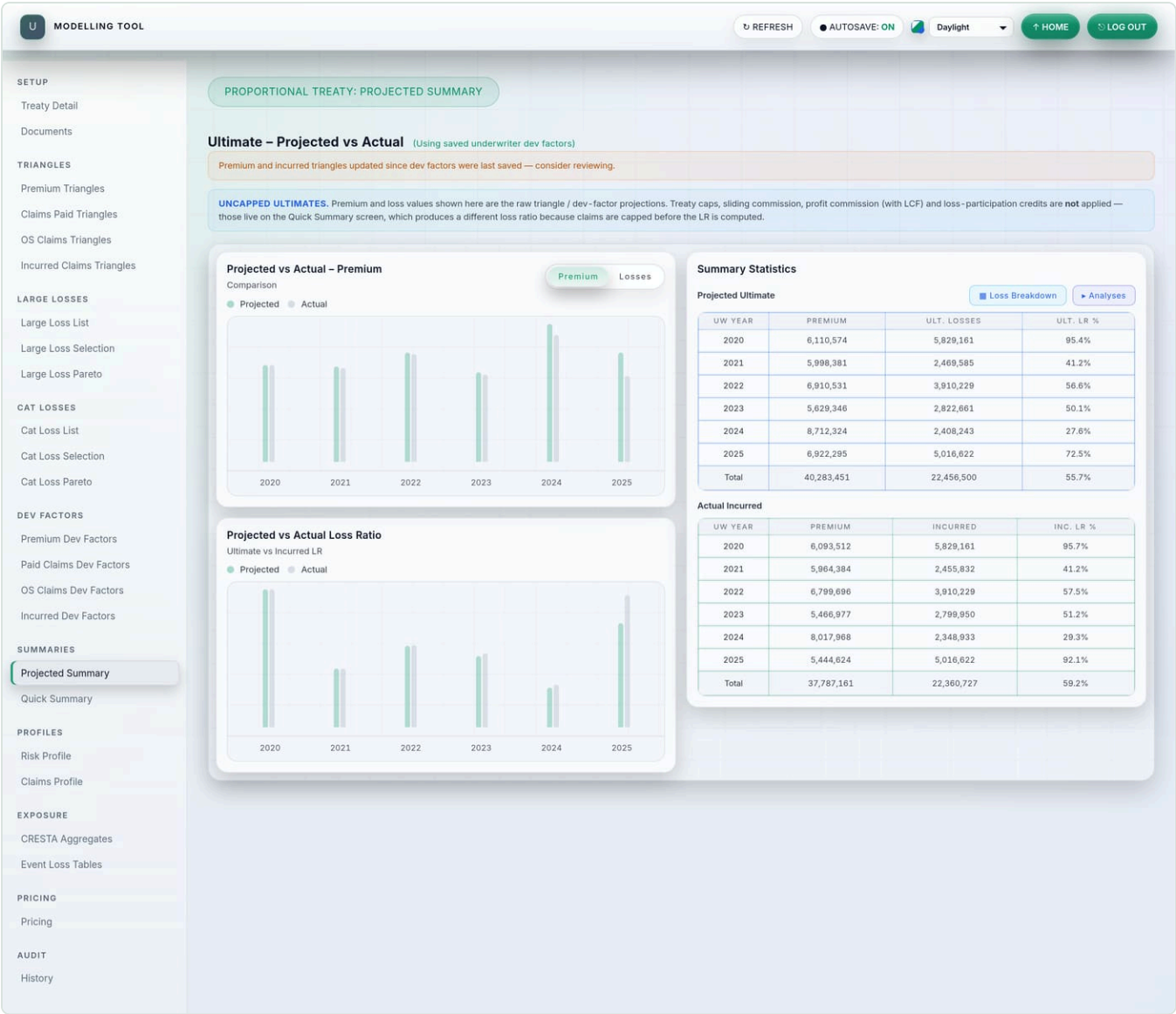
FACTOR	1-2	2-3	3-4	4-5	5-6
Link Ratio (LDF)	1.0229	1.0239	1.0312	1.0635	0.9640
Cumulative (CDF)	1.0924	1.0407	1.0164	1.0094	1.0056
Incl. L/C — LDF (ref)	1.4724	1.1515	1.0622	1.0276	1.0127
Incl. L/C — CDF (ref)	1.8742	1.2729	1.1054	1.0407	1.0127

Incurred dev factors — /prop/incurred-dev-factors

ON THIS SCREEN

- A modal compares the full incurred triangle against the stripped (attritional) one side by side, so the effect of removing the large and cat losses is visible before you commit to a basis.

Ultimates by underwriting year, projected on your saved development factors and set against actuals. Premium, ultimate losses and ultimate loss ratio per year, with projected-vs-actual charts for both premium and loss ratio.

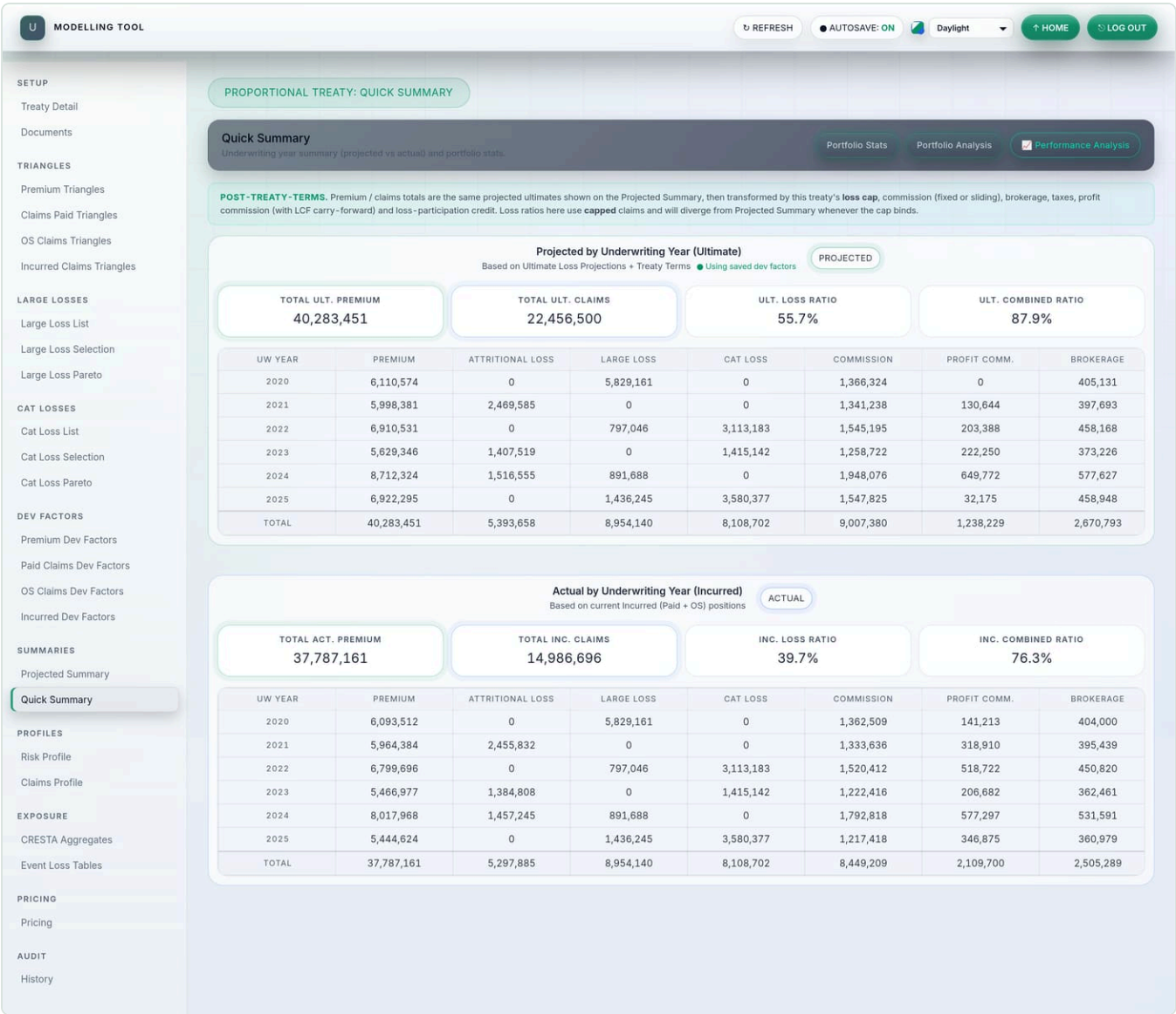


Projected summary — /prop/projected-summary

ON THIS SCREEN

- These are **uncapped** ultimates: the raw triangle × dev-factor projections. Treaty caps, sliding commission, profit commission and loss-participation credits are *not* applied here.
- The screen warns you when the triangles have changed since the dev factors were last saved.
- A loss-breakdown view splits attritional, large and cat.

The same ultimates, now put through the treaty's actual terms. This is the post-terms view, and it is the one that answers "does this treaty make money".

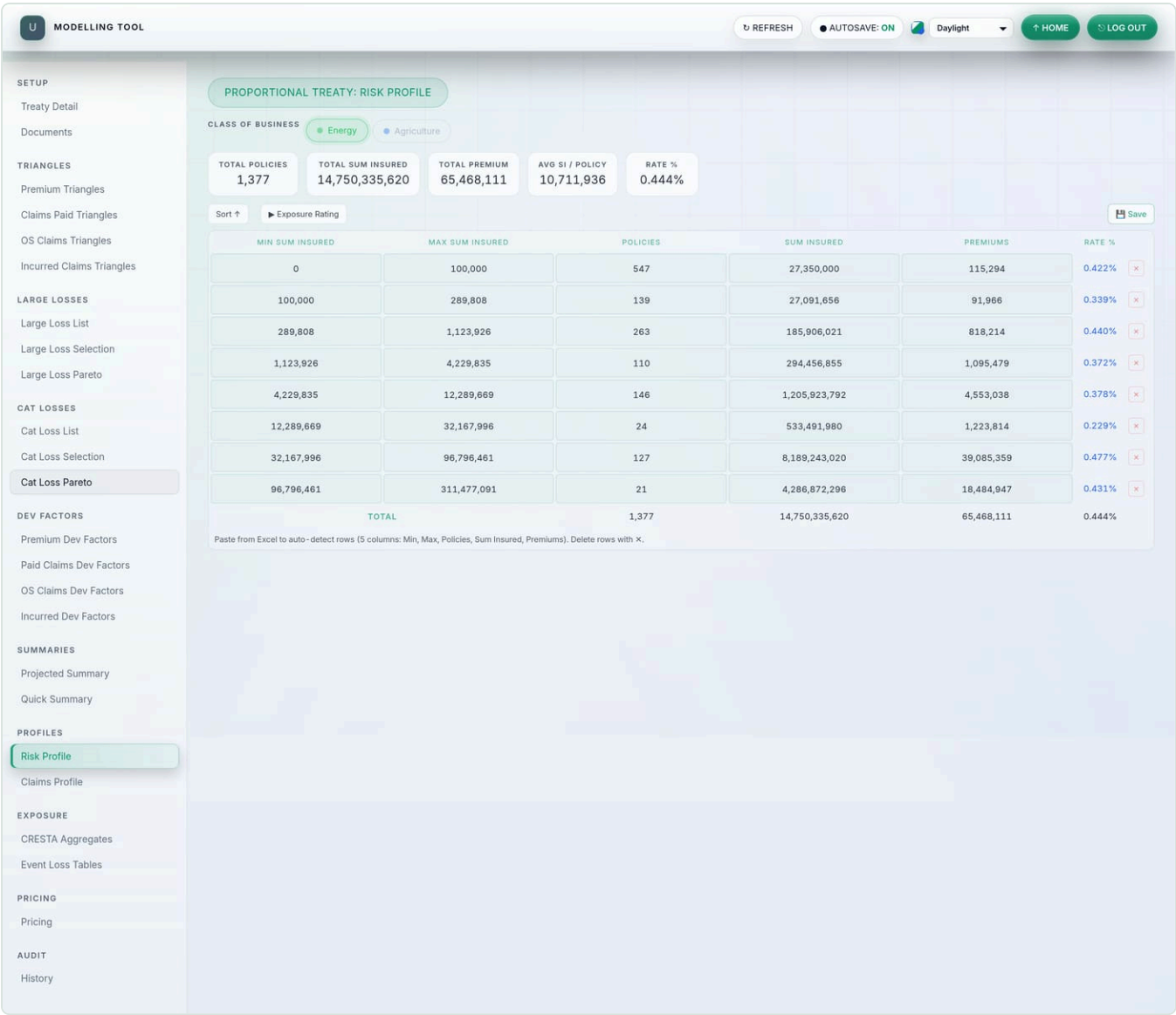


Quick summary — /prop/quick-summary

ON THIS SCREEN

- Per underwriting year: premium, attritional / large / cat loss, commission, profit commission, brokerage, taxes, loss-participation credit, result, cumulative result and cumulative percentage.
- Headline tiles: total ultimate premium, total ultimate claims, ultimate loss ratio and ultimate combined ratio.
- Loss ratios here use **capped** claims and will diverge from the Projected Summary whenever the cap binds — that divergence is the point of having both screens.
- Portfolio stats and performance analysis views sit alongside.

The cedant's portfolio banded by sum insured: policy counts, sum insured and premium per band, with the implied rate.

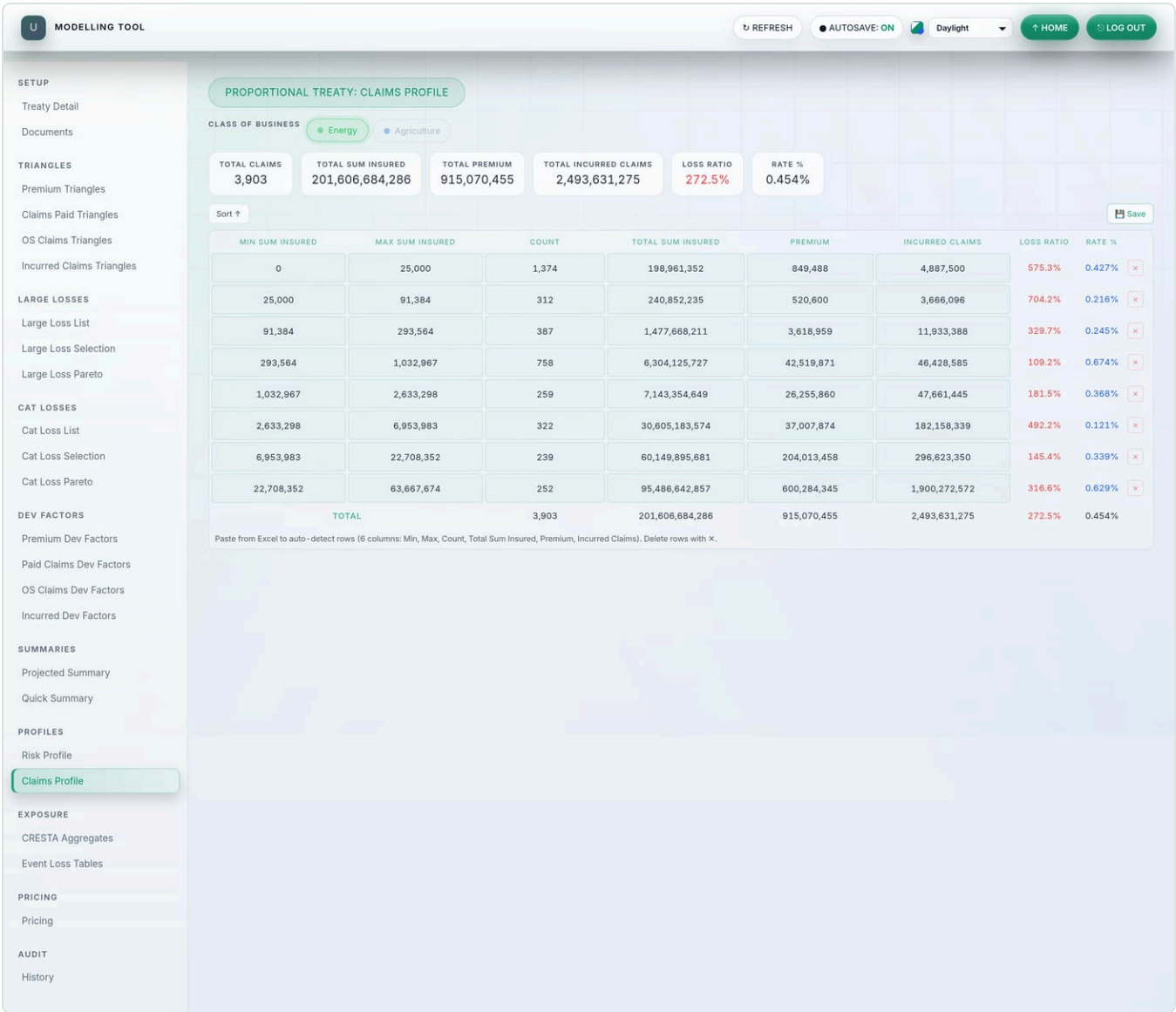


Risk profile — /prop/risk-profile

ON THIS SCREEN

- One tab per class of business on the treaty.
- Paste five columns from Excel — min, max, policies, sum insured, premiums — and rows are detected automatically.
- Totals and the overall rate sit at the foot. **Exposure Rating** runs the profile through the exposure curves.

The claims equivalent of the risk profile: claim counts, sum insured, premium and incurred claims per band, with the loss ratio and rate implied by each band.

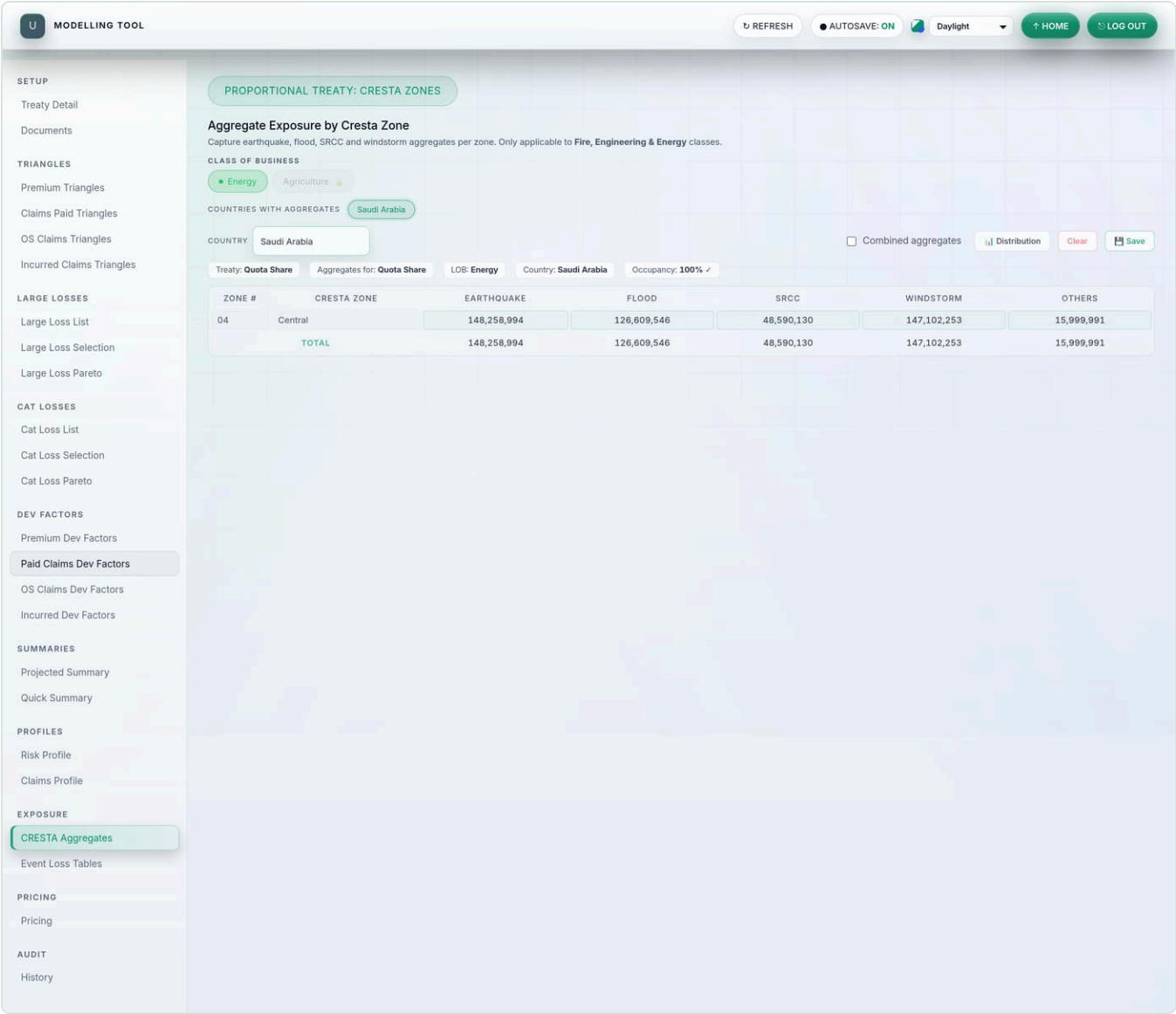


Claims profile — /prop/claims-profile

ON THIS SCREEN

- Paste six columns — min, max, count, total sum insured, premium, incurred claims.
- Reading the loss ratio down the bands is what tells you where the cedant's book is actually losing money.

Accumulated exposure by CRESTA zone and peril — earthquake, flood, SRCC, windstorm and other.

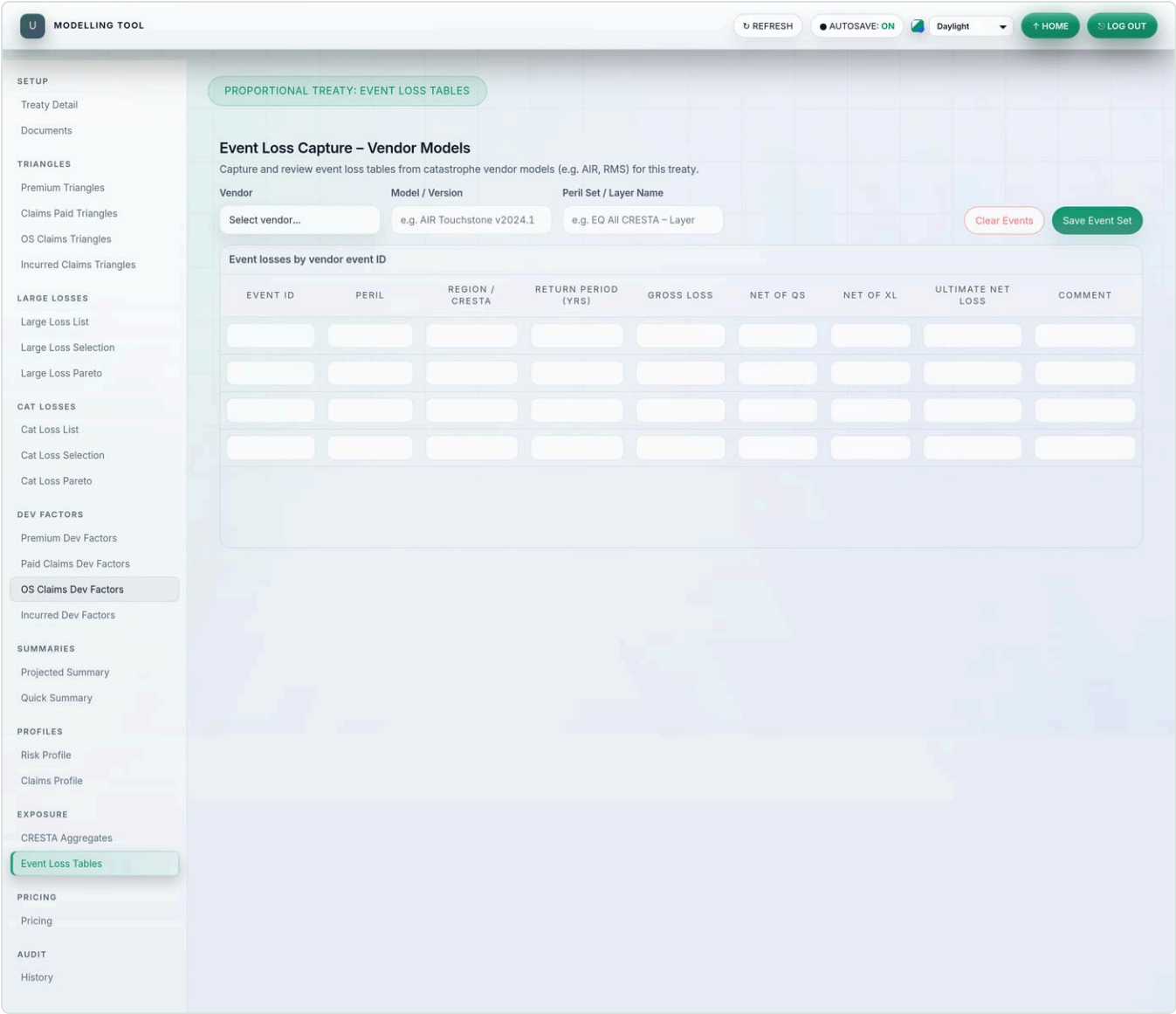


CRESTA aggregates — /prop/cresta-aggregates

ON THIS SCREEN

- Applies to Fire, Engineering and Energy classes; other classes are locked out with a padlock rather than silently accepting numbers that mean nothing.
- Per country, with a combined view across countries that carry aggregates. A distribution chart shows the concentration.
- These aggregates are what the cat exposure rating and the accumulation checks on the pricing screen read.

Capture of vendor catastrophe model output — AIR or RMS — event by event.

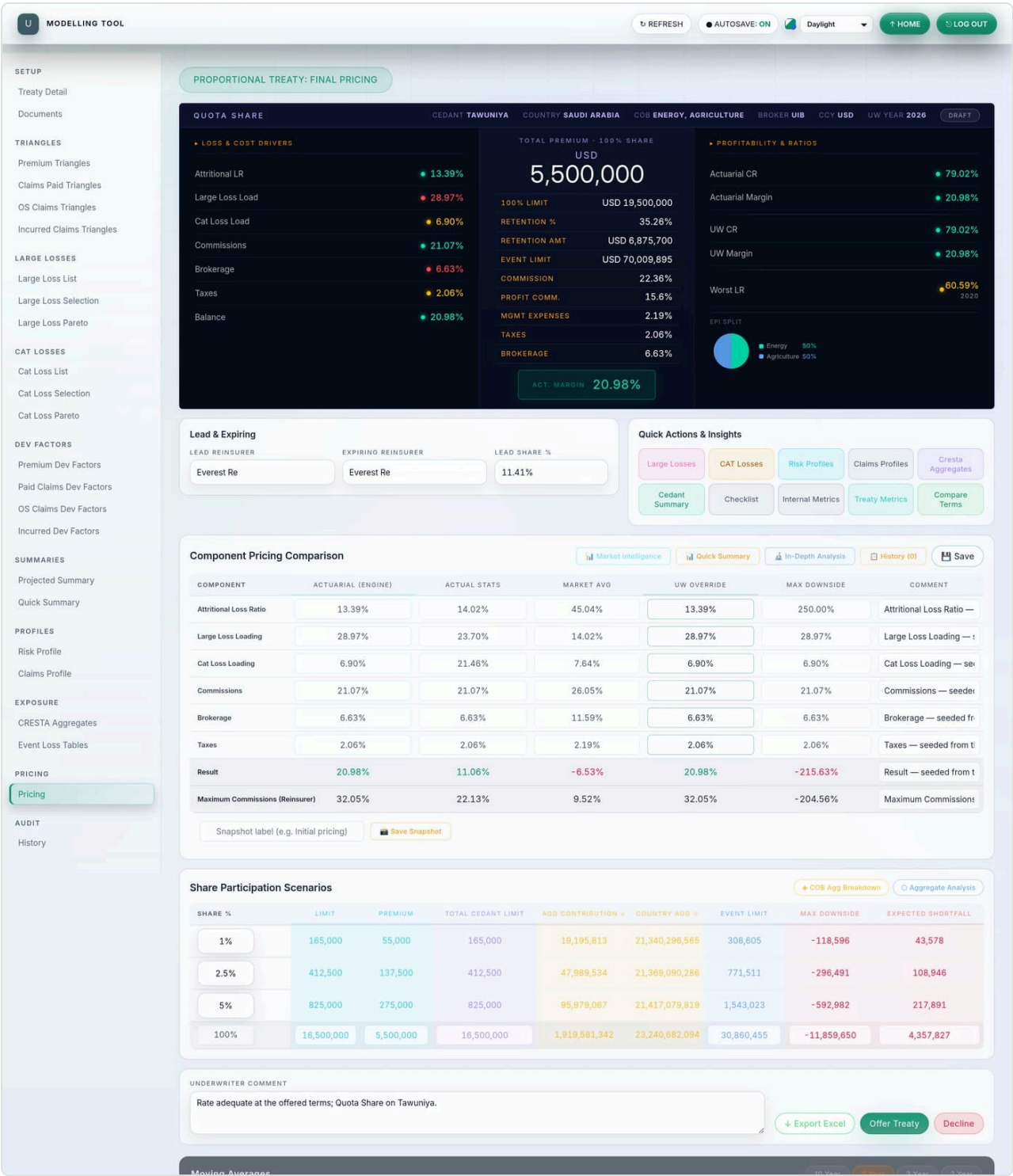


Event loss tables — /prop/event-loss-tables

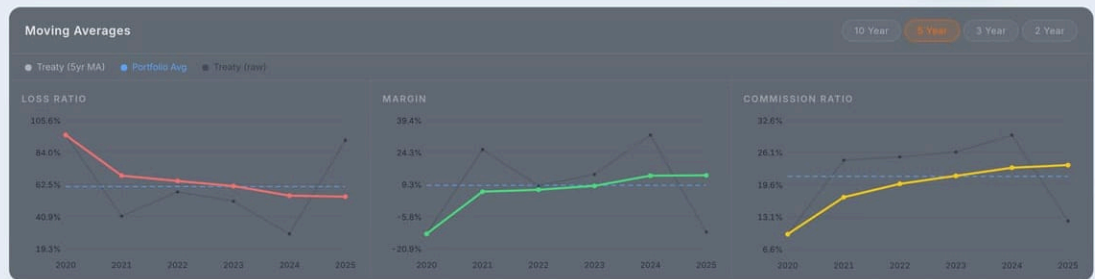
ON THIS SCREEN

- Vendor, model/version and peril set at the top, then one row per event ID: peril, region/CRESTA, return period, gross loss, net of QS, net of XL, ultimate net loss and a comment.
- This is where a modelled AEP/OEP view enters the treaty file.

The decision screen. Everything upstream converges into one rate build-up, one set of ratios, and the Offer / Decline buttons.



Final pricing — /prop/pricing (1 of 2)



Final pricing — /prop/pricing (2 of 2)

ON THIS SCREEN

- **Hero panel** — loss and cost drivers (attritional LR, large loss load, cat loss load, commissions, brokerage, taxes, balance) on the left; total premium at 100% share with limit, retention, event limit, commission, profit commission, management expenses, taxes and brokerage in the middle; profitability and ratios (actuarial CR and margin, UW CR and margin, worst historical LR and the year it came from) on the right, with the EPI split by class.
- **Component pricing comparison** — every component (attritional LR, large loss loading, cat loss loading, commissions, brokerage, taxes, result, maximum reinsurer commission) shown four ways: actuarial (engine), actual stats, market average, and your UW override, with a max-downside column and a comment. The override column is where underwriting judgement is recorded explicitly rather than buried.
- **Share participation scenarios** — the treaty at 1%, 2.5%, 5% and 100%: limit, premium, total cedant limit, aggregate contribution, country aggregate, event limit, maximum downside and expected shortfall. This is the accumulation check before you commit a line.
- **Lead & expiring** — lead reinsurer, expiring reinsurer and lead share.
- **Moving averages** — loss ratio, margin and commission ratio for this treaty against the portfolio average over 2 / 3 / 5 / 10 years.
- **Save snapshot** pins a labelled version of the component table so the history of the negotiation is kept.
- **Offer Treaty** submits into the approval workflow; **Decline** records the decline with its reason. Export Excel produces the pricing pack.

Every change, status transition and approval decision on this treaty, newest first. Read-only.

MODELLING TOOL REFRESH AUTOSAVE: ON Daylight HOME LOG OUT

PROPORTIONAL TREATY: AUDIT & HISTORY

Audit & History
Every change, status transition and approval decision for this treaty, newest first. Read-only.

History

- CU Chief Underwriter CU** UPDATED just now
Treaty details updated
23/08/2026, 07:32:55
- CU Chief Underwriter CU** UPDATED 2m ago
Treaty details updated
23/08/2026, 07:30:36
- CU Chief Underwriter CU** UPDATED 39m ago
Treaty details updated
23/08/2026, 06:54:13
- CU Chief Underwriter CU** UPDATED 41m ago
Treaty details updated
23/08/2026, 06:52:29
- CU Chief Underwriter** STATUS CHANGED 7mo ago
Status changed
20/01/2026, 11:00:00
- CU Chief Underwriter** STATUS AWAITING SIGNED_LINE 7mo ago
Status changed AWAITING_APPROVAL → AWAITING_SIGNED_LINE
"AWAITING_APPROVAL → AWAITING_SIGNED_LINE (seeded lifecycle)"
► 1 field change
20/01/2026, 11:00:00
- U1 Underwriter 1** STATUS CHANGED 7mo ago
Status changed
12/01/2026, 09:00:00
- U1 Underwriter 1** STATUS AWAITING APPROVAL 7mo ago
Status changed DRAFT → AWAITING_APPROVAL
"DRAFT → AWAITING_APPROVAL (seeded lifecycle)"
► 1 field change
12/01/2026, 09:00:00
- CU Chief Underwriter** STATUS CHANGED 8mo ago
Status changed
05/01/2026, 14:00:00
- CU Chief Underwriter** STATUS SIGNED 8mo ago
Status changed AWAITING_SIGNED_LINE → SIGNED
"AWAITING_SIGNED_LINE → SIGNED (seeded lifecycle)"
► 1 field change
05/01/2026, 14:00:00

Audit & history — /prop/history

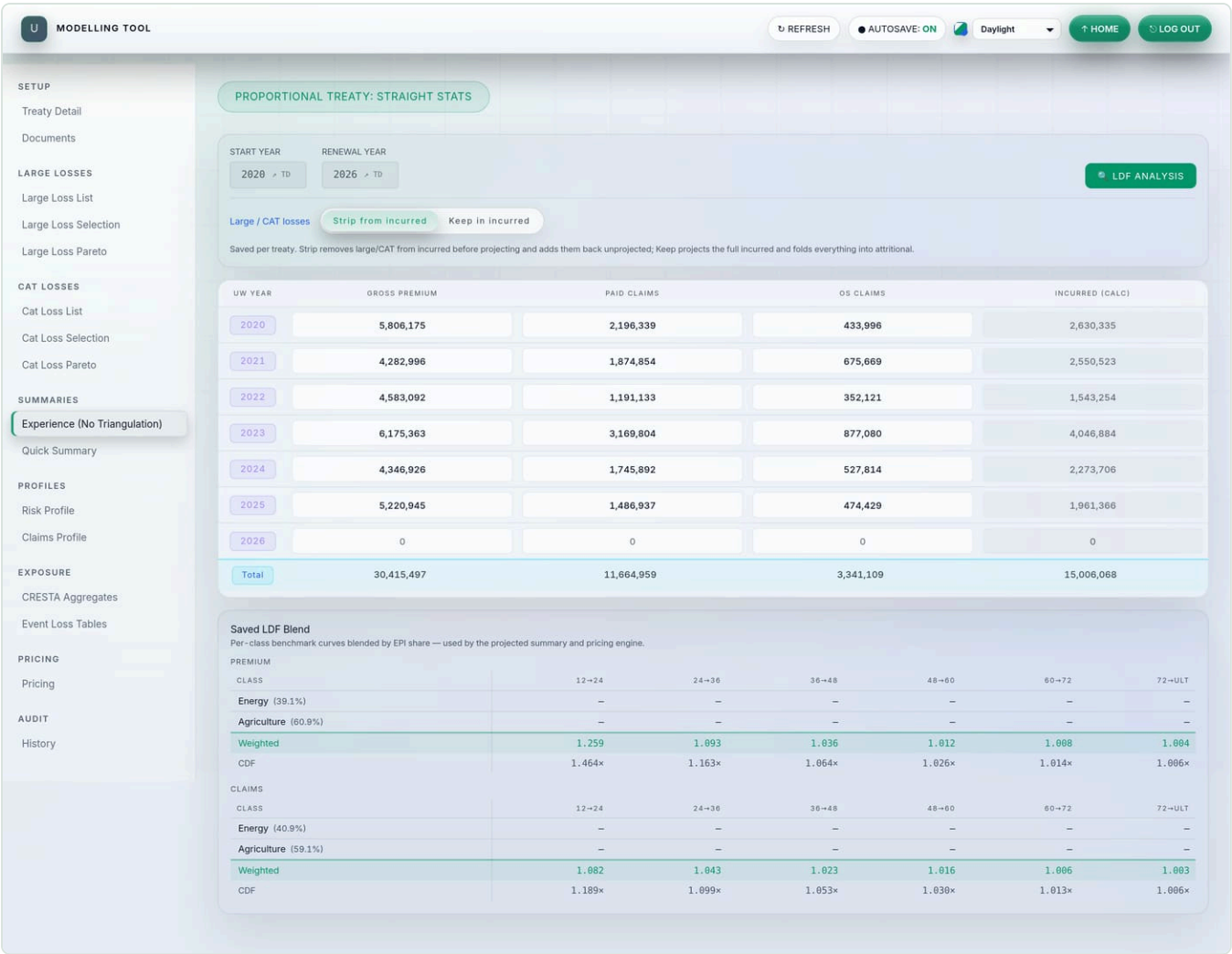
ON THIS SCREEN

- Each entry names the actor and their role, the type of event, when it happened, and — for status changes — the transition and the reason given.
- Field-level changes are expandable inline.

Straight stats (no triangulation)

/prop/no-triangulation

The alternative experience route when the cedant cannot supply triangles. Replaces the triangle, dev-factor and projected-summary steps.



Straight stats (no triangulation) — /prop/no-triangulation

ON THIS SCREEN

- A flat year-by-year table: gross premium, paid claims, OS claims and calculated incurred, from the start year to the renewal year.
- A toggle decides whether large and cat losses are stripped from incurred before projecting (and added back unprojected) or kept in and folded into attritional. The choice is saved per treaty.
- **Saved LDF blend** — per-class benchmark development curves from the formula workbench, blended by EPI share, with the weighted LDF and CDF rows the projection and pricing engine will use.

6. Pricing a non-proportional treaty

The route through

Twenty screens: **Setup** (treaty detail, documents), **Structure** (layer tower, premiums & inflation), **Large losses** and **Cat losses** (list, selection, Pareto and dev factors for each), **Experience** (excess dev factors, historical performance), **Profiles**, **Exposure**, and **Final Pricing**.

The sidebar adapts to the treaty type. A **Risk XL** hides every cat screen plus CRESTA and event loss tables. A **CAT XL** hides the risk and large-loss screens. **Risk & CAT XL** and **Aggregate XL** show everything. **Stop Loss** swaps the layer grid for a loss-ratio layer table and routes pricing to its own dedicated screen.

Header of the XL programme: parties, period, currency, classes, treaty type and the metadata that determines which screens the wizard will show you.

MODELLING TOOL REFRESH AUTOSAVE: ON Daylight HOME LOG OUT

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- Large Loss List
- Large Loss Selection
- Large Loss Pareto
- Large Loss Dev Factors

CAT LOSSES

- Cat Loss List
- Cat Loss Selection
- Cat Loss Pareto
- Cat Loss Dev Factors

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- Excess Dev Factors
- Historical Performance

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- Claims Profile

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- CRESTA Aggregates
- Event Loss Tables

PRICING

- Final Pricing

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- History

NON-PROPORTIONAL TREATY: TREATY DETAIL

Renewal — linked to prior year contract

Cedant: Al Rajhi Takaful - Country: Saudi Arabia - Broker: UIB - Currency: SAR - Type: Risk & CAT XL - COB: Workers Compensation, Medical, Energy, Credit & Surety

Contract: 2026 Al Rajhi Takaful Risk & CAT XL (Workers Compensation, Medical, Energy, Credit & Surety) SA

CONTRACT DETAILS INPUT Upload Slip

Country: Saudi Arabia

Cedant Name: Al Rajhi Takaful

Treaty Type: Risk & CAT XL

Classes of Business: Workers Compensation, Medical +2 more

Broker: UIB

Contract ID: 7272d38e-d771-4170-b21f-b2fd8604bfd3

Currency: SAR

Treaty Inception Date: 04/01/2026

Treaty Renewal Date: 04/01/2027

UW Year: 2026

Experience Start Year: 2020

Contract Description: 2026 Al Rajhi Takaful Risk & CAT XL (Workers Compensation, Medical, Energy, Credit & Surety)

STRUCTURE TERMS

LAYER CONFIGURATION

Number of Layers: 4

Expiring - Number of Layers: 3

Deductible: 16,125,000 SAR

Maximum Retention: 13,712,413 SAR

Accounting Method: Losses Occurring

Type of XL: Gross XL

Accounts: Annual

PREMIUM & COMMISSIONS

Est. GNPI: 118,785,868 SAR

Brokerage %: 11.61%

Taxes %: 4.99%

No Claims Bonus %: 13.18%

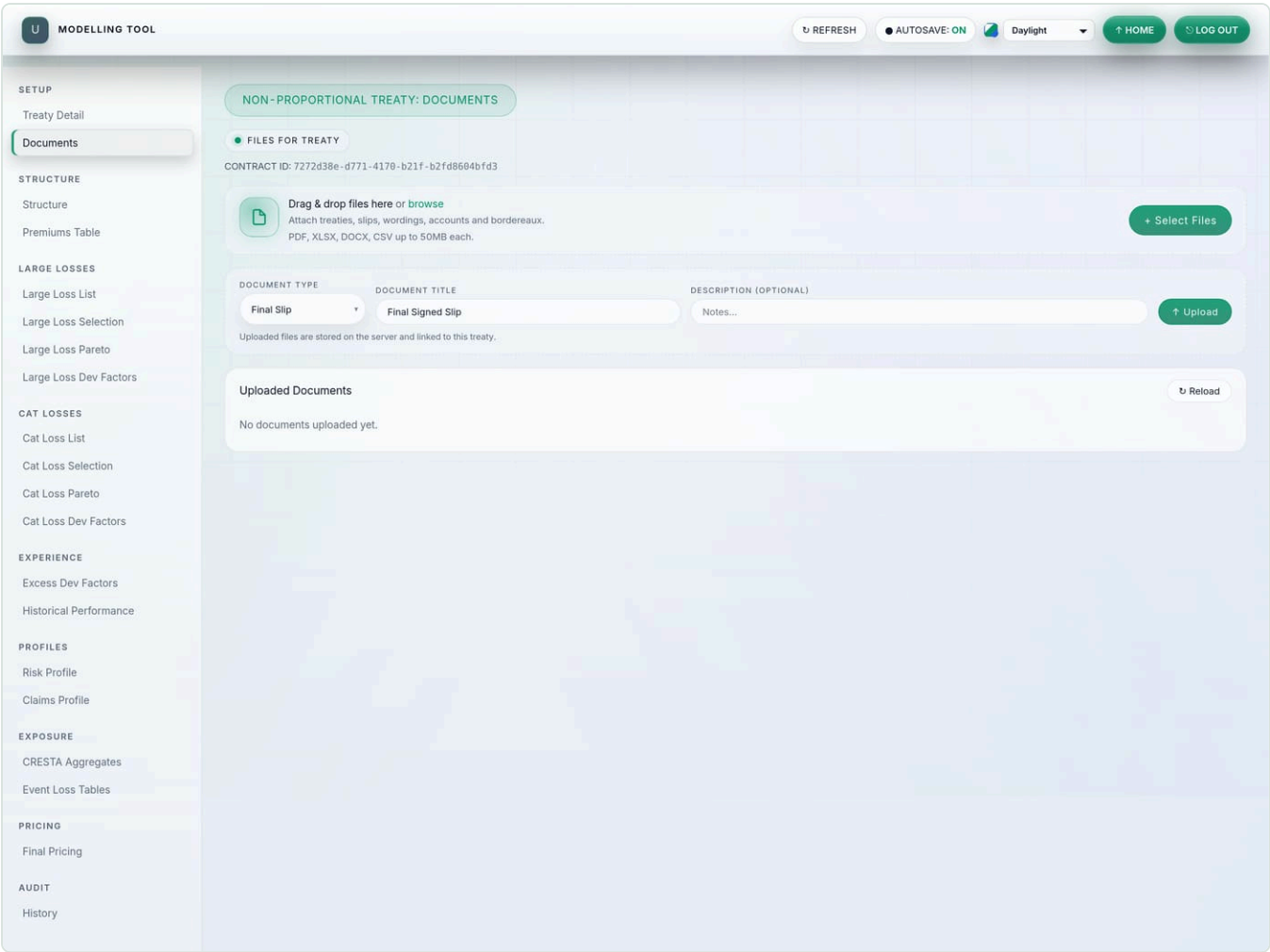
Profit Commission %: 0.29%

Treaty Detail — /np/treaty-detail

ON THIS SCREEN

- Country, cedant, treaty type (Aggregate XL, CAT XL, Risk & CAT XL, Risk XL, Stop Loss), classes of business, broker, currency, dates and underwriting year.
- Layer count and the layer-1 deductible entered here seed the Structure screen.
- Renewals are linked to the prior-year contract, which is what populates the year-on-year columns in Final Pricing.
- Upload Slip** runs the broker slip through the ingestion agent.

The same document store as the proportional wizard: slips, wordings, loss listings, cat modelling output, accounts.



Documents — /np/documents

ON THIS SCREEN

- Typed uploads, stored against this treaty, viewable or downloadable from the list.

The layer tower. One row per layer with limit, deductible/attachment, aggregate limit, EGNPI, rate, earned premium, MDP, reinstatements, AAD and the risk/cat flags.

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Large Loss Pareto

Large Loss Dev Factors

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NON-PROPORTIONAL TREATY: STRUCTURE

TREATY STRUCTURE - LAYERS

RISK XL

CAT XL

+ Add Layer

Delete Layer

LAYER	LIMIT		DEDUCTIBLE / ATTACHMENT		AGGREGATE LIMIT		EGNPI		R
1	77,812,500	SAR	16,125,000	SAR	233,437,500	SAR	144,000,000	SAR	21
2	33,750,000	SAR	93,937,500	SAR	135,000,000	SAR	144,000,000	SAR	6
3	222,187,500	SAR	127,687,500	SAR	666,562,500	SAR	144,000,000	SAR	32
4	168,750,000	SAR	349,875,000	SAR	506,250,000	SAR	144,000,000	SAR	11
TOTAL	502,500,000	SAR	16,125,000	SAR	1,541,250,000	SAR	144,000,000	SAR	72

Enter Rate to calculate Earned Premium = EGNPI x Rate. ROL is implied as Earned Premium x Limit. MDP% is implied as MDP + Earned Premium. Layer 1 deductible is pulled from NP treaty detail; subsequent deductibles auto-calculate as prior deductible + prior limit.

Classes of Business & Layer Participation

Underwriting limits are entered per class. Tick layers that participate for each class.

CLASS OF BUSINESS	UNDERWRITING LIMIT		LAYER 1	LAYER 2	LAYER 3	LAYER 4
Workers Compensation	133,125,000	SAR	☒	☒	☒	☒
Medical	125,625,000	SAR	☒	☒	☒	☒
Energy	76,750,000	SAR	☒	☒	☒	☒
Credit & Surety	63,750,000	SAR	☒	☒	☒	☒

Underwriting limits are captured per class. Participation checkboxes sync with layer COB selections.

Proportional Structure Covered

If this treaty is written on a Net XL basis, summarise the underlying proportional programmes whose net are covered here.

PROPORTIONAL PROGRAMME	QS LIMIT 100%	RI

Populate only when treaty nature is Net XL; leave empty for Gross XL.

Expiring Structure & Terms

✓ Loaded from prior year contract. You may edit fields below.

+ VIEW IMPLIED PRICING CURVE

Layers 4

BROKERAGE % 11.03

NCB % 8.71

PROFIT COMM. % 11.16

LAYER	LIMIT		DEDUCTIBLE / ATTACHMENT		AGGREGATE LIMIT		EGNPI		R
L1	71,587,500	SAR	14,835,000	SAR	214,762,500	SAR	129,600,000	SAR	20
L2	31,050,000	SAR	86,422,500	SAR	124,200,000	SAR	129,600,000	SAR	6
L3	204,412,500	SAR	117,472,500	SAR	613,237,500	SAR	129,600,000	SAR	30
L4	155,250,000	SAR	321,885,000	SAR	465,750,000	SAR	129,600,000	SAR	10
TOTAL	462,300,000	SAR	14,835,000	SAR	1,417,950,000	SAR	129,600,000	SAR	66

Expiring structure is used to auto-populate year-on-year metrics in Final Pricing.

Structure — /np/structure (1 of 2)

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Expiring Proportional Structure Covered

Prior year underlying proportional programmes covered under this Net XL treaty.

PROPORTIONAL PROGRAMME	QS LIMIT 100%	RI

Expiring proportional covered — leave empty for Gross XL.

Structure — /np/structure (2 of 2)

ON THIS SCREEN

- Separate RISK XL and CAT XL tabs where the treaty covers both.
- Earned premium is derived as $EGNPI \times \text{rate}$; ROL is implied as $\text{earned premium} \div \text{limit}$; MDP% as $MDP \div \text{earned premium}$. You type the rate, not the answer.
- Layer 1's deductible comes from Treaty Detail; each subsequent deductible auto-calculates as the prior deductible plus the prior limit, so the tower cannot develop a gap by accident.
- **Classes of business & layer participation** — an underwriting limit per class, with tick boxes for which layers that class participates in.
- **Proportional structure covered** — for a net XL basis, the underlying QS/surplus programmes that inure to this cover, pulled from the database or entered manually.
- Stop Loss and Aggregate XL replace the standard grid with their own surfaces: loss-ratio attach/limit/EPI rows for Stop Loss, absolute aggregate amounts plus franchise-deductible and structured-deal toggles and a class inner-limits table for Aggregate XL.

The prior placement's terms, layer by layer, so Final Pricing can show year-on-year movement. For new business you type the market's expiring terms in by hand.

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NP: EXPIRING STRUCTURE

LAYERS4

BROKERAGE %11.03%

NCB %8.71%

PROFIT COMM. %11.16%

SAVE

EXPIRING TREATY STRUCTURE - LAYERS

New business — enter expiring market terms manually for YoY comparison in pricing metrics.

VIEW IMPLIED PRICING CURVE

RISK XL

CAT XL

LAYER	LIMIT		DEDUCTIBLE / ATTACHMENT		AGGREGATE LIMIT		EGNPI		RATE %
L1	71,587,500	SAR	14,835,000	SAR	214,762,500	SAR	129,600,000	SAR	20.67%
L2	31,050,000	SAR	86,422,500	SAR	124,200,000	SAR	129,600,000	SAR	6.37%
L3	204,412,500	SAR	117,472,500	SAR	613,237,500	SAR	129,600,000	SAR	30.81%
L4	155,250,000	SAR	321,885,000	SAR	465,750,000	SAR	129,600,000	SAR	10.77%
TOTAL	462,300,000	SAR	14,835,000	SAR	1,417,950,000	SAR	518,400,000	SAR	—

Expiring structure is used to auto-populate year-on-year metrics in Final Pricing.

Proportional Structure Covered

If Net XL basis, summarise the underlying proportional programmes covered here.

FROM DATABASE

MANUAL ENTRY

PROGRAMME	QS LIMIT 100%		RETENTION %		RETENTION AMT		SURPLUS LINES		TOTAL CAPACITY
—	—	SAR	—%		—	SAR	—		— SAR
—	—	SAR	—%		—	SAR	—		— SAR
—	—	SAR	—%		—	SAR	—		— SAR
—	—	SAR	—%		—	SAR	—		— SAR

+ ADD ROW

0 programmes - Total capacity: 0 SAR

Expiring structure — /np/expiring-structure

ON THIS SCREEN

- Same columns as the current tower, plus brokerage, no-claims bonus and profit commission percentages.
- **View implied pricing curve** fits a power curve to the expiring layers — display-only analytics that show what the market was charging across the tower.
- The covered-proportional section mirrors the one on Structure for the expiring year.

The subject premium by underwriting year and the inflation assumptions that trend historical losses forward. Everything that on-levels experience starts here.

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NON-PROPORTIONAL TREATY: PREMIUMS & INFLATION

Premiums & Inflation Cockpit

Underwriting years 2020 to 2027 · Al Rajhi Takaful · Risk & CAT XL

UNDERWRITING YEARS · PREMIUM

Rate Changes

PORTFOLIO MOVEMENT

UW	PREMIUM	Δ VS PREV
2020	123,677,474	—
2021	163,958,684	32.57%
2022	123,655,611	-24.58%
2023	139,725,482	13%
2024	101,554,600	-27.32%
2025	118,785,868	16.97%
2026	0	—
2027	0	—

Tip: paste values directly from Excel. We auto-calc change vs previous year.

INFLATION ASSUMPTIONS

Country: Saudi Arabia

COUNTRY

COUNTRY

Use country inflation

Use average inflation

INFLATION UW	INFLATION %	CUMULATIVE
2020	19.39%	1
2021	12.9%	1.13
2022	6.47%	1.2
2023	7.15%	1.29
2024	12.35%	1.45
2025	4.4%	1.51
2026	0.00%	1.51
2027	0.00%	1.51

Cumulative factor applies sequentially year - on - year (base = 1.00).

Premiums & inflation — /np/premiums-table

ON THIS SCREEN

- **Underwriting years · premium** — one premium per year with the portfolio movement (change versus previous year) computed for you. Paste the row straight from Excel.
- **Rate changes** — a compound on-level adjustment modal. The chain of rate changes captured here is what the burning-cost screens divide by.
- **Inflation assumptions** — country CPI or an average, shown as the annual rate and the cumulative factor applied sequentially from a base of 1.00.

The from-the-ground-up large-loss listing for the risk section.

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NON-PROPORTIONAL TREATY: LARGE LOSSES

REPORT DATEmm/dd/yyyy16 records

Loss AnalysisLoadClearSave

#	UW YEAR	INSURED NAME	LOSS / EVENT	DATE OF LOSS	CLASS	PAID	O/S	INCURRED	SAVED
1	2020	Tema Oil Refinery	Credit & Surety loss 202	03/08/2020	Credit & Surety	28,649,902	4,987,597	33,637,499	2020-04-08
2	2020	Ridge Hospital Complex	Credit & Surety loss 202	04/24/2020	Credit & Surety	49,933,997	31,821,957	81,755,954	2020-05-24
3	2020	Ridge Hospital Complex	Credit & Surety loss 202	06/13/2020	Credit & Surety	16,465,981	4,583,000	21,048,981	2020-07-13
4	2020	Kosmos Energy FPSO	Medical loss 2020-8	09/12/2020	Medical	27,651,350	19,330,911	46,982,261	2020-10-12
5	2021	Kumasi Brewery	Workers Compensation	01/12/2021	Workers Compensation	15,567,861	3,260,780	18,828,641	2021-02-12
6	2021	Unilever Ghana	Workers Compensation	08/16/2021	Workers Compensation	63,825,128	6,281,180	70,106,308	2021-09-16
7	2021	Golden Star Resources	Energy loss 2021-2	08/22/2021	Energy	7,646,964	905,445	8,552,409	2021-09-22
8	2022	Ghana Grid Company Subs	Energy loss 2022-12	02/12/2022	Energy	45,648,742	9,902,156	55,550,898	2022-03-12
9	2022	Nestlé Ghana Tema Plant	Energy loss 2022-10	07/08/2022	Energy	19,828,786	3,076,001	22,904,787	2022-08-08
10	2023	Volta Aluminium	Credit & Surety loss 202	02/11/2023	Credit & Surety	10,887,927	6,564,086	17,452,013	2023-03-11
11	2023	Kosmos Energy FPSO	Credit & Surety loss 202	02/11/2023	Credit & Surety	37,084,619	6,250,572	43,335,191	2023-03-11
12	2023	Golden Star Resources	Workers Compensation	03/28/2023	Workers Compensation	8,130,107	1,895,293	10,025,400	2023-04-28
13	2023	Kumasi Brewery	Workers Compensation	05/09/2023	Workers Compensation	23,379,998	11,283,285	34,663,283	2023-06-09
14	2023	Ghana Cocoa Board Wareh	Energy loss 2023-11	05/28/2023	Energy	50,372,711	31,448,571	81,821,282	2023-06-28
15	2023	Kumasi Brewery	Medical loss 2023-13	12/09/2023	Medical	31,946,508	9,978,935	41,925,443	2023-12-09
16	2025	Nestlé Ghana Tema Plant	Medical loss 2025-3	12/08/2025	Medical	51,427,667	8,370,444	59,798,111	2025-12-08
TOTAL							488,448,248	159,940,213	648,388,461

Large loss list — /np/large-loss-list

ON THIS SCREEN

- Year, insured, event, date, class, paid, outstanding, incurred; paste-aware; totals at the foot.

Select and trend the large losses that will drive the risk-section burning cost and severity fit.

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NON-PROPORTIONAL TREATY: LARGE LOSS SELECTION

Losses have been added or edited since the last selection was saved — review and re- save your selection.

TOTAL16

SELECTED16

INCURRED648,388,461

INFLATION-ADJ1,219,986,177

AVG FACTOR1.8755

INCEPTION YR2026

CPI SOURCESaudi Arabia

THRESHOLD

e.g. 500,000

Apply

Inflation (Saudi Arabia)

Factor +10%

Loss Loading

Yearly Aggregates

All

None

Save

✓	YEAR	INSURED / EVENT	LOSS NAME	DATE	PAID	OS	INCURRED	FACTOR	INFLATED
✓	2020	Tema Oil Refinery	Credit & Surety loss 2020-1	2020-03-08	28,649,902	4,987,597	33,637,499	2.5866	87,006,755
✓	2020	Ridge Hospital Complex	Credit & Surety loss 2020-15	2020-04-24	49,933,997	31,821,957	81,755,954	3.2842	268,502,904
✓	2020	Ridge Hospital Complex	Credit & Surety loss 2020-4	2020-06-13	16,465,981	4,583,000	21,048,981	2.2899	48,200,062
✓	2020	Kosmos Energy FPSO	Medical loss 2020-8	2020-09-12	27,651,350	19,330,911	46,982,261	1.4887	69,942,492
✓	2021	Kumasi Brewery	Workers Compensation loss 2021-9	2021-01-12	15,567,861	3,260,780	18,828,641	2.5244	47,531,021
✓	2021	Unilever Ghana	Workers Compensation loss 2021-5	2021-08-16	63,825,128	6,281,180	70,106,308	2.1125	148,099,576
✓	2021	Golden Star Resources	Energy loss 2021-2	2021-08-22	7,646,964	905,445	8,552,409	2.3791	20,347,036
✓	2022	Ghana Grid Company Substation	Energy loss 2022-12	2022-02-12	45,648,742	9,902,156	55,550,898	1.4556	80,859,887
✓	2022	Nestlé Ghana Tema Plant	Energy loss 2022-10	2022-07-08	19,828,786	3,076,001	22,904,787	2.0705	47,424,361
✓	2023	Volta Aluminium	Credit & Surety loss 2023-7	2023-02-11	10,887,927	6,564,086	17,452,013	1.3656	23,832,469
✓	2023	Kosmos Energy FPSO	Credit & Surety loss 2023-14	2023-02-11	37,084,619	6,250,572	43,335,191	1.2635	54,754,014
✓	2023	Golden Star Resources	Workers Compensation loss 2023-16	2023-03-28	8,130,107	1,895,293	10,025,400	1.4829	14,866,666
✓	2023	Kumasi Brewery	Workers Compensation loss 2023-6	2023-05-09	23,379,998	11,283,285	34,663,283	1.6811	58,272,445
✓	2023	Ghana Cocoa Board Warehouse	Energy loss 2023-11	2023-05-28	50,372,711	31,448,571	81,821,282	1.5416	126,135,688
✓	2023	Kumasi Brewery	Medical loss 2023-13	2023-12-09	31,946,508	9,978,935	41,925,443	1.3518	56,674,814
✓	2025	Nestlé Ghana Tema Plant	Medical loss 2025-3	2025-12-08	51,427,667	8,370,444	59,798,111	1.1294	67,535,987

ADDITIONAL LOADINGS

No additional loadings. Click + Add to include IBNR, trend, or other adjustments.

+ Add

Large loss selection — /np/large-loss-selection

ON THIS SCREEN

- Threshold filter, per-loss tick list, CPI trending to the inception year with a visible per-loss factor, optional +10% override and additional loadings.
- Counters: total, selected, incurred, inflation-adjusted, average factor, inception year and the CPI source country.

Severity fit on the selected large losses, and the pure premium it implies for the risk layers.

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NON-PROPORTIONAL TREATY: LARGE LOSS PARETO

Severity Distribution Fit

Fit parametric distributions to large losses. Adjust threshold and limit. Compare Pareto, Lognormal, Exponential, Weibull via KS goodness-of-fit.

Save

Save Curve

View Sev-Freq Curve

PARAMETERS

Inputs auto-recompute stats + pricing

PARETO MIN (THRESHOLD)

14,866,666

Only losses $\geq$ threshold drive the tail.

TREATY LIMIT (MAX)

502,500,000

Auto-set from total Risk/Cat layer limits in Structure. Editable override.

ALPHA (A)

0.722

Higher $\alpha \rightarrow$ lighter tail. Editable.

OBSERVATION YEARS

10

Frequency = tail count + years.

GOODNESS OF FIT

Select a distribution to use for return periods and pricing. KS statistic: lower = better fit.

USE	DISTRIBUTION	PARAMETERS	N	KS STAT	P-VALUE	VERDICT
☒	Pareto	$\alpha=0.722$	16	0.3800	0.0139	Poor
☐	Lognormal	$\mu=17.90 \sigma=0.71$	16	0.1883	0.5907	Good
☐	Exponential	$\lambda=0.000000$	16	0.2241	0.3552	Fair
☐	Weibull BEST	$k=1.208 \lambda=70059232$	15	0.1674	0.7627	Good

LARGE LOSS STATISTICS

COUNT	16
TOTAL INCURRED	1,219,986,177
AVERAGE	76,249,136
MAX / MIN	268,502,904 / 14,866,666
STD DEVIATION	60,268,362
TOP 5 CONCENTRATION	58.2%
AVG YEARLY LOSS	121,998,618
FREQUENCY ($\geq$ XM)	1.60 / yr
RISK PURE PREMIUM	0

$\Delta A \leq 1$: INFINITE MEAN

RETURN PERIODS (PARETO)

Implied severity from fitted tail

RETURN PERIOD	ESTIMATED LOSS
1 IN 500 YR	80,879,765,678 Δ
1 IN 250 YR	30,987,481,690 Δ
1 IN 100 YR	8,717,665,117 Δ
1 IN 50 YR	3,340,000,876 Δ
1 IN 25 YR	1,279,655,240 Δ
1 IN 20 YR	939,637,800 Δ
1 IN 10 YR	360,003,629
1 IN 5 YR	137,928,266
1 IN 2 YR	38,803,167

PARETO RANKING

#	UW YR	INSURED / EVENT	LOSS	PAID	OS	INCURRED	INFL.	INFLATED	CUM%	RETURN PERIOD
1	2020	Ridge Hospital Complex	Credit & Surety loss 2020-15	49,933,997	31,821,957	81,755,954	3.28	268,502,904	22.0%	5.1y
2	2021	Unilever Ghana	Workers Compensation loss 2021-5	63,825,128	6,281,180	70,106,308	2.11	148,099,576	34.1%	3.3y
3	2023	Ghana Cocoa Board Warehouse	Energy loss 2023-11	50,372,711	31,448,571	81,821,282	1.54	126,135,688	44.5%	2.9y
4	2020	Tema Oil Refinery	Credit & Surety loss 2020-1	28,649,902	4,987,597	33,637,499	2.59	87,006,755	51.6%	2.2y
5	2022	Ghana Grid Company Substation	Energy loss 2022-12	45,648,742	9,902,156	55,550,898	1.46	80,859,887	58.2%	2.1y
6	2020	Kosmos Energy FPSO	Medical loss 2020-8	27,651,350	19,330,911	46,982,261	1.49	69,942,492	64.0%	1.9y
7	2025	Nestlé Ghana Tema Plant	Medical loss 2025-3	51,427,667	8,370,444	59,798,111	1.13	67,535,987	69.5%	1.9y
8	2023	Kumasi Brewery	Workers Compensation loss 2023-6	23,379,998	11,283,285	34,663,283	1.68	58,272,445	74.3%	1.7y
9	2023	Kumasi Brewery	Medical loss 2023-13	31,948,508	9,978,935	41,925,443	1.35	56,674,814	78.9%	1.6y
10	2023	Kosmos Energy FPSO	Credit & Surety loss 2023-14	37,084,619	6,250,572	43,335,191	1.26	54,754,014	83.4%	1.6y
11	2020	Ridge Hospital Complex	Credit & Surety loss 2020-4	16,465,981	4,583,000	21,048,981	2.29	48,200,062	87.4%	1.5y
12	2021	Kumasi Brewery	Workers Compensation loss 2021-9	15,567,861	3,260,780	18,828,641	2.52	47,531,021	91.3%	1.4y

Large loss Pareto — /np/large-loss-pareto (1 of 2)

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11	2020	Ridge Hospital Complex	Credit & Surety loss 2020-4	16,465,981	4,583,000	21,048,981	2.29	48,200,062	87.4%	1.5y
12	2021	Kumasi Brewery	Workers Compensation loss 2021-9	15,567,861	3,260,780	18,828,641	2.52	47,531,021	91.3%	1.4y
13	2022	Nestlé Ghana Tema Plant	Energy loss 2022-10	19,828,786	3,076,001	22,904,787	2.07	47,424,361	95.2%	1.4y
14	2023	Volta Aluminium	Credit & Surety loss 2023-7	10,887,927	6,564,086	17,452,013	1.37	23,832,469	97.1%	0.9y
15	2021	Golden Star Resources	Energy loss 2021-2	7,646,964	905,445	8,552,409	2.38	20,347,036	98.8%	0.8y
16	2023	Golden Star Resources	Workers Compensation loss 2023-16	8,130,107	1,895,293	10,025,400	1.48	14,866,666	100.0%	0.6y

LAYER BURNING COST
 Empirical = layerHit per loss + 10 years · Model = Pareto freq × E[loss in layer] · Blend = weighted average

METHOD: Empirical 50 / 50 Model Custom: Empirical % 50 Model % 50

LAYER	DEDUCTIBLE	LIMIT	RP AT ATTACH.	EMPIRICAL ROL	MODEL ROL	BLENDED ROL	ANNUAL LOSS
1	16,125,000	77,812,500	1-in-0.7y	90.258%	— ▲	45.129%	35,115,942
2	93,937,500	33,750,000	1-in-2.4y	29.540%	— ▲	14.770%	4,984,909
3	127,687,500	222,187,500	1-in-3.0y	7.256%	— ▲	3.628%	8,061,374
4	349,875,000	168,750,000	1-in-6.1y	—	— ▲	—	0

Large loss Pareto — /np/large-loss-pareto (2 of 2)

ON THIS SCREEN

- Pareto / lognormal / exponential / Weibull fitted and ranked by KS; you pick the one that drives return periods and pricing.
- Threshold, treaty limit (auto-set from the total risk/cat layer limits on Structure), alpha and observation years are all editable, and the statistics block recomputes as you change them.
- Statistics include top-5 concentration, average yearly loss, frequency above the threshold, and the risk pure premium.

Development of the large losses to ultimate, so that immature years are not understated in the burning cost.

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NON-PROPORTIONAL TREATY: LARGE LOSS DEVELOPMENT FACTORS

SELECTED LOSSES16

ACCIDENT YEARS5

TOTAL REPORTED1,219,986,177

TAIL FACTOR1

Save

Dev Factors

Loss Detail

Development Factors

Enter LDFs to develop large losses to ultimate. Dev periods: 5

FACTOR	1-2	2-3	3-4	4-5	5-6	TAI
LDF	0	0	0	0	0	1
CDF	1.0000	1.0000	1.0000	1.0000	1.0000	1.00

All LDFs are 1.0 — ultimates equal reported losses and IBNR is zero. Enter development factors above to project to ultimate.

Ultimate Loss Summary

Selected losses aggregated by accident year, developed to ultimate using chosen LDFs

ACC. YEAR	NO. LOSSES	REPORTED (INFLATED)	APPLIED CDF	IBNR	ULTIMATE
2020	4	473,652,213	1.0000	0	473,652,213
2021	3	215,977,633	1.0000	0	215,977,633
2022	2	128,284,249	1.0000	0	128,284,249
2023	6	334,536,096	1.0000	0	334,536,096
2025	1	67,535,987	1.0000	0	67,535,987
Total	—	1,219,986,177	—	0	1,219,986,177

Large loss dev factors — /np/large-loss-dev-factors

ON THIS SCREEN

- Factor table, link ratios and ultimate summary; the selected pattern is saved against the treaty.

Universe 3 — Underwriter Manual

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The catastrophe loss listing for the cat section.

U

MODELLING TOOL

REFRESH

AUTOSAVE: ON

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NON-PROPORTIONAL TREATY: CAT LOSSES

REPORT DATE

mm/dd/yyyy

7 records

Loss Analysis

Load

Clear

Save

#	UW YEAR	INSURED NAME	LOSS / EVENT	DATE OF LOSS	CLASS	PAID	O/S	INCURRED	SAVED
1	2021	Volta Aluminium	Earthquake event 2021	06/07/2021	Energy	77,460,445	63,051,055	140,511,500	2021-07-07
2	2023	Accra Mall Holdings	Flood event 2023-4	02/26/2023	Energy	92,308,234	18,033,312	110,341,546	2023-03-26
3	2025	Volta Aluminium	Flood event 2025-5	03/09/2025	Credit & Surety	40,589,318	30,099,067	70,688,385	2025-04-09
4	2025	Kumasi Brewery	Fire following event 20:	06/19/2025	Credit & Surety	18,470,749	10,243,249	28,713,998	2025-07-19
5	2025	Ghana Grid Company Subt	Windstorm event 2025	08/16/2025	Workers Compensation	107,023,820	36,473,638	143,497,458	2025-09-16
6	2025	Ridge Hospital Complex	Fire following event 20:	09/18/2025	Workers Compensation	100,450,267	23,065,156	123,515,423	2025-10-18
7	2025	Ghana Grid Company Subt	Earthquake event 2025	11/23/2025	Medical	30,650,555	12,259,640	42,910,195	2025-12-23
TOTAL							466,953,388	193,225,117	660,178,505

Cat loss list — /np/cat-loss-list

ON THIS SCREEN

- Separate from large losses throughout: separate selection, separate curve, separate development, separate pricing column.

Select and trend the cat losses for the cat-section basis.

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NON-PROPORTIONAL TREATY: CAT LOSS SELECTION

Losses have been added or edited since the last selection was saved — review and re-save your selection.

TOTAL7

SELECTED6

INCURRED631,464,507

INFLATION-ADJ953,061,357

AVG FACTOR1.4354

INCEPTION YR2026

CPI SOURCESaudi Arabia

THRESHOLD

e.g. 500,000

Apply

Inflation (Saudi Arabia)

Factor +10%

Growth

Loss Loading

Yearly Aggregates

All

None

Save

	YEAR	INSURED / EVENT	LOSS NAME	DATE	PAID	OS	INCURRED	FACTOR	INFLATED
☒	2021	Volta Aluminium	Earthquake event 2021-1	2021-06-07	77,460,445	63,051,055	140,511,500	2.3749	333,700,761
☒	2023	Accra Mall Holdings	Flood event 2023-4	2023-02-26	92,308,234	18,033,312	110,341,546	1.6405	181,015,306
☒	2025	Volta Aluminium	Flood event 2025-5	2025-03-09	40,589,318	30,099,067	70,688,385	1.1360	80,302,005
☐	2025	Kumasi Brewery	Fire following event 2025-2	2025-06-19	18,470,749	10,243,249	28,713,998	1.1426	32,808,614
☒	2025	Ghana Grid Company Substation	Windstorm event 2025-3	2025-08-16	107,023,820	36,473,638	143,497,458	1.1677	167,561,982
☒	2025	Ridge Hospital Complex	Fire following event 2025-7	2025-09-18	100,450,267	23,065,156	123,515,423	1.1423	141,091,668
☒	2025	Ghana Grid Company Substation	Earthquake event 2025-6	2025-11-23	30,650,555	12,259,640	42,910,195	1.1510	49,389,634

ADDITIONAL LOADINGS

No additional loadings. Click + Add to include IBNR, trend, or other adjustments.

+ Add

Cat loss selection — /np/cat-loss-selection

ON THIS SCREEN

- Same mechanics as large-loss selection, with a growth adjustment alongside inflation.

Severity fit on the cat basis.

U

MODELLING TOOL

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NON-PROPORTIONAL TREATY: CAT LOSS PARETO

Severity Distribution Fit

Fit parametric distributions to cat losses. Adjust threshold and limit. Compare Pareto, Lognormal, Exponential, Weibull via KS goodness-of-fit.

✓ Saved

Save Curve

View Sev-Freq Curve

PARAMETERS

Inputs auto-recompute stats + pricing

PARETO MIN (THRESHOLD)

49,389,634

Only losses ≥ threshold drive the tail.

TREATY LIMIT (MAX)

502,500,000

Auto-set from total Risk/Cat layer limits in Structure. Editable override.

ALPHA (A)

1.006

Higher α → lighter tail. Editable.

OBSERVATION YEARS

10

Frequency = tail count + years.

GOODNESS OF FIT

Select a distribution to use for return periods and pricing. KS statistic: lower = better fit.

USE	DISTRIBUTION	PARAMETERS	N	KS STAT	P-VALUE	VERDICT
☒	☒ Pareto	α=1.006	6	0.3187	0.4991	Fair
☐	☒ Lognormal BEST	μ=18.71 σ=0.61	6	0.2027	1.0000	Fair
☐	☐ Exponential	λ=0.000000	6	0.2340	0.9461	Fair
☐	☐ Weibull	k=1.637 λ=147332439	5	0.2687	0.7451	Fair

CAT LOSS STATISTICS

	COUNT	6
TOTAL INCURRED	953,061,357	
AVERAGE	158,843,559	
MAX / MIN	333,700,761 / 49,389,634	
STD DEVIATION	90,927,357	
TOP 5 CONCENTRATION	94.8%	
AVG YEARLY LOSS	95,306,136	
FREQUENCY (≥ XM)	0.60 / yr	
RISK PURE PREMIUM	71,044,473	

RETURN PERIODS (PARETO)

Implied severity from fitted tail

COMPARE WITH TP →

RETURN PERIOD	ESTIMATED LOSS
1 IN 500 YR	23,857,121,250 ▲
1 IN 250 YR	11,974,563,906 ▲
1 IN 100 YR	4,814,259,731 ▲
1 IN 50 YR	2,416,413,121 ▲
1 IN 25 YR	1,212,868,089 ▲
1 IN 20 YR	971,495,954 ▲
1 IN 10 YR	487,621,296
1 IN 5 YR	244,750,920
1 IN 2 YR	98,399,784

PARETO RANKING

#	UW YR	INSURED / EVENT	LOSS	PAID	OS	INCURRED	INFL.	INFLATED	CUM%	RETURN PERIOD
1	2021	Volta Aluminium	Earthquake event 2021-1	77,460,445	63,051,055	140,511,500	2.37	333,700,761	35.0%	11y
2	2023	Accra Mall Holdings	Flood event 2023-4	92,308,234	18,033,312	110,341,546	1.64	181,015,306	54.0%	6.2y
3	2025	Ghana Grid Company Substation	Windstorm event 2025-3	107,023,820	36,473,638	143,497,458	1.17	167,561,982	71.6%	5.7y
4	2025	Ridge Hospital Complex	Fire following event 2025-7	100,450,267	23,065,156	123,515,423	1.14	141,091,668	86.4%	4.8y
5	2025	Volta Aluminium	Flood event 2025-5	40,589,318	30,099,067	70,688,385	1.14	80,302,005	94.8%	2.7y
6	2025	Ghana Grid Company Substation	Earthquake event 2025-6	30,650,555	12,259,640	42,910,195	1.15	49,389,634	100.0%	1.7y

LAYER BURNING COST

Empirical = layerHit per loss + 10 years - Model = Pareto freq × E[loss in layer] - Blend = weighted average

METHOD ☐ Empirical ☒ 50 / 50 ☐ Model Custom: Empirical % Model %

LAYER	DEDUCTIBLE	LIMIT	RP AT ATTACH.	EMPIRICAL ROL	MODEL ROL	BLENDED ROL	ANNUAL LOSS
1	18,135,000	77,815,500	1 in 1.7y	65.633%	34.367%	44.538%	34,414,048

LAYER	DEDUCTIBLE	LIMIT	RP AT ATTACH.	EMPIRICAL ROL	MODEL ROL	BLENDED ROL	ANNUAL LOSS
1	16,125,000	77,812,500	1-in-1.7y	52.523%	35.933%	44.228%	34,414,946
2	93,937,500	33,750,000	1-in-3.2y	40.000%	26.832%	33.416%	11,277,956
3	127,687,500	222,187,500	1-in-4.3y	14.070%	13.335%	13.703%	30,445,523
4	349,875,000	168,750,000	1-in-11.9y	—	6.829%	3.415%	5,762,256

CAT MODEL OEP BURNING COST ▼
Enter OEP curve from cat model (RMS / AIR / Verisk). Layer burning cost = $\int [D, D+L] P(\text{occ loss} > x) dx$.

Cat loss Pareto — /np/cat-loss-pareto (2 of 2)

ON THIS SCREEN

- Cat samples are small; the KS verdicts and p-values are there so you can see how much weight the fit deserves before you lean on it.

Development of cat losses to ultimate. Cat years famously keep developing, and the benchmark curves in the workbench reflect that.

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NON-PROPORTIONAL TREATY: CAT LOSS DEVELOPMENT FACTORS

SELECTED LOSSES6

ACCIDENT YEARS3

TOTAL REPORTED953,061,357

TAIL FACTOR1

Save

Dev Factors

Loss Detail

Development Factors

Enter LDFs to develop cat losses to ultimate. Dev periods: 5

FACTOR	1~2	2~3	3~4	4~5	5~6	TAI
LDF	0	0	0	0	0	1
CDF	1.0000	1.0000	1.0000	1.0000	1.0000	1.00

All LDFs are 1.0 — ultimates equal reported losses and IBNR is zero. Enter development factors above to project to ultimate.

Ultimate Loss Summary

Selected losses aggregated by accident year, developed to ultimate using chosen LDFs

ACC. YEAR	NO. LOSSES	REPORTED (INFLATED)	APPLIED CDF	IBNR	ULTIMATE
2021	1	333,700,761	1.0000	0	333,700,761
2023	1	181,015,306	1.0000	0	181,015,306
2025	4	438,345,289	1.0000	0	438,345,289
Total	—	953,061,357	—	0	953,061,357

Cat loss dev factors — /np/cat-loss-dev-factors

ON THIS SCREEN

- Factor table, link ratios, ultimate summary.

Development of losses *in excess of the retention* — the layer's own development, which is slower than the ground-up pattern.

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MODELLING TOOL

REFRESHAUTOSAVE: ONDaylightHOMELOG OUT

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NON-PROPORTIONAL TREATY: EXCESS DEVELOPMENT FACTORS

START YEAR2020DEV YEARS7TAIL FACTOR1DATA MODEDirect EntryTriangle

Development FactorsLink RatiosUltimate SummarySave

Reported Excess Losses

Cumulative reported losses above the retention, per accident year

YEAR	REPORTED XS LOSS
2020	0
2021	0
2022	0
2023	0
2024	0
2025	0
2026	0

Input Development Factors

Enter LDFs from benchmark or market dataDev periods: 5

PERIOD	LDF
1-2	0
2-3	0
3-4	0
4-5	0
5-6	0

Input Development Factors

From manual LDF inputs above

FACTOR	1-2	2-3	3-4	4-5	5-6
LDF	1.0000	1.0000	1.0000	1.0000	1.0000
CDF	1.0000	1.0000	1.0000	1.0000	1.0000

Parametrized Development Factors

Exponential fit to cumulative development

FACTOR	1-2	2-3	3-4	4-5	5-6
LDF	1.0000	1.0000	1.0000	1.0000	1.0000
CDF	1.0000	1.0000	1.0000	1.0000	1.0000

Underwriter Chosen Factors

Base

INPUTPARAMLINK RATIOS

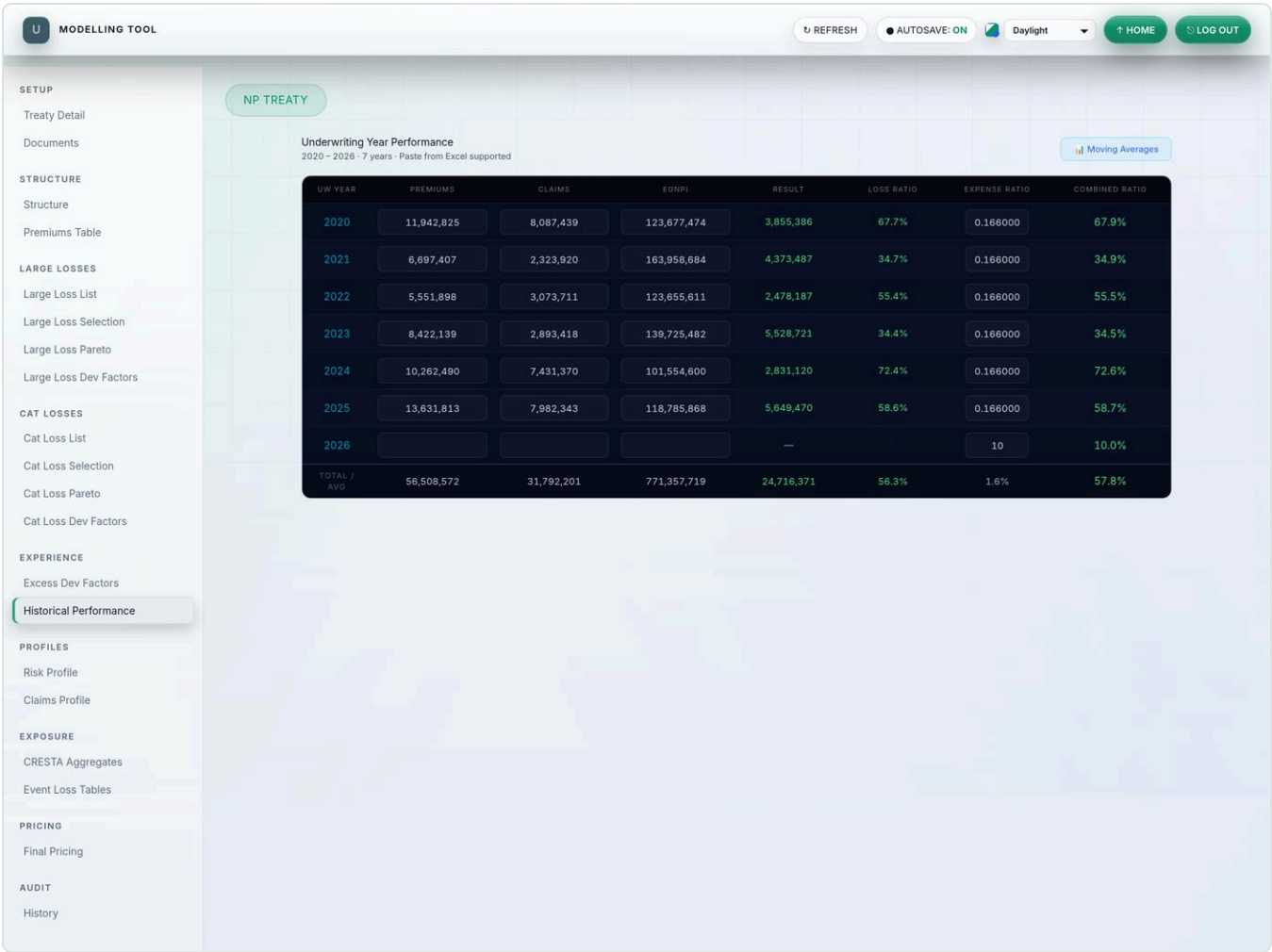
FACTOR	1-2	2-3	3-4	4-5	5-6
LDF	1.0000	1.0000	1.0000	1.0000	1.0000
CDF	1.0000	1.0000	1.0000	1.0000	1.0000

Excess dev factors — /np/excess-dev-factors

ON THIS SCREEN

- **Data mode** — direct entry of reported excess losses per accident year, or a full triangle.
- Three factor rows are offered: your input LDFs, an exponential parametrized fit, and the link ratios from the triangle. The **underwriter chosen** row is what the projection uses.
- Start year, development years and a tail factor are set at the top.
- Views: development factors, link ratios, ultimate summary.

The treaty's own results by underwriting year: premiums, claims, EGNPI, result, loss ratio, expense ratio and combined ratio.

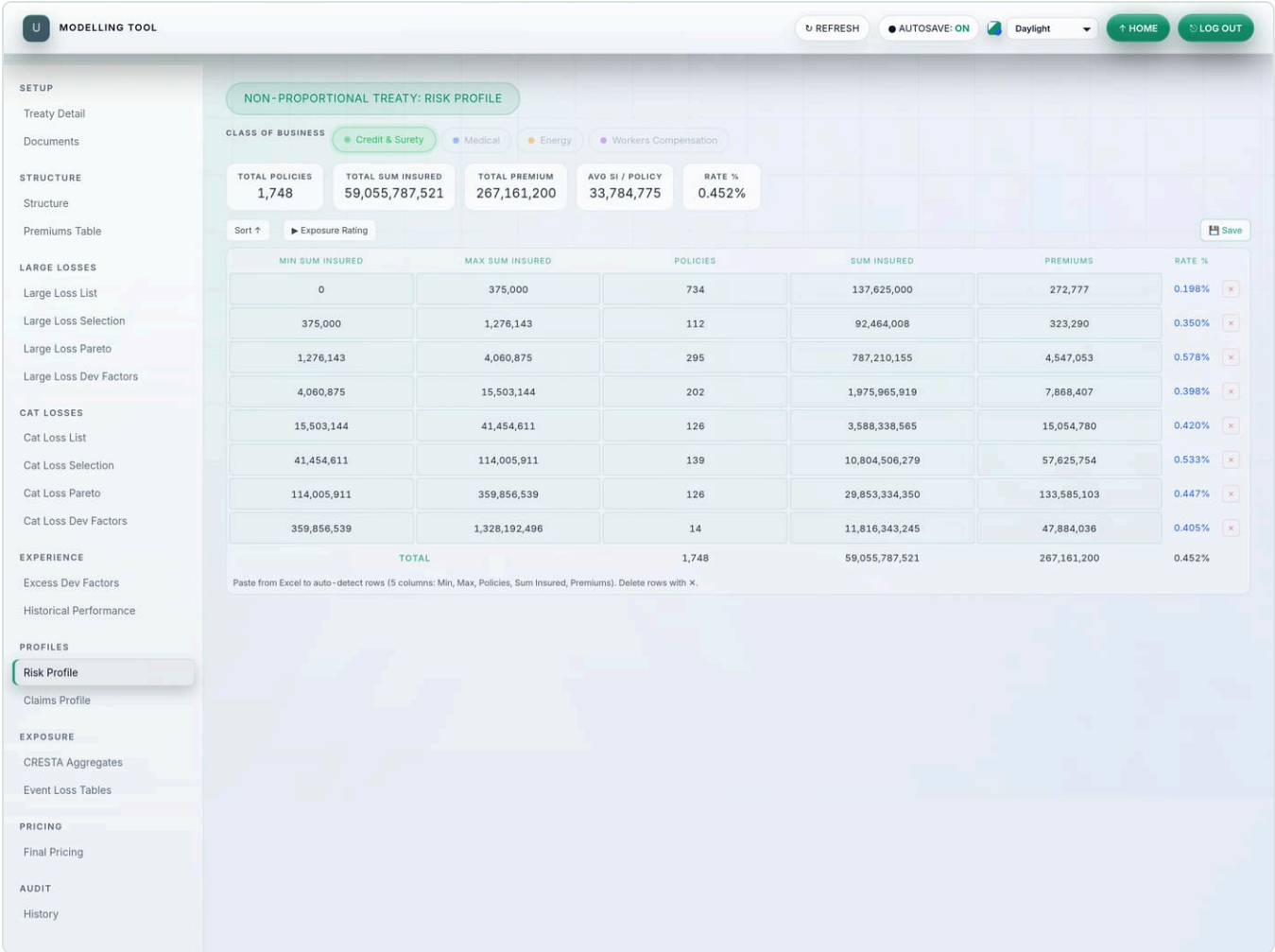


Historical performance — /np/historical-performance

ON THIS SCREEN

- Paste-aware, one row per year across the full experience period.
- **Moving averages** smooths the series so a single bad year does not dominate the read.
- Totals and averages at the foot are what Final Pricing shows as historical margin per layer.

The cedant's portfolio banded by sum insured — the input to exposure rating for the risk layers.



Risk profile — /np/risk-profile

ON THIS SCREEN

- One tab per class. Paste five columns: min, max, policies, sum insured, premiums.
- **Exposure Rating** runs the banded profile through the exposure curves to produce a layer loss cost independent of the loss experience.

Claim counts and incurred by sum-insured band, with the loss ratio and rate each band implies.

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NON-PROPORTIONAL TREATY: CLAIMS PROFILE

CLASS OF BUSINESS

Credit & Surety

Medical

Energy

Workers Compensation

TOTAL CLAIMS

2,521

TOTAL SUM INSURED

289,436,295,708

TOTAL PREMIUM

1,100,804,565

TOTAL INCURRED CLAIMS

4,192,209,075

LOSS RATIO

380.8%

RATE %

0.380%

Sort ↑

Save

MIN SUM INSURED	MAX SUM INSURED	COUNT	TOTAL SUM INSURED	PREMIUM	INCURRED CLAIMS	LOSS RATIO	RATE %
0	93,750	328	223,986,340	922,505	2,484,375	269.3%	0.412%
93,750	336,576	394	1,692,056,734	6,381,219	36,577,710	573.2%	0.377%
336,576	1,000,092	609	4,260,715,516	12,652,491	64,828,398	512.4%	0.297%
1,000,092	3,641,227	592	8,314,476,240	116,539,505	213,500,674	183.2%	1.402%
3,641,227	10,886,457	242	23,765,179,005	144,866,445	159,804,524	110.3%	0.610%
10,886,457	34,702,011	316	99,611,869,406	299,672,909	1,527,213,678	509.6%	0.301%
34,702,011	87,085,411	4	4,739,759,469	13,159,806	60,893,711	462.7%	0.278%
87,085,411	338,295,790	36	146,828,252,998	506,609,685	2,126,906,005	419.8%	0.345%
TOTAL		2,521	289,436,295,708	1,100,804,565	4,192,209,075	380.8%	0.380%

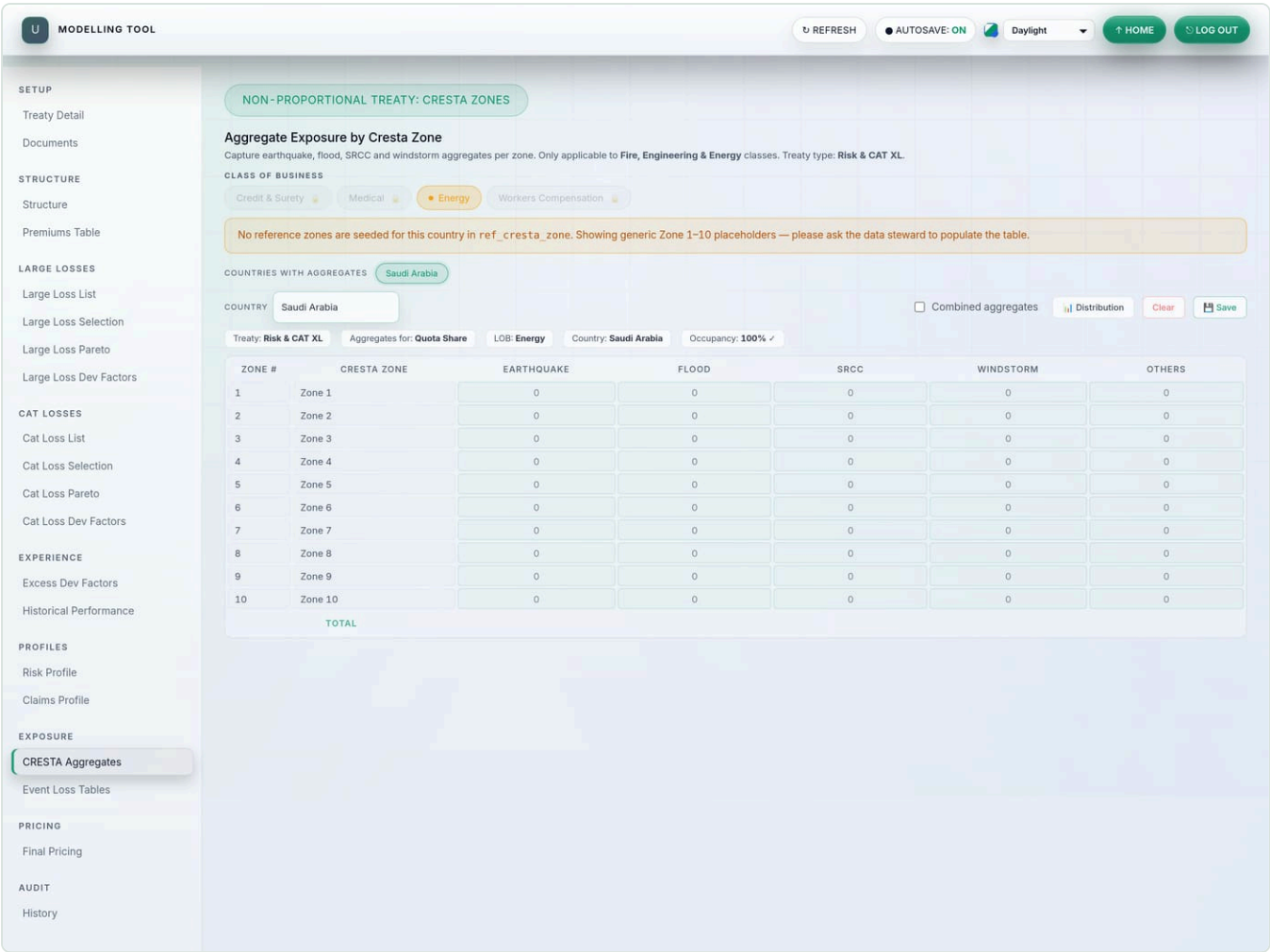
Paste from Excel to auto-detect rows (6 columns: Min, Max, Count, Total Sum Insured, Premium, Incurred Claims). Delete rows with X.

Claims profile — /np/claims-profile

ON THIS SCREEN

- Paste six columns: min, max, count, total sum insured, premium, incurred claims.

Accumulated exposure by CRESTA zone and peril, feeding cat exposure rating and the accumulation checks.

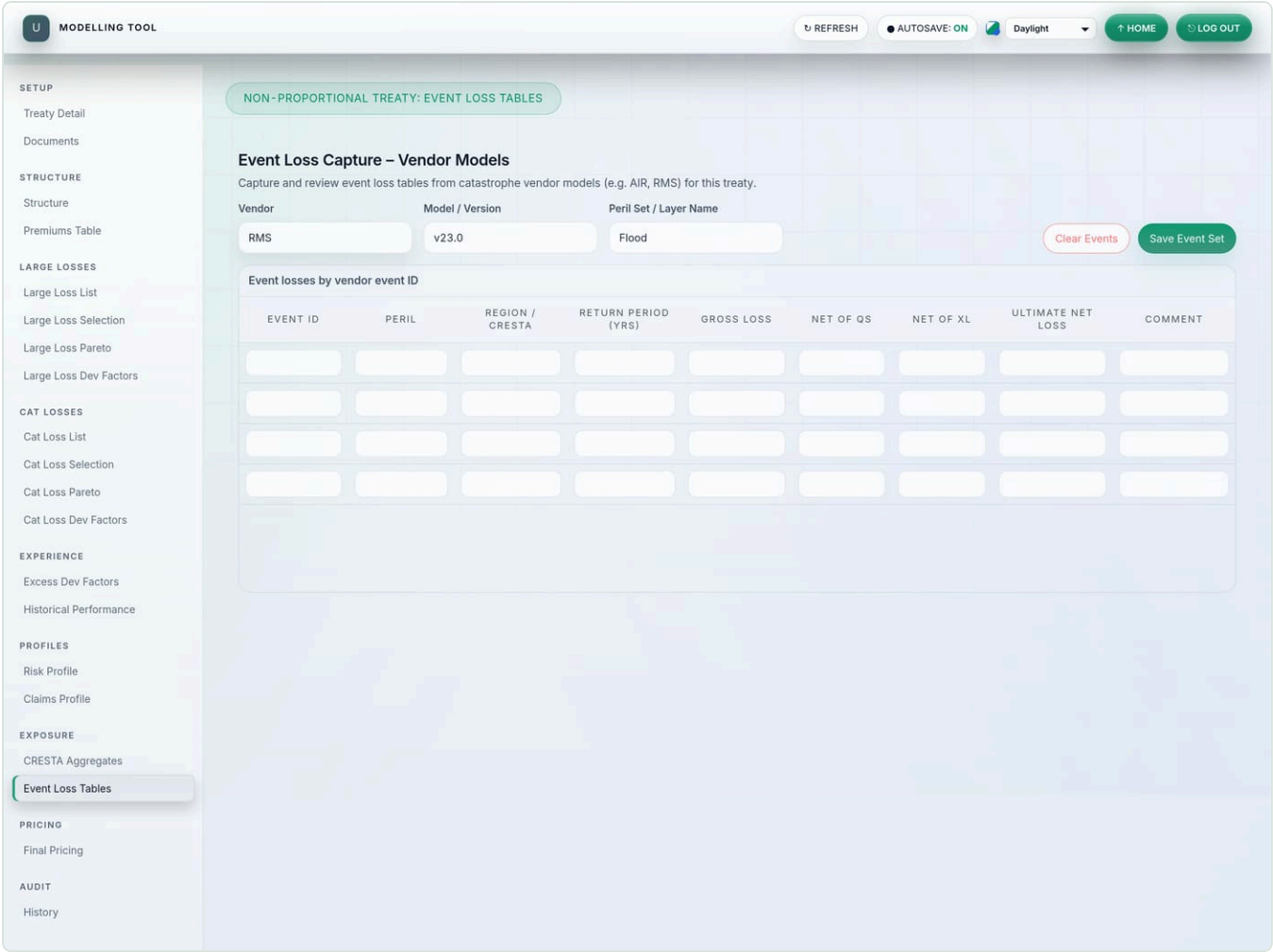


CRESTA aggregates — /np/cresta-aggregates

ON THIS SCREEN

- Applies to Fire, Engineering and Energy classes; the others are locked.
- Where a country has no seeded reference zones the screen says so plainly and falls back to generic Zone 1–10 placeholders rather than pretending the zones are real.

Vendor cat model output — AIR or RMS — event by event, gross through to ultimate net.



Event loss tables — /np/event-loss-tables

ON THIS SCREEN

- Vendor, model version, peril set; then event ID, peril, region/CRESTA, return period and the loss at each net-down step.

The XL pricing workbench. Every method runs per layer, the weights between them are yours, and the result is the price you take to the market.

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NON-PROPORTIONAL TREATY: FINAL PRICING

RISK & CAT XL

CEDANT AL RAJHI TAKAFUL COUNTRY SAUDI ARABIA COB WORKERS COMPENSATION, MEDICAL, ENERGY, CREDIT & SURETY BROKER UIB CCY SAR UW YEAR 2026 SIGNED

STRUCTURE METRICS

Layers4

Total Earned PremSAR 104,011,200

Total ROL20.70%

COST STRUCTURE

Brokerage11.61%

Technical Ratio251.50%

Result-151.50%

Max RetentionSAR 13,712,413

EST. GNPI - 100% TREATY

SAR

118,785,868

TOTAL LIMITSAR 502,500,000

TREATY TYPERISK & CAT XL

XL TYPEGross XL

ACCTG METHODLosses Occurring

LAYERS4

TOTAL ROL20.70%

LAYER STRUCTURE

RISK XL

Layers4

DeductibleSAR 16,125,000

Total LimitSAR 502,500,000

CAT XL

Layers4

DeductibleSAR 16,125,000

Total LimitSAR 502,500,000

Large Losses

CAT Losses

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CAT Modelling

Run Actuarial Engine

Pure Burn - Pareto - MBBEFD - CRESTA Exposure Rating

Risk XL Layers

RISK XL

LAYER	LIMIT	DEDUCTIBLE	PURE BURN	PARETO	BURN + PARETO	EXPOSURE MBBEFD	WT BURN %	WT PARETO %	WT EXP %	LOADING %
1	77,812,500	16,125,000	0.00%	0.00%	0.00%	0.00%	50	0	50	15
2	33,750,000	93,937,500	0.00%	0.00%	0.00%	0.00%	50	0	50	15
3	222,187,500	127,687,500	0.00%	0.00%	0.00%	0.00%	50	0	50	15
4	168,750,000	349,875,000	0.00%	0.00%	0.00%	0.00%	50	0	50	15
TOTAL	502,500,000		0.00%	0.00%	0.00%	0.00%	50	0	50	15

CAT XL Layers

CAT XL

LAYER	LIMIT	DEDUCTIBLE	PURE BURN	PARETO	BURN + PARETO	EXPOSURE DMD RATIO	WT BURN %	WT PARETO %	WT EXP %	LOADING %
1	77,812,500	16,125,000	0.00%	0.00%	0.00%	0.00%	50	0	50	15
2	33,750,000	93,937,500	0.00%	0.00%	0.00%	0.00%	50	0	50	15
3	222,187,500	127,687,500	0.00%	0.00%	0.00%	0.00%	50	0	50	15
4	168,750,000	349,875,000	0.00%	0.00%	0.00%	0.00%	50	0	50	15
TOTAL	502,500,000		0.00%	0.00%	0.00%	0.00%	50	0	50	15

Pricing

COMBINED

Market Intelligence

LAYER	LIMIT	DEDUCTIBLE	REINSTATEMENTS	REINSURER	LEAD	EXPIRING	HIST. MARGIN	MARGIN	TECH RATIO
1	77,812,500	16,125,000	-		40.27%	37.42%	-164.92%	-	248.32%
2	33,750,000	93,937,500	3@50%		28.59%	26.59%	-417.34%	-	500.74%
3	222,187,500	127,687,500	1@100%		21.02%	19.53%	-118.74%	-	202.14%

Final pricing — /np/final-pricing (1 of 2)

Universe 3 — Underwriter Manual

70

Layer	Share	Premium	Per Risk Limit	Cat Limit	Cedant Total Limit	Agg Contribution	Total Country Agg	Annual Aggregate Limit
3	222,187,500	127,687,500	1@100%		21.02%	19.53%	-118.74%	202.14%
4	168,750,000	349,875,000	1@100%		9.68%	8.99%	83.40%	-0.00%
TOTAL	502,500,000				20.70%	19.23%		

Layer	Share	Premium	Per Risk Limit	Cat Limit	Cedant Total Limit	Agg Contribution	Total Country Agg	Annual Aggregate Limit
1	0%	-	-	-	-	-	-	-
2	0%	-	-	-	-	-	-	-
3	0%	-	-	-	-	-	-	-
4	0%	-	-	-	-	-	-	-
TOTAL		-	-	-	-	-	-	-

Buttons: Save, Export Excel, SIGNED, Pricing locked — open offer modal to review, View Offer

Final pricing — /np/final-pricing (2 of 2)

ON THIS SCREEN

- **Header** — structure metrics (layers, total earned premium, total ROL), cost structure (brokerage, technical ratio, result, max retention), estimated GNPI at 100%, total limit, treaty and XL type, accounting method, and the layer structure split risk versus cat.
- **Run Actuarial Engine** — pure burn, Pareto, MBBEFD and CRESTA exposure rating, computed for every layer at once.
- **Layer tables** (one per peril section) — limit, deductible, pure burn, Pareto, burn+Pareto, exposure/MBBEFD, then the weight you give each method (WT burn %, WT Pareto %, WT exposure %), the loading %, the resulting total ROL, your UW price, and the probability of attachment and exhaustion.
- **Pricing** — per layer: reinstatements, reinsurer/lead/expiring rates, historical margin, margin and technical ratio, with the programme total. This is the year-on-year read.
- **Programme limits & downside** — share, premium, per-risk limit, cat limit, cedant total limit, aggregate contribution, total country aggregate and annual aggregate limit per layer, with an optimal-shares view.
- **Analysis tabs** — large losses, cat losses, aggregates, risk profile, cedant summary, reinsurer analysis, technical analysis, historical performance, market analysis, checklist and cat modelling, each opening a full workbench modal.
- **Pricing analysis modal** — five tabs: per-peril blending with the weight cells and metrics table, Pareto simulation (severity/frequency), inflation and loss, cat modelling (GEM/HAZUS earthquake damage ratios), and aggregate analysis over CRESTA zones.
- Save, Export Excel, and the offer workflow. Once the treaty is signed the pricing locks and the button becomes View Offer.

The dedicated screen for stop-loss covers — aggregate protections quoted as percentages of subject premium, for example "20% xs 80% LR". Shown only when the treaty type is Stop Loss.

U

MODELLING TOOL

REFRESH

AUTOSAVE: ON

Daylight

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Treaty Detail

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STRUCTURE

Structure

Premiums Table

LARGE LOSSES

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Large Loss Selection

Large Loss Pareto

Large Loss Dev Factors

CAT LOSSES

Cat Loss List

Cat Loss Selection

Cat Loss Pareto

Cat Loss Dev Factors

EXPERIENCE

Excess Dev Factors

Historical Performance

PROFILES

Risk Profile

Claims Profile

EXPOSURE

CRESTA Aggregates

Event Loss Tables

PRICING

Stop Loss Pricing

AUDIT

History

NP TREATY: STOP LOSS

LAYER COVER - LOSS-RATIO BASIS

Read-only — edit on the Structure page. Stop Loss attaches at a loss ratio of subject premium (e.g. "20% xs 80% LR" pays losses between 80% and 100% LR), 3 layers from Treaty Detail.

LAYER	ATTACH LR %	LIMIT LR %	EPI	RESOLVED - LIMIT XS ATTACH
L1	—	—	—	—
L2	—	—	—	—
L3	—	—	—	—

BURNING COST - LOSS RATIOS BY UW YEAR

Premium comes from the Premiums Table and is oil-revised by the rate changes captured there. Loss Ratio = Aggregate Loss ÷ Premium (adjusted). Layer Hit applies the LR layer to that ratio and converts back to currency at the current EPI. Paste a column of losses from Excel.

UW YEAR	PREMIUM (ADJUSTED)	AGGREGATE LOSS	LOSS RATIO	L1 HIT	L2 HIT	L3 HIT
2017	744,274,510 +1.218	0	—	—	—	—
2018	523,596,803 +1.100	0	—	—	—	—
2019	663,770,466 +1.014	0	—	—	—	—
2020	476,026,543 +0.888	0	—	—	—	—
2021	561,614,963 +1.029	0	—	—	—	—
2022	564,640,726	0	—	—	—	—

Enter at least one yearly aggregate to compute burning cost.

EXPOSURE RATING - COMPOUND POISSON

Frequency A is the expected number of claims per year above the severity threshold. The Normal approximation prices the in-layer expected loss analytically; it breaks down for heavy-tailed severity (Pareto $\alpha \leq 2$) — use Monte Carlo instead. The summary below shows Layer 1; per-layer ROL is in the Blended Result table.

FREQUENCY A

e.g. 20

Claims / year (Poisson rate)

SEVERITY

Lognormal

Provide frequency A, and severity parameters to compute exposure rating.

SEVERITY MEAN

e.g. 200,000

EVs per claim

SEVERITY CV

0.6

σ / mean (e.g. 0.6)

MONTE CARLO - AGGREGATE SIMULATION

Simulates N years of compound Poisson aggregate losses and applies the layer. Uses the same frequency and severity as Exposure Rating. Deterministic — seed in for reproducibility. Summary shows Layer 1 percentiles; per-layer MC ROL is in the Blended Result table.

☐ Run Monte Carlo

METHOD BLEND & LOADING

Weights need not sum to 100 — the engine divides by their sum. Set Monte Carlo to 0 to ignore it even when enabled.

BURNING COST WT (%)

50

EXPOSURE WT (%)

50

MONTE CARLO WT (%)

0

LOADING (%)

20

Internal target loss ratio

BLENDING RESULT - BY LAYER

LAYER	LIMIT XS ATTACH	ANNUAL LAYER LOSS	PURE ROL	TOTAL RATE	PREMIUM
L1	—	—	0.000%	0.000%	—
L2	—	—	0.000%	0.000%	—
L3	—	—	0.000%	0.000%	—
TOTAL	0	0	0.000%	—	0

WARNINGS

- Attachment missing — provide either absolute "attachment" or "attachmentLossRatio" + "epi".
- Limit missing — provide either absolute "limit" or "limitLossRatio" + "epi".
- No pricing method produced output — supply yearlyAggregates or frequency+severity.

Stop loss pricing — /np/stop-loss-pricing

ON THIS SCREEN

- Layer cover** — read-only mirror of the Structure page: attach LR%, limit LR%, EPI, and the resolved "limit xs attach" in currency.

- **Burning cost** — one aggregate loss per underwriting year. Premium comes from the Premiums Table and is on-levelled by the rate-change chain captured there; the loss ratio is aggregate loss ÷ adjusted premium; the layer hit applies the LR layer to that ratio and converts back to currency at the current EPI.
- **Exposure rating** — a compound-Poisson model: frequency λ above a severity threshold with a lognormal or Pareto severity, priced analytically by normal approximation.
- **Monte Carlo** — an optional simulated view with a trial count and a seed, so a run is reproducible.
- **Blend weights** — weights across burning cost, exposure rating and Monte Carlo, plus a loading. The weights need not sum to 100; the engine divides by their sum.
- **Blended result** — per layer: annual layer loss, pure ROL, loaded total rate and premium, with a programme total across layers and any engine warnings.

The full change trail for the XL treaty, newest first. Read-only.

MODELLING TOOL

REFRESHAUTOSAVE: ONDaylightHOMELOG OUT

SETUP

Treaty Detail

Documents

STRUCTURE

Structure

Premiums Table

LARGE LOSSES

Large Loss List

Large Loss Selection

Large Loss Pareto

Large Loss Dev Factors

CAT LOSSES

Cat Loss List

Cat Loss Selection

Cat Loss Pareto

Cat Loss Dev Factors

EXPERIENCE

Excess Dev Factors

Historical Performance

PROFILES

Risk Profile

Claims Profile

EXPOSURE

CRESTA Aggregates

Event Loss Tables

PRICING

Final Pricing

AUDIT

History

NON-PROPORTIONAL TREATY: AUDIT & HISTORY

Audit & History

Every change, status transition and approval decision for this treaty, newest first.
Read-only.

Refresh

CUChief UnderwriterCU

NP SAVEDjust now

Np saved

23/08/2026, 07:35:06

CUChief UnderwriterCU

NP SAVEDjust now

Np saved

23/08/2026, 07:34:35

CUChief UnderwriterCU

DEV FACTORS SAVEDjust now

Dev factors saved

23/08/2026, 07:34:35

CUChief UnderwriterCU

NP SAVED1m ago

Np saved

23/08/2026, 07:34:28

CUChief UnderwriterCU

NP CAT LOSS LDfs SAVED1m ago

Np cat loss ldfs saved

23/08/2026, 07:34:28

CUChief UnderwriterCU

NP SAVED1m ago

Np saved

23/08/2026, 07:34:02

CUChief UnderwriterCU

NP LARGE LOSS LDfs SAVED1m ago

Np large loss ldfs saved

23/08/2026, 07:34:02

CUChief UnderwriterCU

NP SAVED2m ago

Np saved

23/08/2026, 07:33:35

CUChief UnderwriterCU

NP EGNPI SAVED2m ago

Np egmpi saved

23/08/2026, 07:33:35

CUChief UnderwriterCU

NP EXPIRING SAVED2m ago

Np expiring saved

23/08/2026, 07:33:29

CUChief UnderwriterCU

NP SAVED2m ago

Np saved

23/08/2026, 07:33:29

CUChief UnderwriterCU

NP SAVED2m ago

Np saved

23/08/2026, 07:33:14

CUChief UnderwriterCU

UPDATED2m ago

Treaty details updated

23/08/2026, 07:33:14

CUChief UnderwriterCU

NP SAVED39m ago

Audit & history — /np/history (1 of 2)

CU	Chief Underwriter	CU	NP SAVED	39m ago
Np saved				
23/08/2026, 06:55:51				
CU	Chief Underwriter	CU	NP SAVED	40m ago
Np saved				
23/08/2026, 06:55:27				
CU	Chief Underwriter	CU	DEV FACTORS SAVED	40m ago
Dev factors saved				
23/08/2026, 06:55:27				
CU	Chief Underwriter	CU	NP SAVED	40m ago
Np saved				
23/08/2026, 06:55:22				
CU	Chief Underwriter	CU	NP CAT LOSS LDFS SAVED	40m ago
Np cat loss ldfs saved				
23/08/2026, 06:55:22				
CU	Chief Underwriter	CU	NP SAVED	40m ago
Np saved				
23/08/2026, 06:55:02				
CU	Chief Underwriter	CU	NP LARGE LOSS LDFS SAVED	40m ago
Np large loss ldfs saved				
23/08/2026, 06:55:02				
CU	Chief Underwriter	CU	NP SAVED	41m ago
Np saved				
23/08/2026, 06:54:42				
CU	Chief Underwriter	CU	NP EGNPI SAVED	41m ago
Np egmpi saved				
23/08/2026, 06:54:42				
CU	Chief Underwriter	CU	NP EXPIRING SAVED	41m ago
Np expiring saved				
23/08/2026, 06:54:37				
CU	Chief Underwriter	CU	NP SAVED	41m ago
Np saved				
23/08/2026, 06:54:37				
CU	Chief Underwriter	CU	NP SAVED	41m ago
Np saved				
23/08/2026, 06:54:27				
CU	Chief Underwriter	CU	UPDATED	41m ago
Treaty details updated				
23/08/2026, 06:54:27				
CU	Chief Underwriter	CU	NP SAVED	47m ago
Np saved				
23/08/2026, 06:48:10				
CU	Chief Underwriter	CU	NP SAVED	47m ago
Np saved				
23/08/2026, 06:47:46				
CU	Chief Underwriter	CU	DEV FACTORS SAVED	47m ago
Dev factors saved				
23/08/2026, 06:47:46				

Audit & history — /np/history (2 of 2, continues below the fold)

ON THIS SCREEN

- Every save is recorded by area — NP saved, dev factors saved, EGNPI saved, large-loss LDFs saved, expiring saved, and so on — with actor, role and timestamp.
- Status transitions carry the from-state, the to-state and the reason.

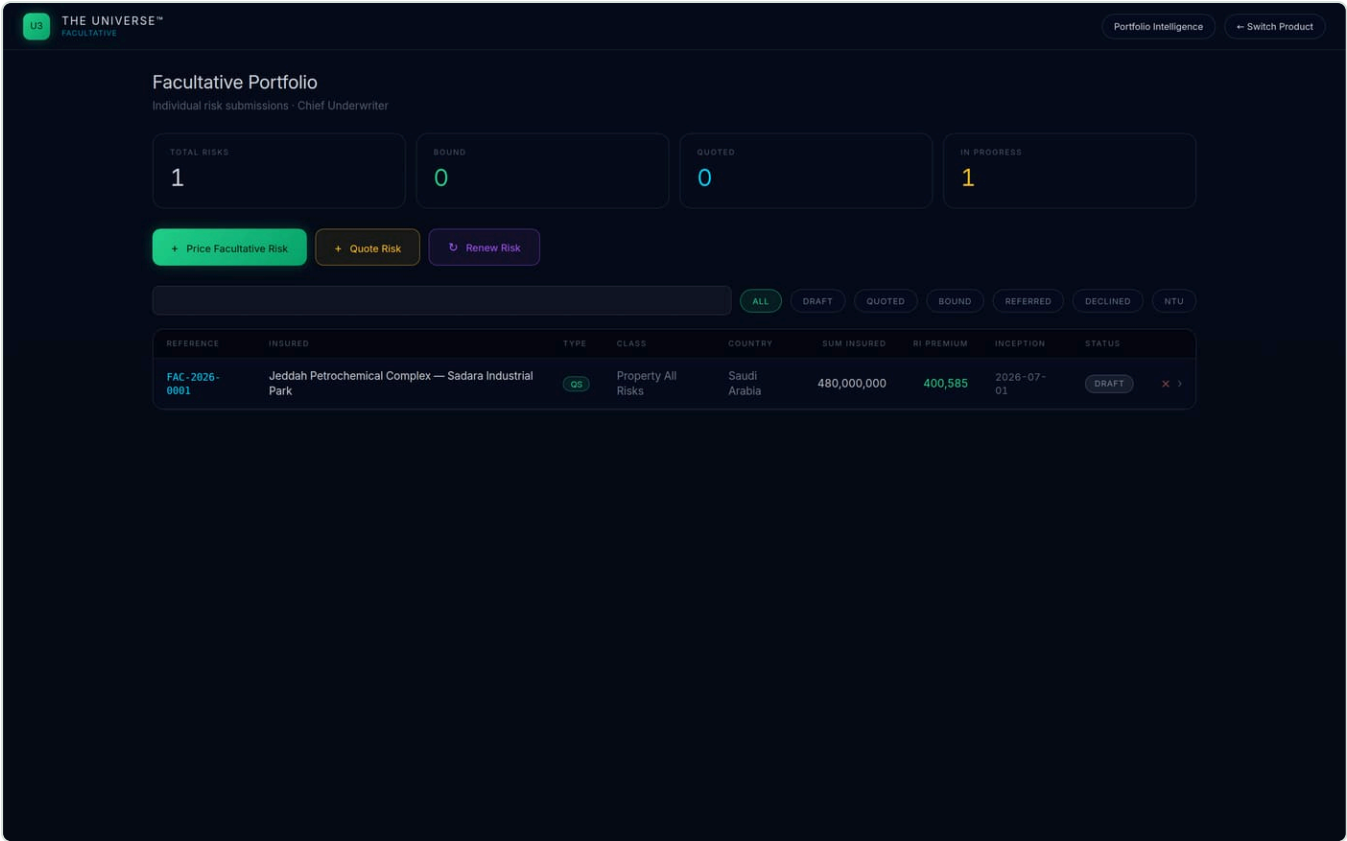
7. Pricing a facultative risk

The route through

Nine screens in four groups: **Submission** (risk detail, documents, locations & SI), **Risk** (COPE, placement structure, deductibles & terms), **Experience** (loss history) and **Pricing** (pricing, summary & approval).

Facultative pricing is engine-led. Rather than projecting the cedant's triangles, it rates the risk from a reference table — occupancy drives a FLEXA base rate, the risk zone drives flood and earthquake loadings, and an underwriting-factor scorecard applies discounts and loadings on top. Which steps appear depends on the rating family of the classes you select: a schedule-property risk gets the location schedule; a liability risk gets limits and ILFs instead.

The fac module home. Portfolio counts and the register of every risk on the desk.



Facultative portfolio — /fac

ON THIS SCREEN

- Tiles: total risks, bound, quoted, in progress.
- Three ways to start: price a facultative risk, quote a risk, or renew an existing one.
- The register filters by status (all, draft, quoted, bound, referred, declined, NTU) and lists reference, insured, placement type, class, country, sum insured, RI premium, inception and status.
- **Portfolio Intelligence** in the header opens the unsupervised analysis of what drives profitability across the decisioned book.

The submission header, and the screen that determines how the risk will be rated.

U

MODELLING TOOL

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Daylight

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SUBMISSION

Risk Detail

Documents

Locations & SI

RISK

COPE Assessment

Placement Structure

Deductibles & Terms

EXPERIENCE

Loss History

PRICING

Pricing

Summary & Approval

FAC-2026-0001

FAC-2026-0001

PARTIES

Country

Saudi Arabia

Cedant

— Select Cedant —

Broker

Marsh

Currency

USD

INSURED RISK

Insured Name

Jeddah Petrochemical Complex — Sadara Industrial Park

Insured Address

Plot 14, Industrial City Phase 3, Jeddah, Saudi Arabia

Nature of Business

Petrochemical manufacturing, storage and distribution

RISK PROFILE

Cedant Region

— Select —

Renewal / New

☐ New ☐ Renewal

Risk Country / Zone

KSA- Western

Multi-Location

☒ Yes ☐ No

Multi-Occupancy

☒ Yes ☐ No

Risk Top Location Address

Plot 14, Industrial City Phase 3, Jeddah

Occupancy

Chemical Manufacturing (others), Pharmaceutical formulation, Toilettry produ

HAZARD GRADE

7

HAZARD CATEGORY

High

RISK CATEGORY

3

FREQUENCY

2

SECTIONS & CLASSES OF BUSINESS

Number of Sections

1

SECTION 1 — CLASSES COVERED

CASUALTY

☐ Fidelity Guarantee

☐ General Liability

☐ Motor

☐ Passenger Liability

☐ Personal Accident

☐ Professional Indemnity

CYBER

☐ Cyber - First Party

☐ Cyber - Third Party

ENERGY

☐ Energy Offshore

☐ Energy Onshore

☐ Renewable Energy PROJECT

ENGINEERING

☐ ALoP / DSU PROJECT

☐ Construction All Risks PROJECT

☐ Contractors Plant & Machinery

☐ Electronic Equipment

☐ Erection All Risks PROJECT

☐ Machinery Breakdown

☐ Plant All Risks

MARINE

☐ Cargo Insurance

☐ Goods in Transit

☐ Hull & Machinery

☐ Marine Liability

☐ Project Cargo PROJECT

☐ Stock Throughput

PROPERTY

☐ Assets All Risks

☐ Buildings Combined

Risk detail — /fac/risk/detail (1 of 2)

PROPERTY

☐ Assets All Risks
 ☐ Buildings Combined

☐ Business Interruption
 ☐ Industrial All Risks

☐ Property All Risks

POLICY PERIOD

UW Year: 2026

Inception Date: 07/01/2026

Period (Months): 12

Expiry Date: 07/01/2027

SUMS INSURED (100% BASIS)

Select classes of business above to enter sums insured

UNDERWRITER NOTES

Renewal of the 2025 placement. Cedant has completed the sprinkler upgrade on the storage warehouses flagged at last renewal; survey report dated March 2026 attached. Seeking the same 12.5% share subject to rate adequacy.

RELATED TREATIES

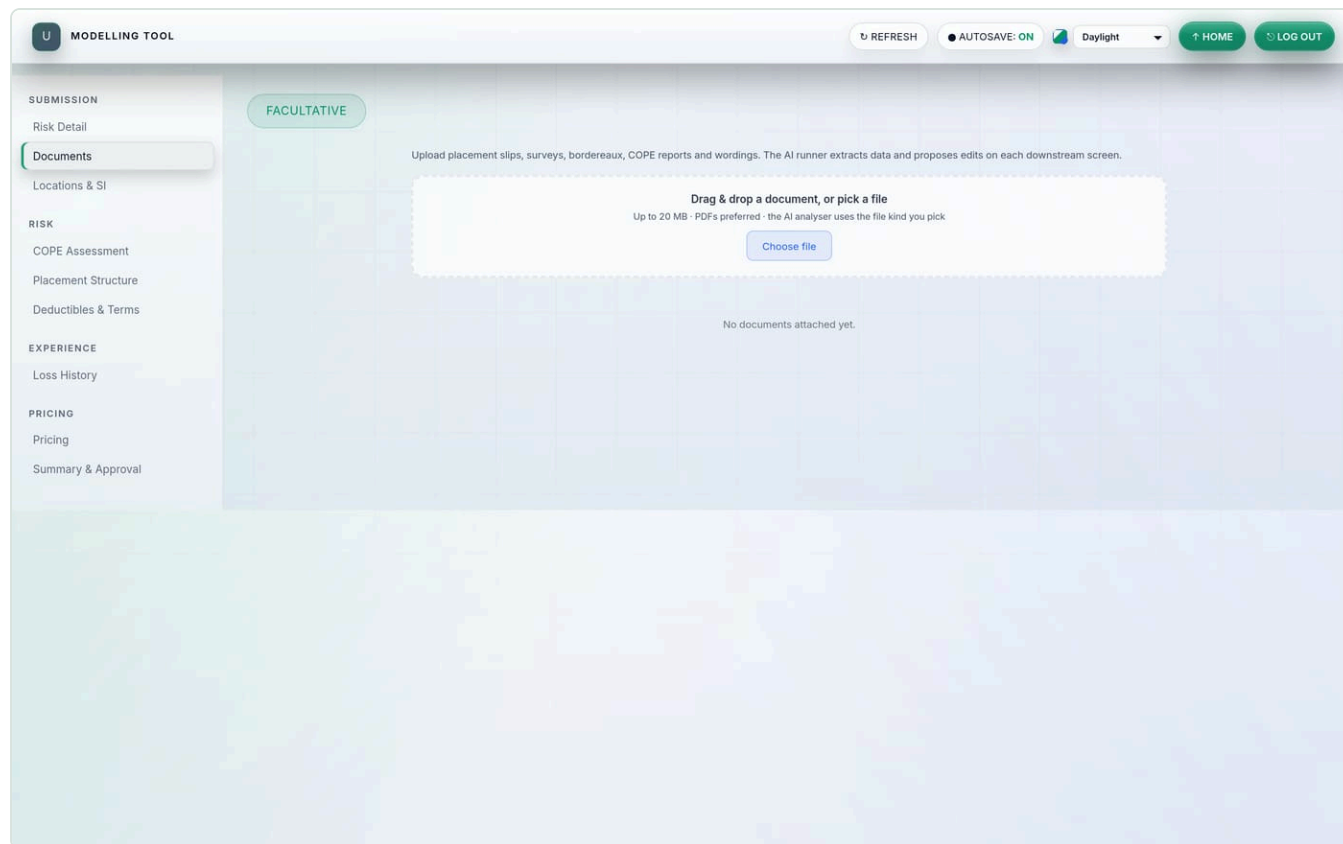
Select a cedant and a class of business above to surface eligible treaties.

Risk detail — /fac/risk/detail (2 of 2)

ON THIS SCREEN

- **Parties** — country (which filters the cedant list), cedant, broker, currency.
- **Insured risk** — insured name, address (with places autocomplete, which also records the geocode), nature of business.
- **Risk profile** — cedant region (used to look up territorial capacity), renewal or new, risk country/zone (which drives the flood and earthquake rates), multi-location and multi-occupancy flags, and the top location address.
- **Occupancy** — the single most consequential field on the screen. It drives the FLEXA base rate, the hazard grade, the risk category and the frequency category, all of which are shown back to you as soon as it is picked. A hazard-grade override is available where the default does not fit the risk.
- **Sections & classes of business** — one section per class with its own sum insured. The rating families of the classes you choose decide which of the later steps appear.
- **Related treaties** — links this risk to the treaties that cover the same cedant and class, so the accumulation view can see both halves.

Placement slips, surveys, bordereaux, COPE reports and wordings — and the entry point for document AI.



Documents — /fac/risk/documents

ON THIS SCREEN

- Drag and drop or pick a file, up to 20 MB; PDFs are preferred. You tag the document kind, which is what the analyser uses.
- **Upload & analyse** runs the extraction, then polls until it finishes. The result opens in a side drawer.
- The drawer lists the AI's proposed edits grouped by the screen they target, each with a confidence bar, a current → suggested diff and its rationale. You **Accept**, **Edit & Accept** or **Reject** each one; accepting is undoable while the flash is still showing.
- A failed analysis exposes a Retry so a transient error does not send you back to the file picker.

The location schedule. One card per location, with separate material-damage and business-interruption rows.

MODELLING TOOL REFRESH AUTOSAVE: ON Daylight HOME LOG OUT

SUBMISSION
Risk Detail
Documents
Locations & SI

RISK
COPE Assessment
Placement Structure
Deductibles & Terms

EXPERIENCE
Loss History

PRICING
Pricing
Summary & Approval

FACULTATIVE

One card per location with separate Material Damage and Business Interruption rows. PML %, FX rate and the carrier's signed share drive the SAR columns automatically.

LOCATION 1: TOP LOCATION

Ethylene Cracker Unit Plot 14A, Industrial City Phase 3, Jeddah 30 USD 3.75000000 12.5

PARTICULARS	SI ORG	PML %	PML ORG	SI SAR	PML SAR	CARRIER SI ORG	CARRIER PML ORG	CARRIER SI SAR	CARRIER PML SAR
MATERIAL DAMAGE	150,000,000	40	60,000,000	562,500,000	225,000,000	18,750,000	7,500,000	70,312,500	28,125,000
BUSINESS INTERRUPTION	30,000,000	35	10,500,000	112,500,000	39,375,000	3,750,000	1,312,500	14,062,500	4,921,875

LOCATION 2

Polyethylene Line 2 Plot 14B, Industrial City Phase 3, Jeddah 30 USD 3.75000000 12.5

PARTICULARS	SI ORG	PML %	PML ORG	SI SAR	PML SAR	CARRIER SI ORG	CARRIER PML ORG	CARRIER SI SAR	CARRIER PML SAR
MATERIAL DAMAGE	95,000,000	35	33,250,000	356,250,000	124,687,500	11,875,000	4,156,250	44,531,250	15,585,937
BUSINESS INTERRUPTION	25,000,000	30	7,500,000	93,750,000	28,125,000	3,125,000	937,500	11,718,750	3,515,625

LOCATION 3

Tank Farm & Loading Ga Plot 15, Industrial City Phase 3, Jeddah 30 USD 3.75000000 12.5

PARTICULARS	SI ORG	PML %	PML ORG	SI SAR	PML SAR	CARRIER SI ORG	CARRIER PML ORG	CARRIER SI SAR	CARRIER PML SAR
MATERIAL DAMAGE	80,000,000	55	44,000,000	300,000,000	165,000,000	10,000,000	5,500,000	37,500,000	20,625,000
BUSINESS INTERRUPTION	15,000,000	25	3,750,000	56,250,000	14,062,500	1,875,000	468,750	7,031,250	1,757,813

LOCATION 4

Finished Goods Warehouse Plot 16, Industrial City Phase 3, Jeddah 30 USD 3.75000000 12.5

PARTICULARS	SI ORG	PML %	PML ORG	SI SAR	PML SAR	CARRIER SI ORG	CARRIER PML ORG	CARRIER SI SAR	CARRIER PML SAR
MATERIAL DAMAGE	40,000,000	60	24,000,000	150,000,000	90,000,000	5,000,000	3,000,000	18,750,000	11,250,000
BUSINESS INTERRUPTION	15,000,000	40	6,000,000	56,250,000	22,500,000	1,875,000	750,000	7,031,250	2,812,500

LOCATION 5

Admin & Control Building Plot 14, Industrial City Phase 3, Jeddah 30 USD 3.75000000 12.5

PARTICULARS	SI ORG	PML %	PML ORG	SI SAR	PML SAR	CARRIER SI ORG	CARRIER PML ORG	CARRIER SI SAR	CARRIER PML SAR
MATERIAL DAMAGE	15,000,000	25	3,750,000	56,250,000	14,062,500	1,875,000	468,750	7,031,250	1,757,813
BUSINESS INTERRUPTION	15,000,000	20	3,000,000	56,250,000	11,250,000	1,875,000	375,000	7,031,250	1,406,250

+ Add Location (5 / 50)

MATERIAL DAMAGE SAR 1,425,000,000 **BUSINESS INTERRUPTION SAR** 375,000,000 **GRAND TOTAL SAR** 1,800,000,000

Locations & SI — /fac/risk/locations

ON THIS SCREEN

- Per row: original-currency sum insured, PML %, PML in original currency, then the SAR-converted sum insured and PML, then the carrier's signed share of each.
- The SAR and carrier columns are derived from original × FX × share — you type the exposure and the schedule does the arithmetic.

- The largest location is badged as the top location; the engine uses it for the capacity check and for the top-location concentration factor in the score.
- A per-location occupancy code can differ from the risk header where a site genuinely has a different hazard.

Construction, Occupation, Protection and Exposure — the four pillars of risk quality, plus the survey record.

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MODELLING TOOL

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FACULTATIVE

Construction · Occupation · Protection · Exposure — the four pillars of risk quality assessment.

CONSTRUCTION

Construction Type

— Select —

Year Built

Roof Material

e.g. Concrete, Metal deck

Wall Material

e.g. Reinforced concrete

Floors

Total Area (m²)

Fire Walls

☐ Present

Fire Doors

☐ Present

Spatial Separation (m)

OCCUPATION

Occupation

Nature of business activity

Processes

Key processes, materials, hazards

Hazard Grade

— Select —

Operating Hours

— Select —

PROTECTION

Sprinkler System

☐ Installed

Fire Alarm

☐ Installed

Fire Brigade Dist. (km)

Extinguishers

☒ Present

Hydrants

☐ Present

CCTV

☐ Installed

Security Guards

☐ On-site

EXPOSURE

Earthquake

☐ Exposed

Flood

☐ Exposed

Windstorm

☐ Exposed

Other NatCat

e.g. Tsunami, Volcanic

Exposure Notes

Additional exposure commentary

SURVEY REPORT

Survey Date

mm/dd/yyyy

Survey Provider

e.g. Risk Engineering Consultants

Survey Rating

— Select —

COPE assessment — /fac/risk/cope

ON THIS SCREEN

- **Construction** — type (fire resistive through wood frame), year built, roof and wall materials, floors, total area, fire walls and doors, spatial separation.
- **Occupation** — occupation and processes, hazard grade, operating hours (day only, shift, 24/7, seasonal).

- **Protection** — sprinklers, fire alarm, distance to the fire brigade, extinguishers, hydrants, CCTV, on-site guards.
- **Exposure** — earthquake, flood, windstorm and other natural catastrophe exposure, with notes.
- **Survey report** — date, provider and rating. The survey rating and its age are both scoring factors on the pricing screen, so a stale survey costs the risk points.

How the placement is shared and what it costs. Proportional (quota share) or non-proportional (excess of loss).

The screenshot shows the 'Placement Structure' screen within the 'MODELLING TOOL'. The interface includes a sidebar with navigation links: SUBMISSION (Risk Detail, Documents, Locations & SI), RISK (COPE Assessment, Placement Structure, Deductibles & Terms), EXPERIENCE (Loss History), and PRICING (Pricing, Summary & Approval). The 'Placement Structure' section is active. The main content area is titled 'FACULTATIVE' and contains several sections: 'PLACEMENT TYPE' with 'Proportional (QS)' selected; 'PROPORTIONAL STRUCTURE' with input fields for 'Cedant Retention %' (25%), 'RI Share %' (75%), and 'Our Share %' (12.5%); 'COMMISSION & COSTS' with input fields for 'Commission %' (15%), 'Brokerage %' (5%), and 'Taxes %' (2.5%); and 'ORIGINAL PREMIUM' with input fields for 'Original Rate (‰)' (0.8345524), 'Original Premium (100%)' (3,120,000), and 'RI Premium (Our Share)' (40,058,515). The top of the screen features a 'REFRESH' button, 'AUTOSAVE: ON' status, a 'Daylight' theme selector, and 'HOME' and 'LOG OUT' buttons.

Placement structure — /fac/risk/structure

ON THIS SCREEN

- **Proportional** — cedant retention %, RI share % (derived), our share %.
- **Non-proportional** — the excess tower: a primary layer and any excess layers, each with its own limit, share, reinstatements and premium. ROL and payback are derived, never typed — ROL is $\text{premium} \div \text{limit}$ and payback is $1 \div \text{ROL}$, the number of clean years a layer needs to repay one total loss.
- **Commission & costs** — commission, brokerage and taxes.
- **Original premium** — original rate per mille and original premium at 100%; the RI premium at our share is filled in by the pricing engine.

The deductible structure and the clauses-and-exclusions checklist.

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MODELLING TOOL

REFRESH

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Risk Detail

Documents

Locations & SI

RISK

COPE Assessment

Placement Structure

Deductibles & Terms

EXPERIENCE

Loss History

PRICING

Pricing

Summary & Approval

FACULTATIVE

DEDUCTIBLE STRUCTURE

Deductible Amount

2,500,000

Description / Schedule

USD 2.5m each and every loss, PD & BI combined; 45 days BI time excess

Deductible Guidance

Deductibles should reflect the cedant's retention appetite and the nature of the risk. Common structures include flat monetary amounts per occurrence, percentage of sum insured with minimum/maximum amounts, and time-based waiting periods for BI covers. NatCat deductibles are typically higher than standard fire perils.

CLAUSES & EXCLUSIONS CHECKLIST

6 MANDATORY MISSING

CLAUSE	MANDATORY	CHECKED	COMMENTS
Wordings - LM7 LM7 - WORDING	Mandatory	☐	Sub-limit / wording note...
Wordings - ABI ABI - WORDING	Mandatory	☐	Sub-limit / wording note...
Sanction Limitation and Exclusion Clause (LMA LMA_3100 - EXCLUSION	Mandatory	☐	Sub-limit / wording note...
Political Risks Exclusion (NMA 2919 amended) NMA_2919 - EXCLUSION	Mandatory	☐	Sub-limit / wording note...
Electronic Data Endorsement B (NMA 2915) NMA_2915 - EXCLUSION	Mandatory	☐	Sub-limit / wording note...
Claims Cooperation Clause (LM3 / NMA 2737) LM3_NMA_2737 - CLAIMS	Mandatory	☐	Sub-limit / wording note...

Deductibles & terms — /fac/risk/deductibles

ON THIS SCREEN

- Deductible amount and a free-text description or schedule, with guidance on the common structures — flat per occurrence, percentage of sum insured with a minimum and maximum, time excess for BI, and higher natural-catastrophe deductibles.
- Clauses & exclusions checklist** — the master list (LM7 and ABI wordings, LMA 3100 sanctions, NMA 2919 political risks, NMA 2915 electronic data, LM3/NMA 2737 claims cooperation, and the rest) with a tick and a comment per clause. Mandatory clauses that are still unticked are counted in a banner at the top of the card.

Loss history

/fac/risk/losses

From-the-ground-up loss experience, minimum three years. Two views: a ten-year matrix and the individual loss records.

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FOU (from the ground up) loss experience — minimum 3 years. Enter all material losses with details and mitigation measures taken.

YEAR	CLAIMS	FOU PAID	FOU O/S	FOU INCURRED	EXPOSURE (\$)	PREMIUM	CLAIM RATIO
2026	—	—	—	—			—
2025	—	—	—	—			—
2024	1	—	—	—			—
2023	—	—	—	—			—
2022	1	—	—	—			—
2021	1	—	—	—			—
2020	—	—	—	—			—
2019	1	—	—	—			—
2018	—	—	—	—			—
2017	—	—	—	—			—

Exposure and premium are the denominator the burning cost divides by. Enter a row for every year on risk — including the clean ones, which are what stop the burn rate coming out too high.

Claims inflation % p.a.
Experience years
Losses are trended from their year to the current one before they are rated.

Paste a block straight from Excel — click the cell to start at, then paste. Rows are added to fit. Columns map in this order: Year - Date - Description - Cause - FOU Paid - FOU O/S - Mitigation

(FOU Incurred is calculated). A clean loss history is positive for pricing.

YEAR	DATE	DESCRIPTION	CAUSE	FOU PAID	FOU O/S	FOU INCURRED	MITIGATION
2024	06/09/2024	Loss details	Cause			—	Actions taken ✖
2022	11/27/2022	Loss details	Cause			—	Actions taken ✖
2021	02/03/2021	Loss details	Cause			—	Actions taken ✖
2018	08/14/2019	Loss details	Cause			—	Actions taken ✖
	mm/dd/yyyy	Loss details	Cause			—	Actions taken ✖
	mm/dd/yyyy	Loss details	Cause			—	Actions taken ✖
	mm/dd/yyyy	Loss details	Cause			—	Actions taken ✖
	mm/dd/yyyy	Loss details	Cause			—	Actions taken ✖
	mm/dd/yyyy	Loss details	Cause			—	Actions taken ✖
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	mm/dd/yyyy	Loss details	Cause			—	Actions taken ✖
	mm/dd/yyyy	Loss details	Cause			—	Actions taken ✖
	mm/dd/yyyy	Loss details	Cause			—	Actions taken ✖
	mm/dd/yyyy	Loss details	Cause			—	Actions taken ✖
	mm/dd/yyyy						

Loss history — /fac/risk/losses

ON THIS SCREEN

- **Ten-year matrix** — per year: claim count, FGU paid, FGU outstanding, FGU incurred, exposure (sum insured), premium and the resulting claim ratio. Exposure and premium are the denominator the burning cost

divides by, so the clean years matter as much as the loss years — leaving them out is what makes a burn rate come out too high.

- **Loss records** — year, date, description, cause, FGU paid, FGU outstanding, derived FGU incurred and the mitigation taken. The grid opens with empty rows so a loss run can be pasted straight in; columns map in that order.
- A claims-inflation rate and an experience-years setting trend the losses from their own year to the current one before they are rated.

One rate, one build-up. The scorecard sets the loadings and discounts; the engine turns them into a rate; there is no second, competing calculation behind it.

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FACULTATIVE

Priced on the location schedule — 480,000,000 (79.17% material damage)

Family: Schedule Property

Rates: FAC-REF-2026.1

Class accumulation — fac + treaty

UNDERWRITING FACTORS — DRIVERS OF RATE & SCORE

SCHEME WITH BI 77.34 SCORE D- ACCEPT

94% of the scoring weight selected

FACTOR	OPTION	SCORE	LOADING
Construction	Class B - Non Combustible Roof and RCC/St	score 80	-5.00%
Age of Risk	Between 15 to 20 Years	score 70	0%
Claim Experience	Below Average but Good due to single large l	score 60	-5.00%
Fire Fighting Equipments	Type 1 - Adequate Sprinkler and Hydrant Sy	score 100	-15.00%
External Exposure	Hazardous Industrial Area Within 30 meters +	score -30	+20.00%
Natural Perils Exposure	Moderate	score 80	score only
Management of Original Insured	Good	score 100	-5.00%
Survey Report Rating	Above Average	score 80	-10.00%
Survey Report Age	3 Years or Less	score 100	score only
Deductible Level	Above Average	score 80	-10.00%
Gross Retention of Cedant	Good	score 85	score only
Number of Locations	Less than 5	score 100	score only
Percentage of Sum Insured of Top Occupancy	More than 70% for same occupancies	score 100	0%
Percentage of Sum Insured of Top Location	Between 40% to 50%	score 65	score only
Underwriter's Overall Perception	Good	score 80	score only
Market Rate vs Technical Rate	Between 70% to 80%	score 90	score only
Client Overall Relationship	Good	score 80	score only
Business Interruption Plan	Has Proper Business Continuity Plan	score 100	-15.00%

Scored at renewal following the March 2026 Charles Taylor survey. Two open recommendations tracked; sprinkler upgrade in the warehouse completed and verified.

ENGINE INPUTS

Indemnity Period (months)

Drives the BI rate multiplier (1-60)

12

Benchmark Market Rate (%)

Sets the market - vs - technical band in the score

0.70000000

Benchmark Source

e.g. Market benchmark 2026, broker indication

Commission %

20%

Margin %

5%

Other Expenses %

0.5%

EXTENSIONS

Filtered by the classes selected on Risk Detail. Each ticked extension is a cover loading on the engine's net rate — there is no second calculation behind them.

Filtered by the classes selected on Risk Detail. Each ticked extension is a cover loading on the engine's net rate — there is no second calculation behind them.

PROPERTY

☐ Earthquake

+15%

☐ Flood

+10%

☐ Windstorm / Typhoon

+12%

☐ Business Interruption

+8%

☐ Terrorism

+5%

☐ Strikes, Riots & Civil Commotion (SRCC)

+4%

☐ Malicious Damage

+3%

☐ Spontaneous Combustion

+6%

☐ Subsidence & Landslide

+5%

☐ Waiver of Subrogation

+2%

☐ Debris Removal

+1%

☐ Architects & Surveyors Fees

+1%

Custom extension name

Loading %

+ Add

UNDERWRITER ADJUSTMENT

UW Adjustment %

+ surcharge / - discount, applied to the technical gross rate

0%

Reason

Required whenever the adjustment is not zero

e.g. Long-standing client, clean 5-year record

LOSS COST

METHOD	RATE %	WEIGHT
Workbook rate exposure	0.6217	100%
Exposure curve exposure	—	—
No exposure bands — enter the sums insured, or a location schedule, first.		
Burning cost experience	—	—
No exposure years recorded. Burning cost needs what was on risk each year, including the years with no losses — enter it on Loss Experience.		
Benchmark reference only	—	reference
No bound comparables yet for this class, region and size band. The benchmark builds itself as business is bound.		
Blended loss cost credibility Z = 0%	0.6217	

METHOD NOTES

Weight override ignored — it needs a reason code from the approved list: EXPERIENCE_UNREPRESENTATIVE, STRUCTURE_CHANGED, EXPOSURE_CHANGED, DATA_QUALITY, LARGE_LOSS_DISTORTION, UNDERWRITER_JUDGEMENT.

CAPACITY CHECK

REFERRAL — CAPACITY BREACH

A breach is a referral, not a refusal. Binding needs the referral approved and the override recorded with a reason.

PER-RISK LINE

The share being written is 1250.0% of the line, above the 75.0% the score's grade allows.

Line size 180,000,000 Max line 135,000,000 Written 2,250,000,000 Headroom -2,115,000,000

ZONE ACCUMULATION - SA-02

SA-02 carries 0 of committed exposure and this risk would add 24,468,750. No budget is set for the zone, so there is nothing to measure that against — load one in fac_zone_budget.

Committed 0 This risk 24,468,750 Would be 24,468,750

Line measured on top location sl (180,000,000). Zones: SA-02.

TECHNICAL BUILD-UP

Blended loss cost	0.6217
= Expected loss	0.6217
= Technical net	0.6217
+ (1 - commission - brokerage - tax - margin) gross-up +1.342	0.8346
= TECHNICAL GROSS	0.8346

WORKBOOK RATE BUILD-UP

FLEXA base rate occupancy	0.8000
Technical rate × (1 + factor discounts / loadings)	0.5600
Flood / storm	0.0350
Earthquake	0.0188
+ NatCat load zone rates, factor-loaded	0.0537
= Total rate	0.6138
BI rate indemnity loading × total × BI plan	0.6521
= Net rate 79.17% MD / 20.83% BI	0.6217
= Technical net	0.6217
+ (1 - commission - margin - expenses) gross-up +1.342	0.8346
= TECHNICAL GROSS	0.8346
QUOTED RATE (%)	0.8346
Quoted premium on 480,000,000	400,585

QUOTED RATE (%)

0.8346

Quoted premium on 480,000,000

400,585

WARNINGS

No territorial capacity for region "Middle East"; not applied.

ENGINE DETAIL

Underwriting Score	78.70	Scoring Completeness	100.00%
Capacity Grade	D	UW Action	ACCEPT
Max Capacity %	75.00%	Max Capacity (SAR)	135,000,000
Market vs Tech %	112.59%	Market vs Tech Band	More than Equal to 80%
Technical Premium	268,800	Expected Premium	400,585
Total Sum Insured	480,000,000		

QUOTED RATE (%)

0.8346

QUOTED PREMIUM

400,585

Pricing — /fac/risk/pricing (3 of 3)

ON THIS SCREEN

- The header states what the rate is being computed on — here, the location schedule at 480,000,000 with 79.17% material damage — the rating family, and the version of the rate table in use.
- Underwriting factors** — around twenty factors (construction, age of risk, claim experience, fire-fighting equipment, external exposure, natural perils, management, survey rating and age, deductible level, cedant's gross retention, BI plan, client relationship, market versus technical rate, number of locations, top-location and top-occupancy concentration, and the underwriter's overall perception). Each option carries a score and a rate loading or discount, both shown on the row.
- A progress bar tracks how much of the scoring weight you have selected; a grade is only issued above 80%. The running score, its grade and the implied action (ACCEPT / REFER / DECLINE) sit in the panel header.
- Engine inputs** — indemnity period in months (which drives the BI multiplier), a benchmark market rate per mille and its source (which sets the market-versus-technical band in the score), commission, margin and other expenses.
- Extensions** — cover extensions filtered by the classes on the risk. Each ticked extension is a loading on the engine's net rate; custom extensions can be added with their own loading.
- Rate build-up** — FLEXA base rate, technical rate before natural catastrophe, the flood/storm and earthquake loadings, total rate, BI rate, net rate, and the final net and gross rates per mille.
- Capacity** — the check at three levels, against the committed book rather than a static territorial budget. A breach is a referral, not a refusal: the underwriter with the authority still binds it and the override goes on the record. "No budget set" is its own state and deliberately neither passes nor fails.
- Class accumulation — fac + treaty** — every bound fac risk and every inforce treaty covering this class, at our share, with the risk on the desk shown separately so its own contribution stays visible.

The one-page read of the risk, the engine output and the capacity decision. Mostly read-only — three fields are yours.

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RISK SNAPSHOT

Cedant

Insured

Region

Country / Zone

Occupancy

Jeddah Petrochemical Complex — Sadara Industrial Park

Middle East

KSA- Western

Chemical Manufacturing (others), Pharmaceutical formulation, Toiletary products

Hazard Grade

Multi-Location

Multi-Occupancy

Inception / Expiry

Renewal / New

7

Yes

Yes

2026-07-01 → 2027-07-01

RENEWAL

UNDERWRITING SCORE

78.70

D

ACCEPT

CAPACITY

Territorial Cap (SAR)

Top Location SI (SAR)

Top Location PML (SAR)

Max Capacity (engine)

0

180,000,000

70,500,000

75.00% = 135,000,000

Carrier Exposure — Top Loc.

Carrier Exposure — All Locs.

Total PD / BI SAR

Total SAR SI

Premium Expected

22,500,000

60,000,000

380,000,000 / 100,000,000

480,000,000

400,585

Capacity Proposed %

e.g. 0.15 = 15% of the layer

RATE

Technical Rate (%a, no NatCat)

Total Rate (%a)

Final Net Rate (%a)

Final Gross Rate (%a)

0.5600

0.6138

0.6217

0.8346

Market vs Tech %

Market vs Tech Band

Technical Premium

Expected Premium

112.59%

More than Equal to 80%

268,800

400,585

Accepted Rate (%a)

The rate actually agreed with

UNDERWRITER NOTE

Narrative on the proposed terms, conditions, or exceptions...

REFERENCE NUMBERS

FAC Reference

Bound Reference

FAC-2026-0001

—

Expiring Reference

Status

FAC/SA/2025/0417

DRAFT

PROVENANCE

Rating Family

Rate Table

SCHEDULE_PROPERTY

FAC-REF-2026.1

Engine Version

Exposure Basis

2.0.0

LOCATIONS

Save Draft

Submit for Approval

Decline

Back

UNDERWRITER

Not yet submitted.

SENIOR UW / CUO

Not yet counter-signed.

← Back to Fac Portfolio

Summary & approval — /fac/risk/summary

ON THIS SCREEN

- **Risk snapshot** — cedant, insured, region, country/zone, occupancy, hazard grade, multi-location and multi-occupancy flags, period, and renewal or new.

- **Underwriting score** — the score, its grade, the action, the scheme used (with or without BI), the market-versus-technical percentage and band, and what proportion of the scoring weight was actually completed.
- **Capacity** — territorial cap, top-location sum insured and PML, the engine's maximum capacity in both percentage and amount, carrier exposure on the top location and across all locations, total PD/BI, total sum insured and the expected premium. **Capacity proposed %** is editable.
- **Rate** — technical rate without natural catastrophe, total rate, final net and final gross rates, market-versus-technical, technical premium and expected premium. **Accepted rate** is editable.
- **Underwriter note** — editable, and it is the note that travels with the referral.
- **Reference numbers** and **provenance** — the fac reference, the expiring reference, the bound reference, the status, and which engine version and rate table produced these numbers.
- If the engine has not produced a score yet the screen says so and sends you back to Pricing rather than rendering a half-priced summary.

8. Quote mode

What changes

A quote is a pre-bind working record with its own reference (QT-YYYY-NNNN) and its own version chain, so an amended quote keeps its lineage back to the original.

The non-proportional quote wizard is a slimmed-down subset of the treaty wizard: treaty detail (itself cut down), documents, the premiums and inflation cockpit, the four-step large-loss and cat-loss workflows, large- and cat-loss development factors, historical performance, risk profile, CRESTA aggregates, and a Final Quote screen in place of Final Pricing. Structure, expiring structure, excess dev factors, claims profile, event loss tables and the audit trail are not part of it.

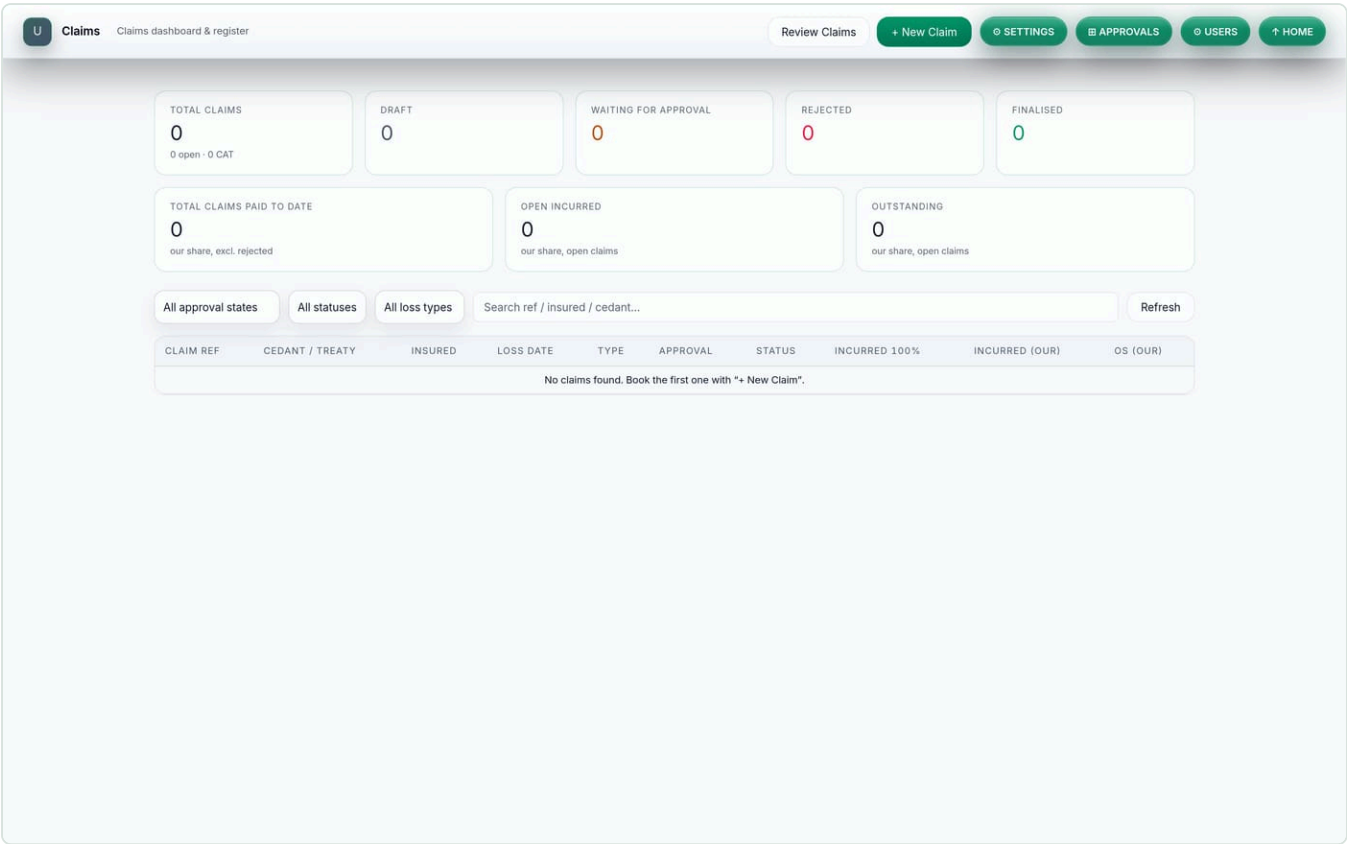
One thing to be aware of: in this build a quote is a standalone artefact. Binding a quote into a contract is intentionally disabled, and a signed quote does not create a contract and does not contribute to portfolio exposure — the dashboard and every accumulation figure aggregate bound contracts only.

9. The supporting modules

Claims

/claims

The treaty claims register. Cedant advices, the movement ledger, reserves and payments — all shown at 100% and at our share.

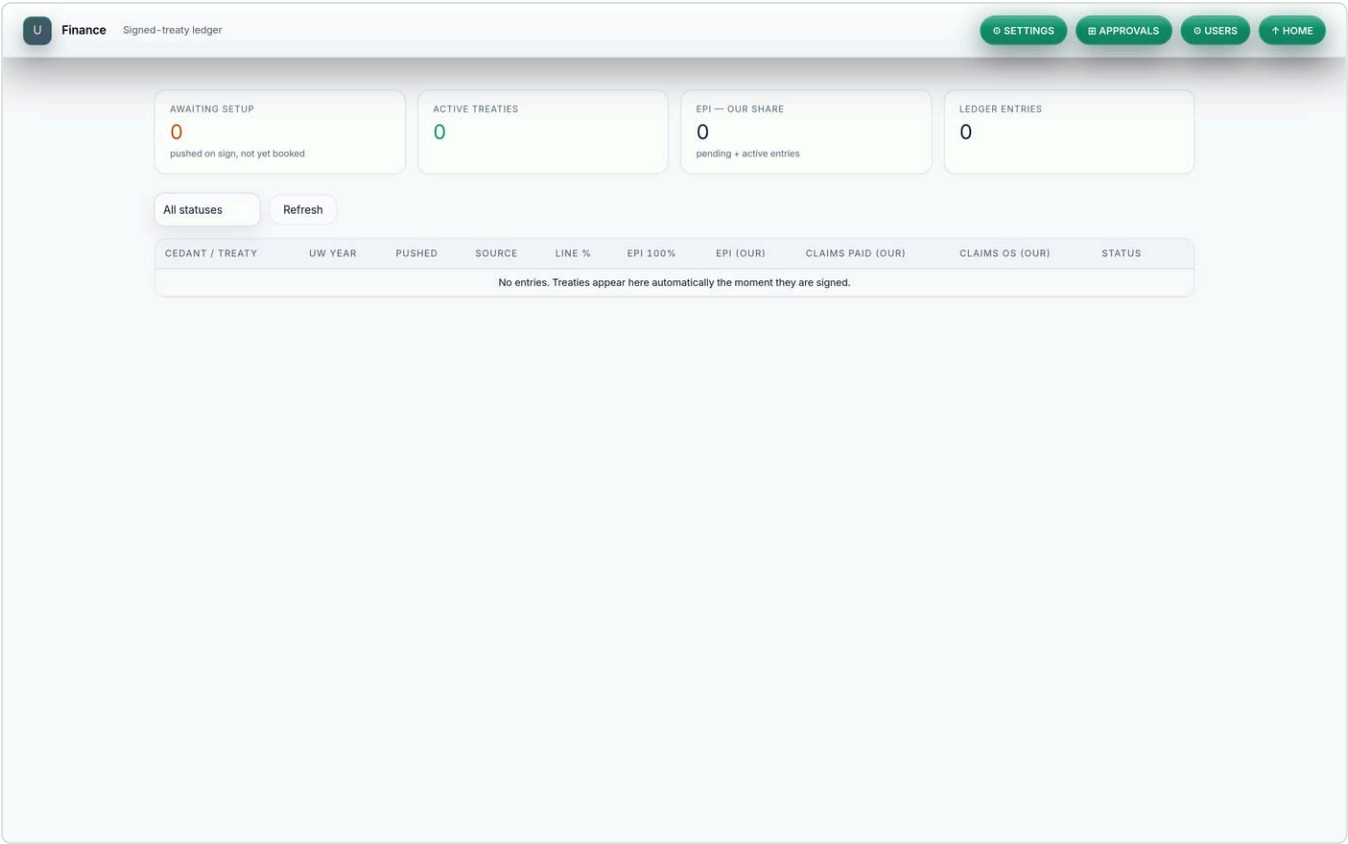


Claims — /claims

ON THIS SCREEN

- Tiles: total claims (open and cat), draft, waiting for approval, rejected, finalised, total paid to date at our share, open incurred and outstanding.
- Filters by approval state, status (open, reopened, closed, declined) and loss type (attritional, large, cat).
- The register lists claim reference, cedant and treaty, insured, loss date, type, approval state, status, incurred at 100%, incurred at our share and outstanding at our share.
- **+ New Claim** books a claim — only against treaties that are SIGNED or BOUND, enforced on the server and mirrored in the contract dropdown.
- Opening a claim gives the header context, the current position, the immutable movement ledger, movement booking, the lifecycle actions (close, decline, reopen) and working notes.

The signed-treaty ledger. Every contract that reaches SIGNED, or is bound from a quote, lands here automatically — the entry is written in the same transaction as the status change, so nothing can be signed and then missed.

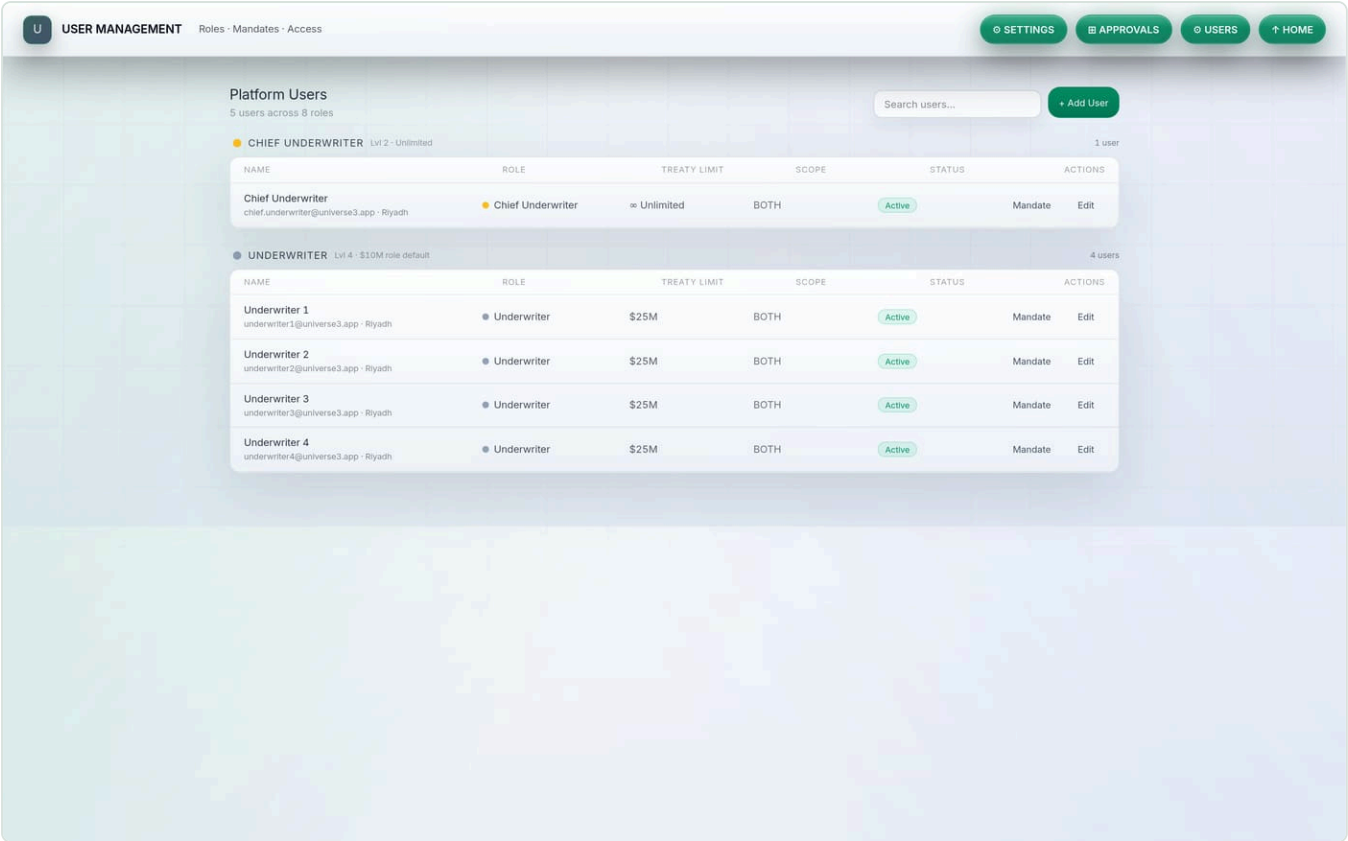


Finance — /finance

ON THIS SCREEN

- Tiles: awaiting setup (pushed on sign, not yet booked), active treaties, EPI at our share, and total ledger entries.
- Entries carry cedant and treaty, underwriting year, when they were pushed, the source, the signed line %, EPI at 100% and at our share, and claims paid and outstanding at our share.
- Finance acknowledges each entry once it is booked in the general ledger (PENDING_SETUP → ACTIVE) and can suspend or close entries thereafter.

Users, roles, mandates and access. Restricted to chief executive and chief underwriter roles.

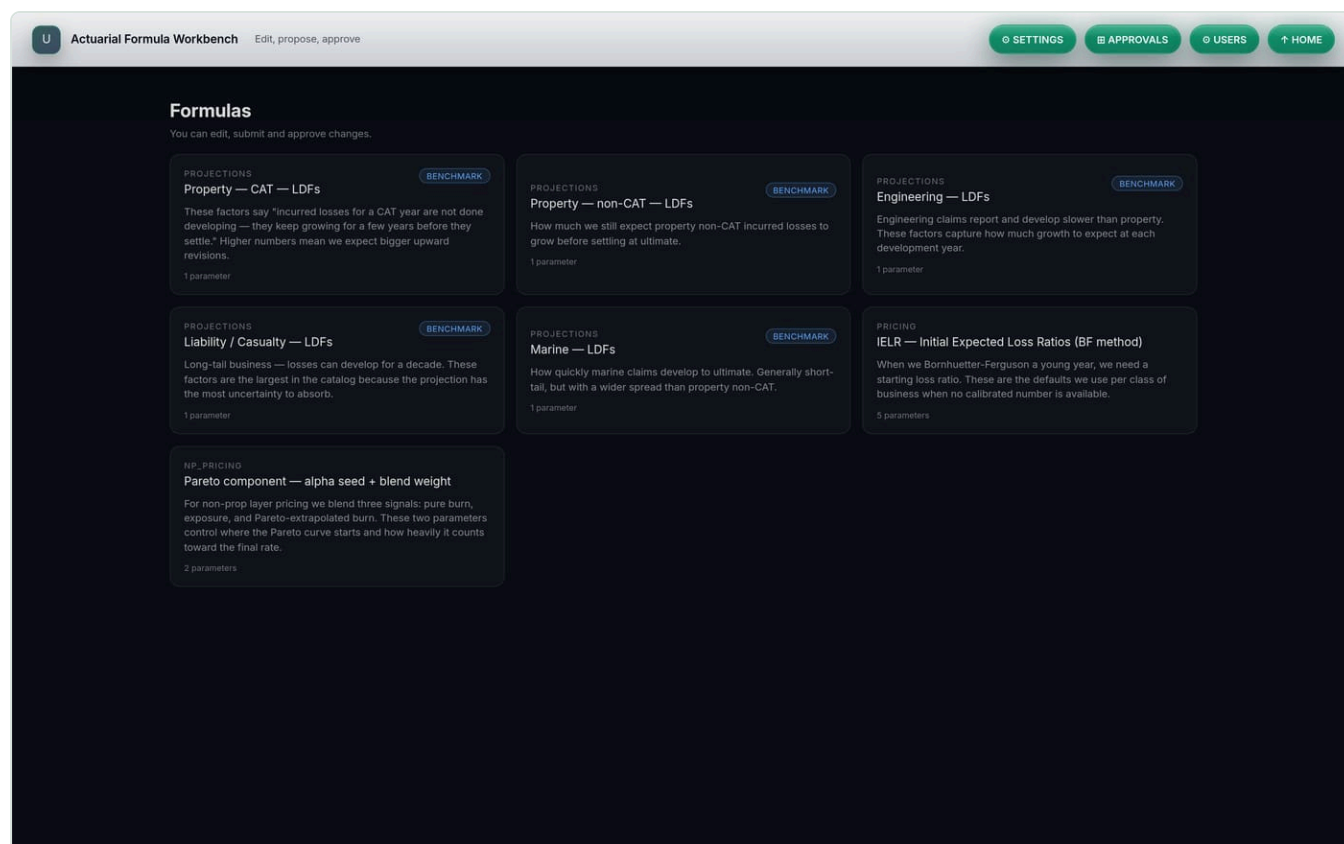


User management — /admin/users

ON THIS SCREEN

- Users are grouped by role with the role's hierarchy level and default authority limit shown on the group header.
- Per user: name, email, office, role, treaty limit, treaty-type scope and active status.
- **Mandate** opens the per-user mandate — the effective authority limit, single-risk limit, treaty-type scope, how many approvals their offers require, and any class-of-business or country restrictions.
- The role ladder runs Chief Executive → Chief Underwriter → Treaty Director → Treaty Manager → Treaty Underwriter, with authority limits attached. An offer above the submitter's limit is routed to the approvals queue.
- New accounts are created with a generated one-time password and must change it at first login.

The parameter catalogue behind the pricing screens — the benchmark development curves, the initial expected loss ratios, the Pareto seeds. Every entry is editable, proposable and approvable, and each one carries a plain-English explanation of what it does.



Actuarial formula workbench — /workbench

ON THIS SCREEN

- **Projections** — benchmark LDF curves per class: property cat, property non-cat, engineering, liability/casualty and marine. Long-tail classes carry the largest factors because their projections have the most uncertainty to absorb.
- **Pricing** — initial expected loss ratios per class, used as the BF starting point when no calibrated number is available.
- **NP pricing** — the Pareto alpha seed and blend weight, and the other non-proportional layer parameters.
- Changes go through propose-and-approve rather than being edited in place, so the parameters that priced last month's treaty can still be reconstructed.

A standalone XL pricing sketchpad: price a tower against a power-law curve calibrated from known expiring terms, without creating a treaty.

US

QUICK BENCHMARK

3rd-Point Geomean Power-Law

Home

Dashboard

TREATY DETAILS

COUNTRY

—Select country—

CEDANT

Select country first

CLASS OF BUSINESS

—Select COB—

TREATY TYPE

Aggregate XL

EGNPI (100%)

e.g. 50,000,000

All

UW YEAR

2026

STRUCTURES TO BENCHMARK

1 structure

Market default (a=0.108, b=-1.074) a=0.10889 b=-1.0748 Add n2 expiring layers to calibrate

EXPIRING STRUCTURE

Known market terms — calibrates the pricing curve. Leave blank to use default coefficients.

+ Layer

#	LIMIT	ATTACHMENT	EGNPI	RATE %	ROL %	PREMIUM	MDP	REINST.	GEOMEAN	X=O/E	
E1				—%	—	—			—	—	x
E2				—%	—	—			—	—	x
E3				—%	—	—			—	—	x

IMPLIED PRICING CURVE

Violet = expiring · Coloured dots = new structure layers priced on curve

Enter layer data to see the pricing curve

STRUCTURE 1

ROL and premium auto-priced from curve · Rate = Premium / EGNPI

+ Layer

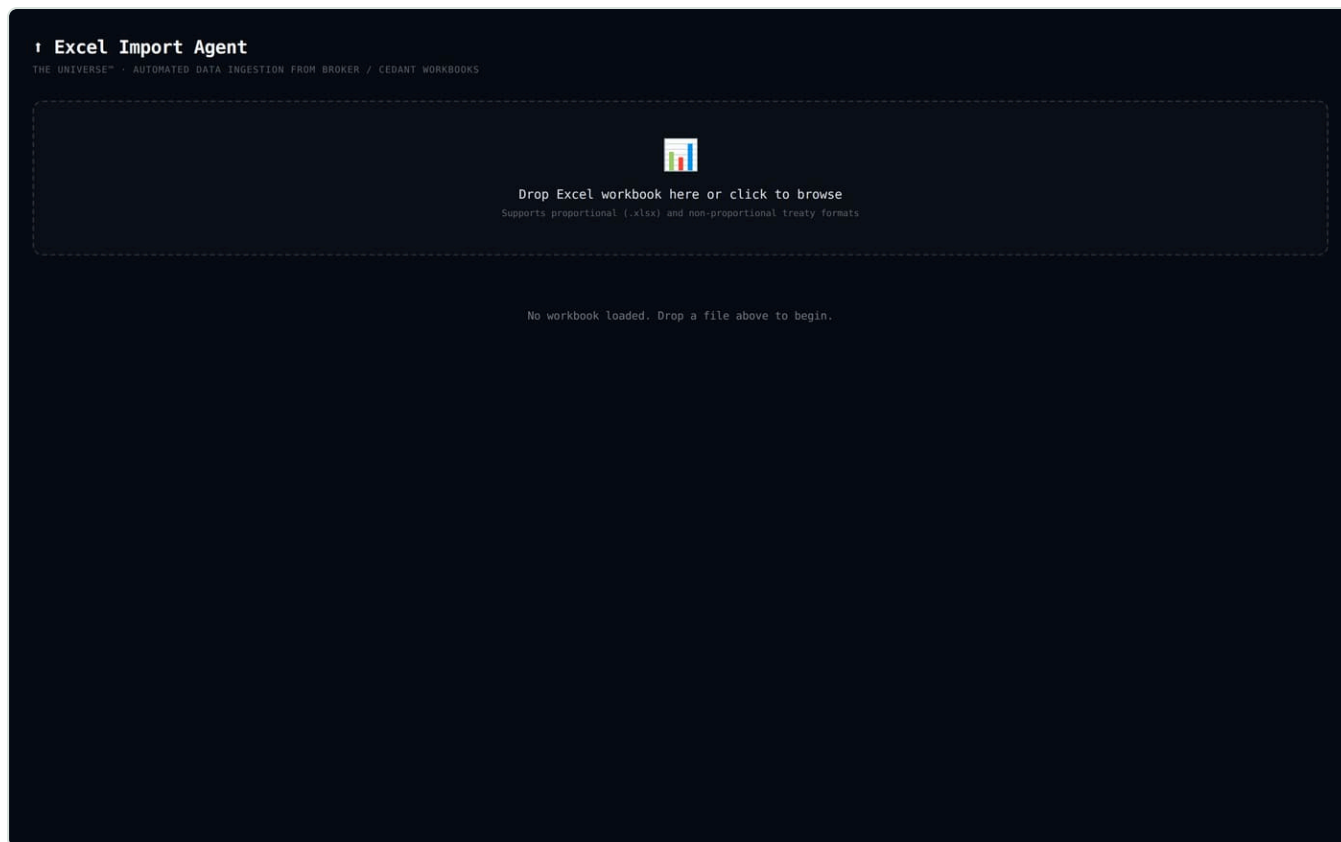
#	LIMIT	ATTACHMENT	PREMIUM	GEOMEAN	X=O/E	ROL % ↗	PREMIUM ↗	RATE % ↗	
L1				—	—	—	—	—	x
L2				—	—	—	—	—	x
L3				—	—	—	—	—	x

Quick benchmark — /benchmark

ON THIS SCREEN

- Set country, cedant, class, treaty type, EGNPI at 100% and underwriting year, and choose how many structures to compare (up to five).
- **Expiring structure** — enter the known market terms layer by layer. Two or more layers calibrate the curve; with fewer, the market default coefficients (a = 0.108, b = -1.074) are used and the screen says so.
- The maths is explicit: $\text{geomean} = \sqrt{((\text{limit} + \text{attachment}) \times \text{attachment})}$; $x = \text{geomean} \div \text{EGNPI}$; expiring $\text{ROL} = \text{rate} \times \text{EGNPI} \div \text{limit}$, which calibrates the curve; priced $\text{ROL} = a \cdot x^b$; $\text{premium} = \text{ROL} \times \text{limit}$; and the weighted ROL is the limit-weighted average across the tower.
- The implied pricing curve is plotted, and the candidate structures are priced against it side by side.

Automated ingestion of broker and cedant workbooks. Drop a file and the agent reads it into the treaty structures rather than making you re-key it.



Excel import agent — /import

ON THIS SCREEN

- Supports proportional and non-proportional treaty workbook formats.
- Progress is reported as it works, and what it produces is a snapshot you review and confirm before it is applied.
- Import snapshots are retained for 30 days and then purged by a scheduled job, so a bad import can always be rolled back within that window.

10. The approval workflow

From draft to signed

A treaty moves through a fixed set of states, and every transition is recorded on the audit screen with its actor and reason.

DRAFT — being modelled. Only the assigned underwriter can edit it.

AWAITING_APPROVAL — an offer has been made from the pricing screen and is sitting in the approvals queue because it exceeds the submitter's authority.

AWAITING_SIGNED_LINE — approved and offered to the cedant or broker; waiting on the market's response.

SIGNED — the line is confirmed. The treaty is pushed into the finance ledger in the same transaction, and it starts contributing to portfolio exposure and accumulation.

NTU — not taken up.

DECLINED — declined, with the reason recorded.

Authority

Each role carries an authority limit and each user can carry a mandate that narrows or widens it, including restrictions by class of business and by country. An offer within your authority proceeds; one above it is routed for approval, and the approvals screen opens the treaty at its pricing screen so the decision is made against the actual numbers.

Peer review sits alongside: an offer can require one or two peers plus an arbiter, and who reviewed what is part of the record.

11. Glossary

Terms used in the interface

AAD	Annual aggregate deductible — an aggregate retention that must be eroded before the layer responds.
AAL	Average annual loss.
Attritional	The working-layer loss experience left once large and catastrophe losses are removed.
BF / Bornhuetter–Ferguson	A projection method that blends the chain-ladder development of reported losses with an a-priori expected loss ratio. Used for immature years where the development alone is unreliable.
Burning cost	The historical loss cost of a layer expressed as a rate on subject premium, on-levelled for rate changes and trended for inflation.
CDF	Cumulative development factor — the product of the LDFs from a given development point to ultimate.
COPE	Construction, Occupation, Protection, Exposure — the four pillars of property risk-quality assessment.
CRESTA	The standard geographic zoning scheme used to accumulate catastrophe exposure.
EGNPI	Estimated gross net premium income — the subject premium an XL layer rates against.
EPI	Estimated premium income.
FGU	From the ground up — losses stated before any reinsurance deduction.
FLEXA	Fire, lightning, explosion, aircraft — the base peril set from which a property rate is built.
ILF	Increased limit factor — how a liability rate scales with the limit purchased.
KS statistic	Kolmogorov–Smirnov — the goodness-of-fit measure used to rank candidate severity distributions. Lower is better.
LCF	Loss carry forward — a deficit carried into the next period's profit-commission calculation.
LDF	Loss development factor — the link ratio from one development period to the next.
MBBEFD	The Maxwell–Boltzmann / Bose–Einstein / Fermi–Dirac curve family used for exposure rating.
MDP	Minimum and deposit premium.
MFL	Maximum foreseeable loss.
NTU	Not taken up.
On-levelling	Restating historical premium at current rate levels by applying the chain of rate changes since.
Payback	The number of clean years a layer needs to repay one total loss — the reciprocal of ROL.
PML	Probable maximum loss.
ROL	Rate on line — premium divided by limit.
SRCC	Strikes, riots and civil commotion.
Stop loss	An aggregate cover attaching at a loss ratio of subject premium rather than at a currency amount.